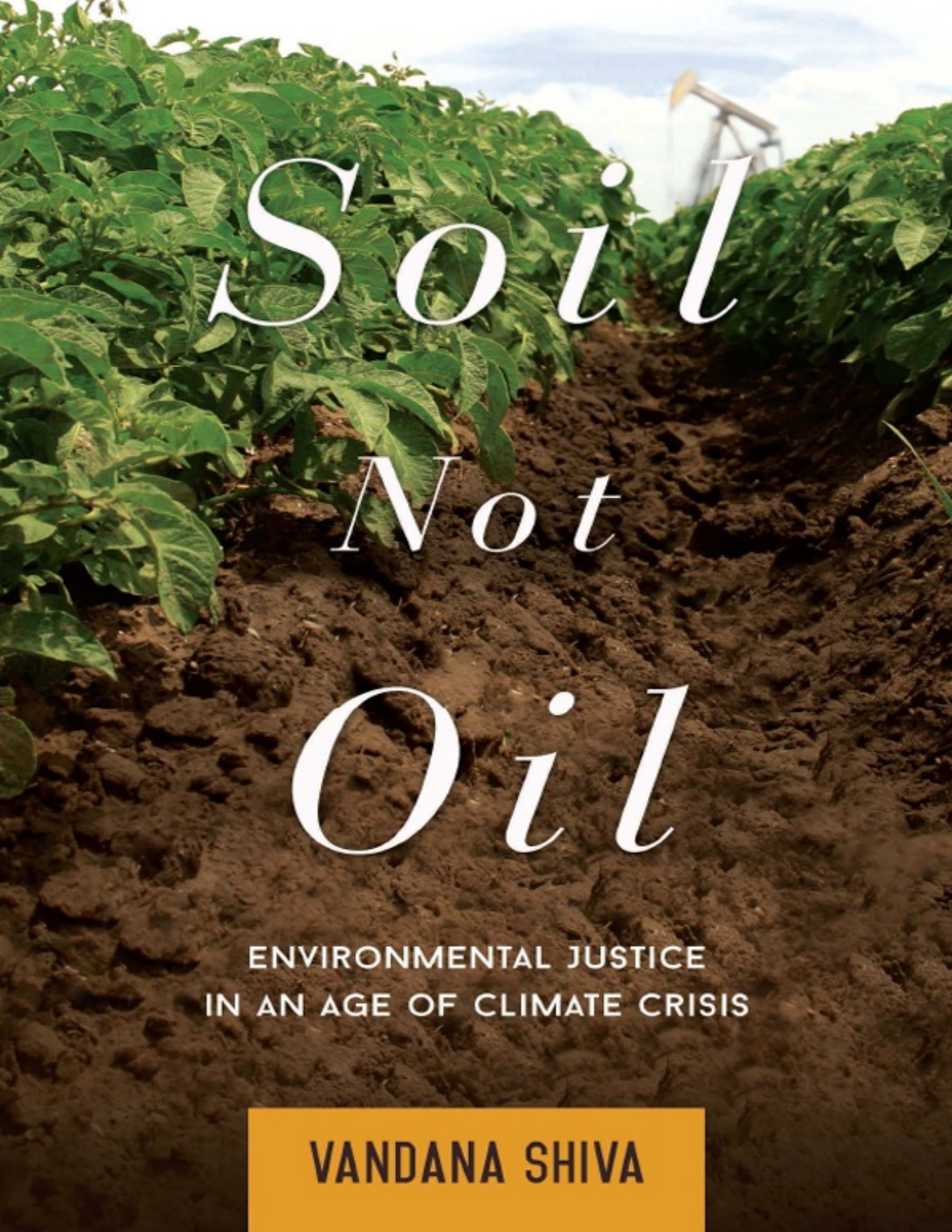




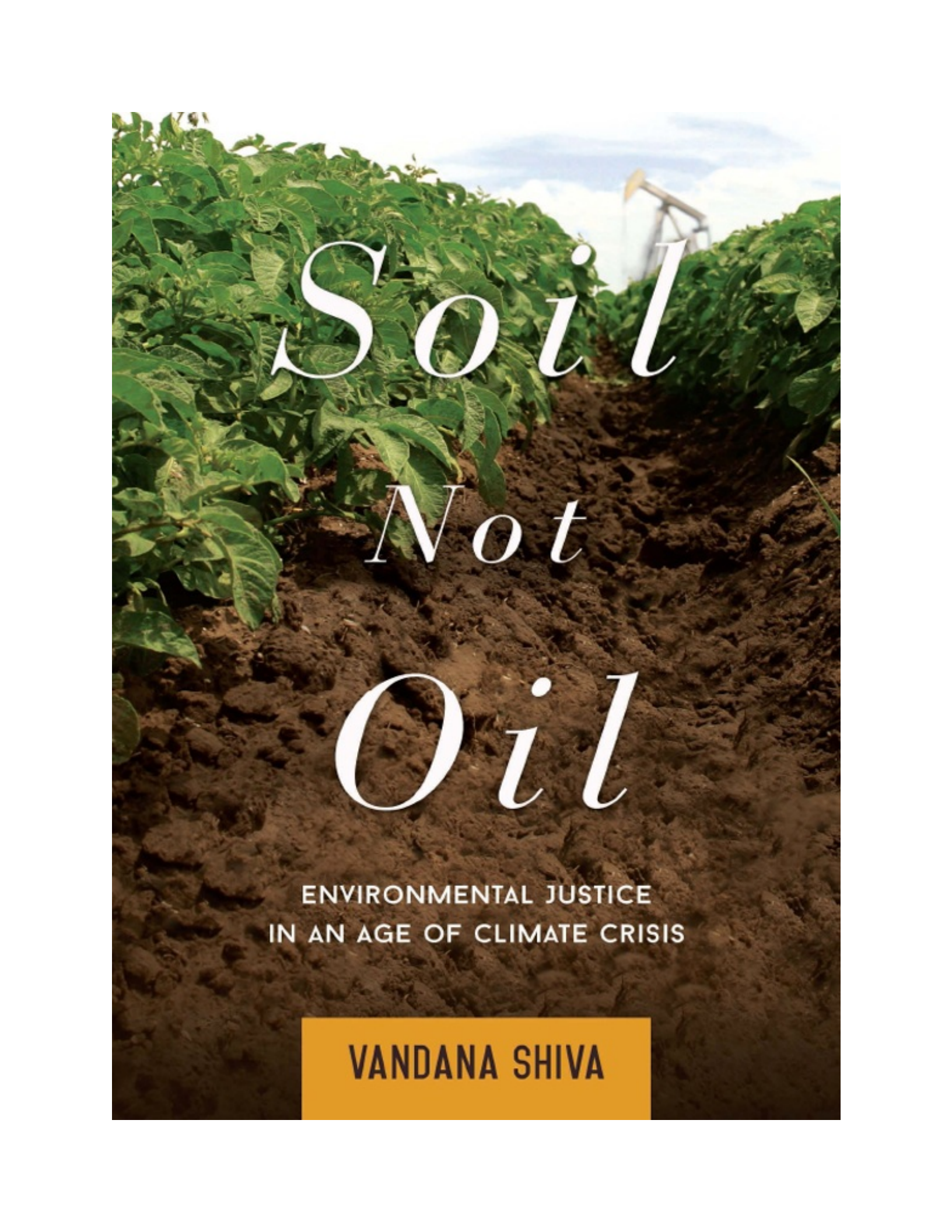
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Oil

ENVIRONMENTAL JUSTICE
IN AN AGE OF CLIMATE CRISIS

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Berkeley, California

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INTRODUCTION

Triple Crisis, Triple Opportunity

Two hundred years into the fossil fuel era, CO₂ emissions have created a greenhouse effect that is responsible for global warming and is leading to a climate crisis. An all too likely increase in temperatures of 3 to 5 degrees Celcius will result in the melting of the polar ice caps and glaciers, and the intensification of floods, droughts, and cyclones. Some of these effects are already being felt. If we do not halt the temperature increase the climate crisis will dramatically change how we live.

And it will decide whether we live or perish.

Besides the problematic “gift” of climate chaos, the age of oil is up against another limit—peak oil. Conceptualized by M. King Hubbert in 1956, peak oil refers to the point at which the world reaches the highest possible level of oil production. After that, oil production must necessarily decrease.¹ Decreasing production will mean increasing prices. The unprecedented increase in oil prices in 2008 is a sign of an emerging oil crisis. Experts such as Jeremy Leggett and Dr. Colin Campbell of the Association for the Study of Peak Oil say we may have already reached the peak, but even if it is a few years off, it *will* occur.² As Heinberg has put it, “the party is over.”

Peak oil and the end of cheap oil make it imperative that we change the way we live. We need to move beyond oil. We need to reinvent society, technology, and economy. We need to do it fast and we need to do it creatively. We can.

Climate chaos and peak oil are converging with a third crisis— the food crisis. The food crisis results from the combined impacts of the industrialization and globalization of agriculture. The very forces and processes that have promised cheap food are pushing food beyond people’s reach. Prices of food are rising worldwide. More than 33 countries have witnessed food riots.

In early June 2008 an emergency meeting of the UN was called to address the crisis of climate change and the food crisis. As expected, the same corporate interests that have created the two crises tried to offer the disease as the cure—more fossil fuel-based chemical fertilizers, more non-renewable genetically engineered and hybrid seeds bred to respond to the intensive use of chemicals, more corporate control of food, and more globalized trade.

The food crisis reflects a deeper crisis—the creation of “redundant” or disposable people and, alongside them, the potential for violence and social and political instability.

Disposability of people is built into the denial of food to millions as well as the destruction of rural livelihoods by the substitution of human energy with machines powered by fossil fuels. The very definition of productivity in the industrial paradigm is labor productivity, i.e., the fewer human beings involved in production, the more “productive” a process is, even if it uses more energy and more resources and produces less per unit of energy and resource inputs.

While wide-ranging wars, colonial expansion, and slavery—among other things—have long resulted in human-generated misery and destruction, never before have the actions of one part of humanity threatened the existence of the entire human species. We are now facing a triple convergence of crises, each of which threatens our survival.

Climate: Global warming threatens our very survival as a species.

Energy: Peak oil spells the end of the cheap oil that has fueled the industrialization of production and the globalization of consumerism.

Food: A food crisis is emerging as a result of the convergence of climate change, peak oil, and the impact of globalization on the rights of the poor to food and livelihood.

Of the three crises, the emerging food crisis poses the most immediate threat to the survival of the poor. The food crisis emerges from two historical processes, one long-term—the industrialization of agriculture and the uprooting of peasants and family farmers from the land—and one more recent—the effects of globalization and trade liberalization of agriculture on food security and food sovereignty. The impact of climate change on agricultural production, along with such false solutions to climate change as industrial biofuels, which divert food and land from the poor to the non-sustainable energy needs of the rich, further exacerbate the food crisis.

We can and must respond creatively to the triple crisis and simultaneously

overcome dehumanization, economic inequality, and, ecological catastrophe.

The energy and climate-change crisis stands as a unique social and ecological challenge. First, the very survival of the human species as a species is threatened. Second, no other challenge is so global in scope. There is no place to hide. Third, climate change is impacted by diverse human activities—how we shop, how we move, how we live, how we eat. Solutions cannot be restricted to one or two sectors. They will touch all aspects of our lives. Mitigation and adaptation must happen across all aspects of our lives. Fourth, climate change results from what is done to the land, and its impacts transform the land. Air, water, land, biodiversity, and energy are intertwined elements of climate change—its cause and solutions. Fifth, those least responsible for climate change are worst affected by it. Peasants, indigenous peoples, and artisans who live outside the industrialized globalized economy, who have caused no harm to the earth or other people, are the worst victims of climate chaos. Over 96 percent of disaster-related deaths in recent years have taken place in developing countries. In 2001, there were 170 million people affected by disasters around the world, of which 97 percent were climate related.³ Sixth, resistance to the limitless destructiveness of the industrialized globalized economy is coming precisely from those least responsible for climate change, the women, the hawkers, and street vendors who stand in front of the juggernaut of fossil fuel-driven, energy- and resource-intensive “development,” refusing to be uprooted, refusing to be turned into disposable people, offering another paradigm and world view—of power and wealth, of nature and culture.

Climate change demands that we reduce fossil fuel use and CO₂ emissions. It also demands that we “power down” through decentralized and decreased energy use. Peak oil and the end of cheap oil demand a paradigm shift in our conception of human progress—we need to imagine how we can live better without oil. The emerging food crisis will add another billion people to the billion who are already denied their right to food and condemned to hunger and malnutrition. The disposability and dehumanization of the poor and the marginal demand that we focus on the dignity of work and the relevance of ecological work. The dominant model of development and globalization is inherently violent because it deprives the poor of their fundamental right to food, land, and livelihoods. By bringing back dignified work based on human energy and living energies we can mitigate climate change and make a transition to a society beyond oil, while ensuring food security and good food for all.

We must, therefore:

- Power down energy and resource consumption

- Power up creative, productive human energy and collective democratic energy to make the necessary transition

Unfortunately, the forces that have given us climate change are using the crisis to further inequality by robbing the poor of their last morsel of food and last inch of land without achieving sustainability.

We have two choices: we can make a nature-centered, people-centered transition to a fossil fuel-free future, with meaningful work and decent and dignified living for all; or we can continue on our current path toward a market-centered future, which will make the crisis deeper for the poor and the marginalized and provide a *temporary* escape for the privileged. The first part of this book deals with the pseudo-solutions; the second part offers equitable and lasting solutions from a people's perspective in the Global South.

Most of the discussions and negotiations on climate change have been restricted to the commercial, consumption-oriented energy paradigm rooted in a reductive, mechanistic worldview and consumerist culture. Within this paradigm there are two dominant approaches: the approach of global business, especially the corporations that have promoted the fossil fuel economy, and the approach of those seeking renewable alternatives to support an energy-intensive consumerist society.

This paradigm, which began in the industrial countries two centuries ago and which is being spread to countries like India through globalization, has given us disposable people, hunger, poverty, cultures of fear and insecurity, and climate chaos.

Fossil fuel-based businesses and industries first tried their best to deny any link between climate change and the economies they built for maximizing their profits. When scientific evidence and people's experience removed the blinders they attempted to put on humanity, they started to propose "solutions." However, these pseudo-solutions will make the crisis worse. Among them are the promotion of non-sustainable energy options such as nuclear energy and industrial biofuels, carbon "offsets," trading and markets in pollution, and reckless technological adventurism, such as placing reflectors in the sky and metal fillings in the ocean, which would further disrupt ecological processes in an attempt to sequester carbon dioxide.

These solutions are designed to maintain the energy-intensive systems of rich industrial societies. The energy transitions they propose work in a limited context—they work in rich countries and they work to maintain systems that are inequitable and non-sustainable in the larger context. They work by increasing the burden on the poor and the burden on the planet. Shifting to industrial

biofuels is a clear example of such misplaced “solutions.” Increasing biofuel production worsens the food crisis by taking land and food from the people in order to produce “feedstock” for the insatiable appetite of the fossil fuel infrastructure and the limitless consumption it requires. From the perspective of the planet and the poor, we need to make a paradigm shift from consumptive energy to productive and regenerative energy, from capital-intensive energy to low-cost energy, from labor-displacing energy to livelihood-generating energy. In other words, the energy transition that the poor need increases the possibilities for meaningful work and decreases the use of fossil fuels. A transition that brings people back into the sustenance economy helps the poor by increasing their security of livelihood and putting more resources back in their hands and helps the planet by reducing carbon dioxide emissions.

“Energy” in the mechanical paradigm, which views nature as a machine and not a living system, is defined as the capacity to do work. However, increasingly, humans are being removed from their role of providing energy, doing work. The poor are victims three times over of the fossil fuel-driven industrial system. First they are displaced from work; then they bear a disproportionate burden of the costs of climate chaos through extreme droughts, floods, and cyclones; and then they lose once more when pseudosolutions like industrial biofuels divert their land and their food. Whether it is industrial agriculture or industrial biofuels, car factories or superhighways, displacement and forced evictions of indigenous peoples and peasants from the land are an inevitable consequence of an economic model that creates growth by extinguishing people’s rights.

Strategies that affirm the rights of the poor to their land and livelihoods are also strategies that reduce our dependence on oil. They help mitigate climate change and help us adapt to climate chaos. Addressing issues of poverty, equity, and justice on a small and finite planet simultaneously addresses peak oil and climate catastrophe.

The most creative and necessary work that humans do is to work with the soil as coproducers with nature. Human effort and knowledge based on care for the soil prevents and reverses desertification, the root of collapse of so many historical civilizations.⁴ Rebuilding soil fertility is the very basis of sustainable food production and food security. There is no alternative to fertile soil to sustain life, including human life, on earth. And, as I show in this book, it is our work with living soil that provides sustainable alternatives to the triple crisis of climate, energy, and food. No matter how many songs on your iPod, cars in your garage, or books on your shelf, it is plants’ ability to capture solar energy that is the root of it all. Without fertile soil, what is life?

The transition from oil to soil is a multidimensional transition of economy,

politics, and culture.

First, it is an economic transition from a fossil fuel-driven, globalized economy—one that favors corporations by subsidizing oil and outsourcing costs—to a network of renewable energy-driven, climate change-resilient, local economies. These living economies are grounded in the soil, literally and metaphorically. They are localized, which reduces our ecological footprint on the planet while enhancing our well-being. Economies rooted in the soil are centered on nature and people. The driving force is maintenance of life, today and in the future. Their currency is not money but life itself.

Second, the transition from oil to soil is a political transition. It is a transition from undemocratic political structures—which impose globalization and a fossil fuel infrastructure on society and force the large-scale uprooting of peasants and indigenous peoples—to a decentralized democracy in which local communities have a say in what happens to their land and their lives. In this sense, soil is a metaphor of decentralized and deep democracy. As David Bosshart writes, “Consumer democracy is the gasoline for the bulldozer of globalization.”⁵ Consumer democracy is a pseudodemocracy associated with economic dictatorship; it desertifies the soil of real democracy. Authentic democracy, like plants, grows from the ground up. It is fertilized by people’s participation.

Third, the transition from oil to soil is a cultural transition—from a deadly consumerism to reclamation of our rightful place as cocreators and coproducers with nature. The shopping mall and the supermarket are temples of consumerism through which global corporations seduce us into participating in the destruction of our productive capacities, our ecological rights, and our responsibilities as earth citizens. Soil teaches us how to be earth citizens. And for the half of humanity that works the soil as peasants, the soil is also protector. As globalization violently pushes peasants off the land, the soil symbolizes another culture, a culture of non-violence, a culture of permanence, a culture of dignity in work, a living culture for the protection and renewal of life.

The convergence of these three crises provides us with the convergence of three opportunities—to create living economies, living democracies, and living cultures. Earth Democracy grows in the fertile soil shaped by the earth, the human imagination, and human action.

The age of oil has symbolized a rule of capital, of centralized control and coercive government, of pollution and non-sustainability, of injustice and inequity, of violence and war. The age of soil symbolizes the age of Gaia, of the flowering of diversity and democracy, of justice, sustainability, and peace.

We will either make a democratic transition from oil to soil or we will perish. The poor, the weak, the excluded, the marginalized are threatened today. In the

short term, we can continue to extend the profits and consumerism of the privileged by further dispossessing the poor. But tomorrow even the rich and the powerful will not be immune from Gaia's revenge and the revenge of the billions of dispossessed. We will either have justice, sustainability, and peace together or we will descend into ecological catastrophe, social chaos, and conflict.

Soil, not oil, offers a framework for converting the ecological catastrophe and human brutalization we face into an opportunity to reclaim our humanity and our future.

CHAPTER ONE

Politics of Climate Change

Eco-Imperialism vs. Earth Democracy

Climate Change Is Happening

The Intergovernmental Panel on Climate Change (IPCC) has recognized that since 1750 the net effect of human activities on the earth's climate has been one of warming. Certainty about the anthropogenic basis for climate change has gone from greater than 66 percent to greater than 90 percent. The 2007 report of the IPCC, which had the participation of 2,500 scientists from 130 countries, confirmed that man-made climate change is threatening life on Earth.¹ As the IPCC report states, the "warming of the climate system is unequivocal, as is now evident from observations of increases in global average air and ocean temperatures, widespread melting of snow and ice and rising average sea level."²

The changing climate is a result of air pollution from greenhouse gases (GHGs). "Global GHG emissions have grown since prehistoric times, with an average increase of 70 percent between 1970 and 2004."³ Carbon dioxide (CO₂) is the most important anthropogenic greenhouse gas. Its annual emissions have increased from 21 to 38 gigatonnes (GT) between 1970 and 2004, an 80 percent increase. CO₂ pollution from fossil fuels accounted for 77 percent of total GHG emission in 2004. The largest growth of emissions has come from the energy supply, transportation, and industrial sectors. Other significant greenhouse gases include methane (CH₄) and nitrous oxide (N₂O).⁴

By studying ice cores spanning thousands of years, scientists have been able to determine historic pollution levels. Atmospheric concentration of CO₂ increased from a pre-industrial concentration of about 280 parts per million to

379 parts per million in 2005.

The global atmospheric concentration of CH₄ has increased from a pre-industrial concentration of 715 parts per billion to 1,774 parts per billion in 2005. The global atmospheric concentration of N₂O, largely due to use of chemical fertilizers in agriculture, increased from about 270 parts per billion to 319 parts per billion in 2005.⁵

The impacts of this atmospheric pollution are already being felt. The decade spanning from 1998 to 2007 is the warmest on record and the eleven warmest years since surface temperatures began to be measured in 1850 have occurred in the past 13 years.⁶ As average temperatures rise, ice melt increases, leading to sea-level rise. Globally, the sea level rose at an average rate of 1.8 millimeters per year from 1961 to 2003 with an average rise of about 3.1 millimeters per year from 1993 to 2003.⁷

Satellite data shows that since 1978 arctic ice has shrunk by 2.7 percent per decade. Worldwide mountain glaciers and snow are experiencing less accumulation and higher rates of melting.⁸ The Greenland ice sheet, which extends 1.3 million square kilometers and has an average thickness of 2 kilometers, could melt completely if the global average temperatures were to increase by 1.9 to 4.6 degrees Celsius. Such melting could lead to a sea-level rise of 7 meters.⁹ The resulting displacement of coastal and island communities would be a human tragedy on an unimaginable scale.

Glaciers that feed many rivers of the world are melting. When they disappear, rivers will run dry.

This Is What Climate Change Looks Like

Climate change is not just a problem for the future. It is impacting us every day, everywhere. Climate change has resulted in an increase in droughts, floods, and tropical cyclones—what are known as “extreme events.” This increase has already begun. The intensity and frequency of hurricanes and cyclones is increasing. Asia was hit by the Orissa supercyclone in 1998; it killed 30,000. In November 2007, Cyclone Sidr hit Bangladesh with wind speeds of 260 kilometers per hour; it killed 4,400 and displaced 4 million. In May 2008, Cyclone Nargis devastated Myanmar. At least 84,537 people were killed and another 53,836 went missing. The intensity of Hurricane Katrina, which devastated the Gulf Coast in August 2005, is also linked to climate change.¹⁰

The higher the speed of a cyclone, the more destructive its force. Globally, category 4 and 5 storms were 50 percent more frequent between 1990 and 2004 than they were between 1975 and 1989.¹¹

In 2003, a heat wave in August led to the deaths of 50,000 people in Europe, including 13,000 in France. Nine hundred died in England, more than 1,300 in Portugal, 8,600 in Spain, 4,600 in Netherlands, and 1,000 in Germany and Switzerland. Italy, which initially reported 8,000 deaths, later raised its death toll to 20,000.¹²

In January 2008, unprecedented snowstorms crippled China. Sixty centimeters of snow covered parts of Xinjiang. More than 100,000 people were evacuated after their homes collapsed under heavy snow. Temperatures plummeted to 43 degrees below zero Celcius. Twenty-one people died; 5,000 people were treated for frostbite.

Extreme droughts, extreme floods, and extreme cyclones are part of the destabilization of the climate due to greenhouse gases.

Global warming is already affecting glaciers in the Himalayas. The Pindari glacier is retreating at a rate of 13 meters a year, and the Gangetic glacier, the source of the Ganges, is retreating at a rate of 30 meters annually. In 13 years it has receded by one third of a kilometer. In a few decades there will be no glacial melt to feed the Himalayan rivers in the peak of the summer, further aggravating drought.¹³

Extreme rain events have grown both in number and intensity over large parts of Central India in the space of half a century. An analysis of rainfall during monsoon season from 1951 to 2000 in an area of about 1.4 million square kilometers in Central India shows that episodes of “heavy” rainfall (10 centimeters or more of rain in a single day) had increased at the rate of 10 percent per decade. Instances of “very heavy” rainfall (15 centimeters or more of rain in a day) more than doubled between 1951 and 2000. In addition, the average intensity of the four heaviest rain events during each monsoon had grown from 18 centimeters in 1951 to 26 centimeters in 2000. These empirical trends have been modeled by the Pune-based Indian Institute of Tropical Meteorology and they fit with climate change projections.¹⁴

While rainfall is becoming more extreme, so are droughts. Bundelkhand, a region divided between Uttar Pradesh and Madhya Pradesh in Central India, recorded only 350 to 500 millimeters of rainfall annually over the last five years a decrease from 1,000 millimeters annually.¹⁵ Annual rainfall has decreased and whatever rainfall does occur is concentrated in fewer days. Intensive rain leads to intensive flooding. Furthermore, rainfall patterns have shifted away from the

periods when they are most beneficial for crops.

The current dry spell in the region is the longest and most severe in the region's recorded history. Government records show that Bundelkhand only experienced 12 years of drought total during the entire 19th and 20th centuries. But the region has experienced drought for the past five years.

Seventy-five percent of the people in Bundelkhand depend on agriculture, and more than 80 percent are small and marginal farmers. Due to the extended and extreme drought, agriculture has virtually come to a halt. Locked houses are a common sight. In the past five years crop failure and debt have driven more than 400 farmers to commit suicide in the seven districts that make up the Uttar Pradesh part of Bundelkhand. Starvation deaths have been reported.¹⁶

Eighty-five year-old Ram Khilawan, a resident of Bundelkhand, says he has never seen such drought in his life. He owns 1 acre of land, which had allowed enough production to feed his entire family and earn Rs 75,000 from the sale of the produce. Five years of drought have left him and his family without food and livelihood.

Five years of drought means no crops and no food. No food means malnutrition and hunger. People are surviving on a roti with salt or onion. The consumption of vegetables, pulses, and dairy items has vanished. Earlier people had chhach (cultured buttermilk) in plenty. With the drought, there is no fodder, therefore no dairy animals and no dairy products. According to a field survey carried out by Navdanya, more than 95 percent of the people in the villages surveyed in Bundelkhand were not getting enough food to eat.¹⁷

- In the monsoon of 2007, the heaviest in 30 years, more than 1,900 people died in floods in South Asia; more than 12 million were displaced in the Indian state of Bihar and 30 million across 20 states. India's death toll between June and early August was more than 1,500. Sixty-eight thousand livestock animals perished. Total value of the losses came to Rs 12 billion.
- On July 26, 2005, Mumbai received 94.4 centimeters of rain in 24 hours. Twenty-five people were drowned after being trapped in their cars and more than 100 were killed during the flooding. Civic and public-transport services collapsed, and 5,000 army troops were deployed. Financial losses amounted to Rs 17 billion.
- Jaisalmer, a town in the Thar Desert in Rajasthan, normally receives 165 millimeters of rain annually. In 2005, the rainfall was only 148.1

millimeters, which forced the state government to declare drought. However, in August 2006, Jaisalmer received 411 millimeters of rain, 207 of which were received during a three-hour period. Barmer, which receives 277 millimeters annually, received 700 millimeters in 36 hours, which submerged more than 800 villages, affecting 2 million people. And while the desert was flooding, Cherrapunji, the wettest place on earth, was suffering a drought.

The people of Rajasthan and Bihar did not contribute to climate change. Yet they are its victims. The catastrophic impacts of climate change are not only going to take place in the distant future. They are taking place now.

Eco-Imperialism and False and Unjust Solutions to Climate Change

The approaching climate crisis is a consequence of nearly two centuries of dependence by industrialized countries on fossil fuels. Industrialization, in fact, is equated with a transition to a fossil fuel economy. Unfortunately, development too has been defined as industrialization.

In biology, the term *development* refers to self-directed, self-regulated, and self-organized evolution from within. In the terms of Chilean scientists Humberto Maturana and Francisco Varela, systems that self-organize and self-renew are *autopoietic* systems.^{18 19} And in the language of David Pimentel of Cornell, autopoietic systems are based on endosomatic or metabolic energy. If the economic domain were to think of development in the same way, it would lead to a flourishing of biodiversity and cultural diversity. Development would conserve resources and energy while improving human well-being and human welfare.

Unfortunately, *development* in economics has the opposite meaning. In economics, development is an externally driven process. It refers to self-organizing, self-regulating systems as “undeveloped” and “underdeveloped” and suggests that they should be made dependent on external inputs—external resources, energy, and money. Living systems, living societies, living cultures are thus transformed into mechanical systems, or, in Maturana and Varela’s terms, into *allopoietic* systems—systems run from external sources. In energy terms these are based on exosomatic energy. Systems that are autopoietic and endosomatic need no external energy inputs. They are self-organizing and self-

generative. They are models for a post-oil future.

Indigenous communities are fully aware of the authentic meaning of development. Years ago, I worked with communities resisting displacement from dams being built on the Suvernarekha River in what is now the largely tribal state of Jharkhand. Niti Mai, a tribal woman, had this to say about the dams as development: “Real development only happens when the people exercise their rights. We shouldn’t have to give up our rights for someone else’s benefits. We want development, not destruction.”

Development cannot be defined by the colonizer, by those imposing allopoetic systems on society for their own ends—profits and power. Development *must* be defined autopoetically, from within. Local communities must define what they see as their development. When street vendors resist Wal-Mart, it is their definition of improvement of retail that should count, not Wal-Mart’s. When peasants resist Special Economic Zones and indigenous people resist mines, it is they who should define development, not the automobile, real estate, and mining corporations. This has always been a justice imperative. It is now also a climate imperative.

In a period of peak oil and climate change, we need to delink development from industrialization. We need to de-link it from oil. We need to relink it to the maintenance of living systems on the basis of living energies. We need to relink it to soil.

We need to do this not just to mitigate climate change. We need to do it so we can adapt. And we need to do it to establish social and ecological justice.

Self-organizing, self-regulated autopoetic systems are diverse and multidimensional. They display structural and functional diversity. They can heal themselves and adapt to changing environmental conditions. Mechanically organized industrial systems are designed externally. They are structurally uniform and functionally one-dimensional. Mechanically organized systems do not heal or adapt; they break down under stress. When a twig of a tree breaks, the branch heals itself. When a part of a car breaks, the car cannot repair itself. It must be taken to the mechanic.

Mechanical-industrial systems are responsible for the increasing amount of greenhouse gases in the atmosphere and the decreasing ability of societies and ecosystems to adapt to the resulting climate change.

The imposition of the mechanical-industrial paradigm for production and distribution of food, clothing, and other basic needs in the South was initially carried out by the World Bank and IMF through “development” aid. It is now imposed through World Bank/IMF structural adjustment programs (SAPs) and the WTO rules of so-called free trade, which are, in effect, rules for the freedom

to destroy resources, deplete energy, and pollute the atmosphere.

Globalization is, in effect, the globalization of energy-intensive, resource-wasteful, fossil fuel-driven industrialization of our production and consumption patterns. Globalization forces non-sustainability on the world. The players are neither individual citizens nor individual countries. The players are global corporations who move production of goods around the globe to where they can obtain the highest profit margins by bearing the lowest costs.

In the face of climate change, corporate globalization advocates business as usual: allow corporations to gain increasing control of the earth's resources—energy, water, air, land, and biodiversity—to continue to run the industrialized globalized economy. This path can only last a few decades, and at very high social, economic, and ecological costs. I call this path eco-imperialism and it is totally undesirable.

Eco-imperialism is a complex dynamic. It includes the control over the economies of the world through corporate globalization and transforms the resources and ecosystems of the world into feedstock for an industrialized globalized economy. It contains the oil wars being fought in the Middle East, Central Asia, and Africa and the new land and food wars triggered by the emergence of industrial biofuels. Eco-imperialism also characterizes the control over the foreign policy and strategic security policy of countries like India through the recent US-India nuclear agreement. And it refers to the attempts to engineer the planet.

Eco-imperialism is a mechanistic paradigm, based on industrial technologies and economies that assume limitless growth. It is the poor and other species who, in a world of limited resources, lose their share of the earth's resources through overexploitation by the rich and powerful. Instead of restraint and limits, the imperialist project seeks to increase corporate control over resources. Ecoimperialism is intolerant of the freedom and sovereignty of the other, be it other communities, other countries, or other species. In the contemporary context of globalization, peak oil, and climate change, eco-imperialist responses do not solve the problem of climate chaos. Rather, they allow corporations and rich countries to take over the resources and policy spaces of the poor.

Carbon Trading: Privatizing the Atmospheric Commons, Creating a Supermarket of Pollution

Polluting the atmosphere is an enclosure of the commons. What was once a

resource available to all, the atmosphere, has been privatized by the oil and coal companies, the automobile and power companies, as a place to dump their pollutants. The buildup of carbon dioxide from coal and oil has deprived humans and other animals of their share of a clean, unpolluted atmosphere. It is the poor, those who have contributed the least to the degradation of the atmosphere and the destruction of its capacity to recycle carbon that bear the heaviest costs.

In the wake of the first privatization of the atmosphere, we now face a second privatization—this one offered as a “solution” to the first. The second enclosure is the carbon and emissions trading schemes that were the main outcomes recommended by the Kyoto Protocol and the Stern report. In one of the most widely read and discussed economic assessments of global warming, Sir Nicholas Stern, a one-time chief economist at the World Bank, stated, “The very basis of emissions trading is assigning property rights to emitters and then allowing these to be traded.”¹⁹ Larry Lohmann, the author of *Carbon Trading*, shows how market solutions to climate change, whether embodied in US pollution trading programs, the Kyoto Protocol, or the EU Emissions Trading Scheme, give “property rights” to the atmosphere “to a selection of historical polluters—wealthy countries and companies—for free.”²⁰

Indiana law professor Daniel Cole notes, “The allocation of marketable pollution permits constitutes a form of limited privatization as the government conveys to private parties limited entitlements to use the public atmosphere.”²¹ PricewaterhouseCoopers, notorious for working with the World Bank to privatize Delhi’s water (a move defeated by a massive mobilization of Citizens Against Privatization through the Water Democracy Movement), states that “trade in CO₂ emissions is equated with the transfer of similar rights such as copyrights, patents, licensing rights and commercial and industrial trademarks.”²²

The Kyoto Protocol allows industrialized countries to trade their allocation of carbon emissions among themselves (Article 17). It also allows an “investor” in an industrialized country (industry or government) to invest in an eligible carbon-mitigation project in a developing country in exchange for Certified Emission Reduction Units that can be used to meet its obligations to reduce greenhouse gas emissions. By developing green abroad, corporations get credits that allow them to pollute at home. This is referred to as the Clean Development Mechanism (COM) under Article 12 of the Kyoto Protocol. The Kyoto Protocol gave emissions rights to the 38 industrialized countries that are the worst historical polluters. The European Union Emissions Trading Scheme (ETS) rewarded 11,428 industrial installations with carbon dioxide emissions rights.

Through emissions trading, Larry Lohmann observes, “rights to the earth’s carbon cycling capacity are gravitating into the hands of those who have the most power to appropriate them and the most financial interest to do so.”²³ That such schemes are more about privatizing the atmosphere than preventing climate change is made clear by the fact that the emission rights given away in the Kyoto Protocol were several times higher than the levels needed to prevent a 2 degree Celcius rise in global temperatures.²⁴

Just as patents generate supeprofits for pharmaceutical and seed corporations, emissions rights generate superprofits for polluters. The ETS granted allowances of 10 percent more than 2005 emission levels; this translated to 150 million tons of surplus carbon credits, which with the 2005 average price of \$7.23 per ton translates to over \$1 billion in free money.

The UK allocations for British industry added up to 736 million tons of carbon dioxide and required no commitment to reducing emissions. Since the restrictions being put on Northern industrial polluters do not call for a reduction in emissions, they will continue to pollute and there will be no reduction in CO₂ emissions. Market solutions in the form of emissions trading run counter to the bedrock environmental principle that the polluter should pay. Through emissions trading, private polluters are getting more rights and more control over the atmosphere, which rightfully belongs to all life on the planet. Emissions trading “solutions” pay the polluter.

Carbon trading is based on inequality because it privatizes the commons and because it claims the resources of poorer people and poorer regions as “offsets.” In the global market it is considered to be 50 to 200 times cheaper to plant trees in poorer countries to absorb CO₂ than to reduce emissions at the source.²⁵ And the *Stern Review* states, “Emissions trading schemes can deliver least cost emissions reductions by allowing reductions to occur wherever they are cheapest.”²⁶ In other words, the burden of “cleanup” falls on the poor. In a market calculus, this might appear efficient. In an ecological calculus, it would be far more effective to reduce emissions at the source. And from an environmental justice perspective, it is perverse to burden the poor twice—first with the externalized costs and climate disasters caused by the pollution of others and then with the burden of remediating the pollution of the rich and powerful.

Only 2 percent of the CDMs of the Kyoto Protocol covers renewable energy projects. Seventy-two percent of the projects are based on carbon capture and 21 percent on biomass. Solving the problem of air pollution has thus translated into a land grab. As Lohmann points out, “The new carbon dumps that large polluters

need usually have to appropriate someone else's land, someone else's water or someone else's future."²⁷

Since the land and biomass that carbon traders seek to appropriate perform life-support and livelihood functions for local communities, emissions trades are not just based on the theft of the air as commons, they are also based on the theft of people's land and biodiversity. This is what is happening with biofuel plantations. First the poor lost climate security, now they are losing their livelihood security through market solutions to climate change.

The *Stern Review* is clear on the severity of climate change: "Climate change is a serious global threat, and it demands an urgent global response." However, the "urgent global response" Stern proposes is emissions trading. "Expanding and linking the growing number of emissions trading schemes around the world is a powerful way to promote cost-effective reductions in emissions and to bring forward action in developing countries: strong targets in rich countries could drive flows amounting to tens of billions of dollars each year to support the transition to low-carbon development paths."²⁸

There are two flaws to this response. First, the *Stern Review* continues to depend on market solutions when market failure has helped create the crisis. Second, by simply focusing on "carbon," the report fails to differentiate between the dead carbon in fossil fuels and the living carbon of biodiversity and renewable resources. It focuses on carbon emissions, rather than addressing the health of the carbon cycle more broadly. It does not differentiate between the high carbon dioxide buildup when fossilized carbon is burned and the recycling of carbon fixed through photosynthesis in plants and the food chain. And the limited focus on carbon emissions allows ecologically unsustainable technology such as nuclear energy and hydroelectric energy based on large dams to be offered as a "low carbon development path."

The *Stern Review* is focused on what is good for business, not on what is good for the planet and the poor. It therefore proposes business as usual for the rich and powerful. The carbon allocations of the ETS are allocations to companies. They are not allocations to all human beings. Stern has called the ETS the "world's largest greenhouse gas emissions market."

Creating a market in pollution is ethically perverse. Some things should not be tradable—water and biodiversity are too valuable to be reduced to marketable commodities. Other things, like toxic waste and greenhouse gases, should not be generated. To turn them into tradable commodities ensures that they will continue to be produced. Instead of putting a value on clean air, emissions trading schemes value pollution. The European market for carbon trade pollution

is estimated at \$115 billion over the three-year period from 2005 to 2007. Instead of defending the rights of all citizens to clean air and climate security, emissions trading defends the rights of corporations to pollute the atmosphere and destabilize the climate.

Market fundamentalists are so committed to defending and enlarging markets that even when it is clear that climate change demands a radical change in our ways of thinking and our patterns of production and consumption, their preoccupation is with protecting market mechanisms, not Gaia's ecological processes. They are busy creating a supermarket of pollution while glaciers and ice caps melt, hurricanes and tornadoes tear apart coastal ecosystems and communities, and temperatures rise.

Carbon trading is also ecologically perverse. The approaching climate crisis demands that we adopt post-fossil fuel, postindustrial pathways for production and consumption of goods and services. Emissions trading and carbon trading promote fossil-fuel dependence and industrial patterns of production instead of ecological patterns.

Carbon trading allows polluters to keep polluting by buying credits. The credits do not value the least polluting activity or pathway available but instead reward tiny reductions in heavily polluting activity. These credits can in turn be sold to a buyer who can then use them to pollute over its allocation. Factories and plants that exceed their allocation are required to buy credits, but not cut their pollution.

Ecologically, carbon trading is a false solution at many levels. It does not begin with policies and laws that protect and support nonpolluting patterns of production, distribution, and consumption. In fact, ecological options disappear from the "carbon market." The "trade" is between polluters—those who continue to pollute, and those who have partially reduced pollution. Nonpolluters are excluded. Economic actors that *never* polluted were never allocated credits and therefore are never able to sell them. There is nothing to encourage truly sustainable development.

Carbon trading transforms the ethical and ecological problem of reducing our carbon footprint into a financial opportunity to make billions. In 2006 the carbon market was worth \$30 billion and could grow to \$1 trillion within a decade. According to Louis Bradshaw, a banker with Barclays, "Carbon will be the world's biggest commodity market, and it could become the world's biggest market overall."²⁹ Nature's economy has been forgotten. Carbon circulates on Wall Street, not in a carbon cycle.

Carbon credits have been given by the governments of Europe to the five dirty industries—electricity, oil, metals, building materials, and paper—through

the ETS. They have also been allocated through the CDMs set up under the Kyoto Protocol. These schemes seek to reduce greenhouse gases by adopting greener and more efficient technologies while also providing incentives for developing countries to improve their technologies. Under the CDM, for each ton of emissions reduced, the developing country gets a carbon emission certificate, which it is then able to sell through free market channels. The developed countries, instead of reducing their own emissions, can buy these certificates to meet their targets under the international climate change treaty.

The global carbon market was \$30 billion in 2006, a massive increase from \$11 billion in 2005. According to the World Bank \$5 billion of that trade came from CDMs.³⁰ But the activities that generate this trade, such as factory farming in Brazil and refrigerator production in China, are in fact highly polluting.

The waste that factory farms produce pollutes water and land. They also pollute the air with methane emissions. Bunge, a corporation that sells feed to factory farms, has earned carbon credits by using the methane to generate electricity. Bunge has 40 such “carbon credit” projects in Brazil and is planning to expand to Mexico, Guatemala, Peru, and the Philippines. And while some carbon emissions are mitigated, the perversion that is built into carbon trading is that it subsidizes the development of polluting industries by allowing slightly greener industries to profit from selling credits to even more polluting industries. It’s a polluter-gets-paid system. The “clean” production that should earn the credits is the small holder raising free-range pigs, not the highly polluting hog farms of Bunge.

China has received the majority of emissions-trading funds. One controversial Chinese CDM project destroys HFC-23, a potent greenhouse gas whose effect on global warming is, ton for ton, 11,700 times greater than that of carbon dioxide. HFC-23 is a byproduct of the production of HCFC-22, a refrigerant. The UN CDM board has registered 16 HFC-23 projects, the majority in China, with the potential to generate 65 million carbon credits per year. It is extremely cheap to destroy the HFC-23 and the polluting gas has become more profitable than the actual product. There is no incentive to develop clean technologies.

Furthermore, since HFC destruction earns credits, it makes sense to not ban HFC. This undermines the 1987 Montreal Protocol, which mandates phasing out HFCs in Southern countries by 2010. As Lohmann observes, “The CDM has now provided a perverse incentive to hike production of HFC’s in order to cash in as much as possible on credit sales.”³¹

While China receives the most emissions-trading funds, India leads the

world in the number of CDM projects. By August 2006, India had registered 82 CDM projects, with 350 more in the pipeline. The CDM National Strategy study conducted by the government of India predicts that India could take 10 to 15 percent of the global CDM market. But these projects are not “clean,” nor are the Indian people the recipients of the credits. The financial transfers are to corporations. The giants of the corporate world—Reliance, Tata, Birla, Ambuja, ITC—are moving to get their dirty industries certified as “clean.”

The sponge iron industry is another dirty industry that has been labeled “clean” through CDMs. Jindal Steel and Power Ltd (JSPL) in Chhattisgarh is building one of the world’s largest CDM operations. A noxious, dirty, polluting sponge iron plant spread over 320 hectares—which has destroyed the village of Patrapati, privatized the rivers, killed the forests—is somehow a “Clean Development Mechanism.”

Sponge iron is obtained by removing the oxygen from iron ore, a process that demands large amounts of water and gas or coal energy. In Chhattisgarh, Orissa, and Jharkhand hundreds of sponge iron plants have been set up, resulting in the contamination of drinking water and groundwater. Sponge iron plants also pollute the air. Vegetables are becoming reddish due to air pollution, and rice-paddy seeds have stopped germinating in the region that is the main center for the indica rice (of which we had more than 200,000 varieties before the Green Revolution).

After the Kyoto meeting identified steel as a major source of pollution, sponge iron was made a candidate for CDMs. It will reduce direct CO₂ emissions, but the cost of polluting the land and water does not enter the equation. Compared with blast furnace production, sponge iron is cheaper, more profitable, can be made on a smaller scale, and avoids environmental regulation. Within a year or two, a company can recover its investment.

Thirty percent of India’s iron output is now produced in sponge iron plants, a figure that is expected to go up to 50 percent over the next ten years. Production has increased from less than 1 million tons in 2001 to 19 million tons in 2006. India today produces 20 percent of the world’s sponge iron, which means that the tribals of Orissa, Jharkhand, and Chhattisgarh are paying the price for “clean” carbon trading.

It is these tribal communities that should be candidates for CDMs. The potter making *gharas* and *surahis* (pots that keep water cool) and the farmers markets that sell fresh vegetables should have been counted as clean development, not HCFC factories, which only serve to increase the use of energy-intensive refrigeration.

The problem with the reductive market-industrial model is that genuine alternatives are not even considered. Instead of clean technological pathways that avoid CO₂ emissions at every step of the production chain, the focus is on dirty technological patterns that partially reduce emissions in a small link of the chain. The activity promoted under the CDM is not clean when compared to the ecological alternatives. It is only clean when compared to its own most polluting version.

CDM projects in India do not support ecological farming, which works to mitigate climate change and adapt to it. They do not include handmade-paper producers or artisanal cold-press oil mills, which use no fossil fuel, generate no pollution, provide rural employment, and produce high-quality products for a high quality of life. CDM projects *do* include coal-fired power plants and automobile factories, steel plants and sponge iron plants, pulp and paper industry and distilleries, petrochemical plants and chemical factories, cement industry, and sugar factories—all of which are polluting and resource- and energy-intensive.

Emissions trading does not reduce emissions. The solution to climate chaos will not come from the distorted markets of globalization. It will come from a deep commitment to Earth Democracy. Carbon trading might benefit the market economy, but it is destroying nature's economy by increasing pollution and destroying the forests and biodiversity that convert dead carbon that pollutes into living carbon that cleans up the atmosphere.

The bundle of technologies recognized by the COM does not include nonindustrial alternatives, of which there are a wealth in the South. There is no COM incentive to protect ecologically sustainable technologies and systems of production.

However, the real tension is not between rich countries and poor countries. It is between corporate industry in the North and farmers, indigenous people, and vulnerable communities. Corporations in the North and the South have now formed partnerships, and the corporations in the South must first pollute and then reduce pollution to get credits. The result of these incentives is an overall increase in pollution when measured against ecological alternatives. Nonpolluting, nonindustrial activity does not even figure in Kyoto's COM. To be counted as clean, you must first be dirty.

Nuclear Energy is neither Clean nor Sustainable

Nuclear energy has suddenly emerged as a “clean” energy option. Governments are promoting it. The French company Areva, the world’s largest builder of nuclear reactors, forecasts that 150 to 300 new nuclear reactors will be built around the world by 2030. At least 50 of them will be in China or India. The *International Herald Tribune* has called it “a nuclear renaissance.”³² Even James Lovelock, the developer of the Gaia hypothesis, has described nuclear energy as a “Gaia friendly energy source.”³³ But this position is shortsighted.

In a study on the environmental impact of nuclear power the *Ecologist* found that each nuclear reactor emits 20 million tons of CO₂ in its construction. The act of mining uranium also causes pollution. Because most uranium deposits are low-grade, almost 100,000 tons of rock have to be mined to produce a ton of uranium. A standard reactor needs 100 tons of uranium. The conversion and enrichment uses halogenated compounds, which are 10,000 times more potent greenhouse gases than CO₂. Once all factors are accounted for, writes Jon Hughes in the *Ecologist*, “mining and milling uranium is uneconomic and uses more energy to recover than it will ultimately produce.”³⁴

Worldwide, cancer rates are higher among people who live near nuclear plants than among the rest of the population.³⁵ And the hazards do not stop at the nuclear power plant itself. The waste generated holds 95 percent of the radioactivity, and each 1,000-megawatt nuclear power reactor produces about 30 metric tons of high-level waste per year. The cost of disposing nuclear waste in the UK has been put at \$108 billion. In the US, \$18 billion has been collected from consumers of nuclear generated electricity for storing nuclear waste.³⁶

All of India’s uranium comes from a single processing plant in Jaduguda, in the Singhbhum district, inhabited by the Ho and Santhali tribes. The mine is located in the heavily forested catchments of the Suvernarekha River (which means the “line of gold”). The plant consists of three underground mines located 1,600 to 2,000 feet below the surface. From the Jaduguda mines, yellowcake (U₃O₈) is transferred to the nuclear fuel complex in Hyderabad for fabrication into fuel rods. Yellowcake contains only about 0.7 percent uranium-235, which is the primary ingredient for nuclear fission reactors. To increase the concentrate to the necessary 3 to 5 percent, the yellowcake is mixed with fluorine and heated. The lighter U-235 molecules are separated from the heavier U-238 molecules. The enriched uranium is used for fuel rods. The remaining 85 percent becomes depleted uranium.

The mine, operated by the Uranium Corporation of India Ltd. (UCIL), was opened in 1967. It impacts the 30,000 people living in 15 villages within a 5-

kilometer radius of the complex. The ore is first crushed into a fine powder, and then chemically treated to remove the uranium. After uranium extraction, 99.94 percent of the mined rock is left as waste. Jaduguda processes 1,000 tons of ore per day and produces 200 tons of uranium in the form of yellowcake every year. Some 350,000 tons of rock are being mined, crushed, and dumped in Jaduguda every year.

This crushed rock, or uranium tailings, contains more than a dozen radioactive materials, including thorium-230, radium-226, and the gas radon-222. If the tailings are allowed to remain on the surface and dry out, they can be carried by the wind onto faraway vegetation, entering the radioactive material into the food chain, or be washed into rivers and lakes, contaminating the water supply.³⁷ At the Jaduguda mine the coarse tailings are dumped back into the mine cavities and the fine tailings are mixed with water and pumped via a pipeline to the tailing dams near Jaduguda village.

The three tailing dams contain millions of tons of radioactive waste covering more than 100 acres. Additional nuclear waste generated throughout India—at the nuclear fuel complex in Hyderabad, the Bhabha Atomic Research Centre in Mumbai, and other complexes—is also dumped in Jaduguda. The waste has contaminated the water and living resources in the area.

A study of people living within 1 kilometer of the tailing dam showed that 47 percent of women had developed menstrual problems, 18 percent had suffered miscarriages or had given birth to stillborn babies, and 30 percent had other fertility problems. Children born in the area are born with deformities, skeletal distortions, partly deformed skulls and organs.³⁸ The more than 7,000 mine workers are also continuously exposed to radiation hazards.

The local tribals have been resisting the uranium mining and its hazards. Some of the demands of the local movement follow:

- An end to the dumping of nuclear waste in the area
- The implementation of international safety norms on radioactive waste
- Resettlement of the villages near the tailing dams
- Treatment and medical facilities for those affected by radiation hazards.

What is happening to the Ho and the Santhals in Jaduguda happened to the Navajo in the US. Joe Shirley Jr., the Navajo tribal president, has referred to uranium mining as “genocide.” Robert Stewart Sr., a Navajo who worked in the uranium mines for five years, said, “You look around the reservation and see so many elderly people who are crippled and can barely breathe.”³⁹

In 2005, the Navajo banned mining on their reservation, which covers 27,000 square miles across parts of Arizona, New Mexico, and Utah. In Australia,

where the world's largest deposits of uranium are located (40 percent of the 3.5 million tons known to exist), movements have forced companies to restrict mining to 10 percent of the reserves and the Australian government has recognized the aboriginal owners' right to veto mining on their land.⁴⁰

"Clean" cannot be measured only in terms of CO₂ emissions. It has to include all ecological risks from cradle to grave. Climate change does not make nuclear energy safe. It forces us to look at energy alternatives in terms of holistic and long-term safety and in terms of decentralized options.

Nuclear energy is neither clean nor cheap. In the United States the per-unit cost of nuclear energy is \$.067, compared with \$.042 for coal and \$.038 to \$.056 for gas. In India, gas power costs Rs 1.20 and nuclear power will cost Rs 30 per unit. And even with all the nuclear plants India will build, the energy will only supply 6.7 percent of the country's energy needs, up from 3 percent today.

For nuclear energy to replace fossil fuels, the world would have to build one reactor per week for the next 60 years.⁴¹

The US-India Nuclear Agreement: Nuclear Winter is Not an Alternative to Global Warming

India, a civilization built on renewable energy and biodiversity economies, is at a crossroads—will she continue on her renewable energy path based on biodiversity and energy equity, or will she follow the non-sustainable energy path of the West, based on fossil fuels and nuclear power?

Historically, India is not among the carbon dioxide polluters of the world. By virtue of culture and economic policy, preference was given to localized, decentralized, labor-intensive economies. However, with globalization and neoliberal economic reform, the renewable is being replaced by the non-renewable, people are being displaced by fossil fuels, decentralized and diverse systems are being replaced by corporate-controlled centralized monocultures of transport, manufacture, and agriculture. Not only does this trend add to the risk of climate change, it also usurps the ecological space for tribals, small-scale farmers, and women.

In 1973, I was training to be a nuclear physicist. Some of my most exciting times were the periods I spent at the Bhabha Atomic Research Center in Bombay, working in the experimental fast-breeder reactor. But I gave up a career in nuclear physics after my sister, Mira, a medical doctor, humbled me. She

pointed out that while I was trained in all the sophisticated minutiae of energy transitions and chain reactions, I was illiterate when it came to nuclear hazards. It was that lesson in humility that precipitated my shift toward sciences that defend life and away from those that annihilate life. It also made me more conscious of the links between knowledge and power, the construction of social irresponsibility built into war- and profit-centered science, and the field's willful mystification regarding matters of social consequence, which shuts out democratic control of dangerous technologies.

The US-India nuclear agreement, which is being offered as a “clean energy” agreement, is an expansion of nuclear power and nuclear fuel sales to India and, in effect, a reversal of the 32-year isolation India has faced on the nuclear front. In the agreement, signed in 2005 and finalized during President Bush's India visit in 2006, the US committed to working with India to achieve full civil nuclear energy cooperation and to working with allies to enter into nuclear trade with India. Nuclearization of India's energy is nuclearization of India's military options. Nuclear power is being offered as “clean energy” because the process does not directly release any CO₂ into the atmosphere. Nuclear waste is being ignored, as is the increased potential of nuclear war. The agreement, in effect, suggests that a nuclear winter is better than global warming. However, for a sustainable world, we need alternatives to both, and these will come from renewable energy.

As a result of the treaty, US companies are anticipating contracts worth \$100 billion for a target of generating 60,000 megawatts by 2030. Under the Hyde Act, the US law that enacts the US-India agreement, India will not be allowed to be a technology developer, only a recipient of nuclear technology and fuels. India's nuclear scientists are worried that the country will lose the freedom to decide on strategic choices. The Hyde Act stipulates that if India detonates a nuclear device, the US has the right to take back all its equipment and any nuclear material produced with that equipment. Such a restriction threatens India's stand on nuclear disarmament, making it subservient to US goals.⁴²

The troublesome aspect of the US-India Nuclear Agreement is Section 123a(4). It states, “except in the case of those agreements for cooperation with nuclear weapons states, a stipulation that the United States shall have the right to require the return of any nuclear materials and equipment transferred pursuant thereto and any special nuclear material produced through the use thereof if the cooperating party detonates a nuclear explosive device.”

That much-debated stipulation, in effect, undoes the separation of civilian and military facilities that was the core of the US-India nuclear agreement

framework agreed to by prime minister Manmohan Singh and President Bush.

In 2007, the US and India completed the final negotiations on the text of the 123 agreement. Despite previous assurances to the Indian parliament, the negotiations were completed without parliamentary approval. The key issue of sovereignty of decision-making has clearly been sacrificed. The US holds the right to withdraw the supply of nuclear fuels and technology if India carries out a nuclear test. There is no prohibition on US nuclear tests. In effect, the US will control India's nuclear decisions. As US Under Secretary of State Nicholas Burns said in an interview:

On the issue of the so called right-of-return, of course, the American president under our Atomic Energy Act has the right to ask for the return of nuclear fuel and nuclear technologies if there is a test. That right-of-return has been, of course, preserved as it must be under our law, and there has been no change in how we understand the rights of the American president and the American government. It has been fully respected in this.⁴³

Not only has the US managed to turn India into its nuclear satellite, it also sees India as a big market for weapons sales. Though the Indian foreign secretary, Shivshankar Menon, denied that the door was being opened to arms trade—"This is an agreement in energy; we did not negotiate anything else"—India's chief negotiator, the national security advisor M. K. Narayanan, hinted otherwise: "I presume the outcome of the nuclear deal would be a transformed relationship, and this would lead to other things." Burns confidently declared, "And I think now that we have consummated the civil nuclear trade between us, if we look down the road in the future, we are going to see far greater defense cooperation between the United States and India: training exercises; we hope defense sales of American military technology to the Indian armed forces."⁴⁴

Using a false excuse of providing "clean" energy to deal with climate change, the US is seeking to contain India's strategic designs for an empire with global reach while simultaneously promoting the buildup of nuclear arms. Under the agreement, the fissile material that will be produced in Dhruva, the weaponsgrade plutonium production facility, the uranium enrichment plant near Mysore, and the prototype fast-breeder reactor under construction, along with eight power reactors, will be able to yield 1,250 kilograms of weapons-grade plutonium a year, enough to build 250 Nagasaki-type bombs, according to the International Panel on Fissile Material.⁴⁵ The US-India agreement is not about energy, it is about creating a nuclear arms race with the US in the driver's seat. Clearly the agreement is not about just energy. It is about foreign policy and security policy. And ultimately it is about democracy and sovereignty.

Engineering the Planet: Mechanistic “Solutions” to Problems Caused by the Mechanical Age

Fossil fuels fueled the mechanical age. When machines driven by non-renewable energy replaced renewable human and animal energy, the atmospheric carbon cycle started to get disrupted. Carbon trapped in fossils over millions of years began to be pumped into the atmosphere. Over a few hundred years, fossil fuel use spread from the making of textiles to every sphere of production. And as “development” became defined through the use of fossil fuels, societies based on renewable energy have been increasingly pushed into fossil fuel dependence.

The accumulation of carbon dioxide in the atmosphere has led to climate imbalance. The mechanical mind that has caused the problem of climate change is now applying the same mechanistic and reductionist thinking to solutions to climate change. These mechanical interventions threaten to cause even more ecological dislocations and disruptions to our fragile biosphere.

In a memo sent by the US government to the IPCC, the Bush administration describes “modifying solar radiance” as “important insurance” against the threat of climate change.⁴⁶ Interfering in Gaia’s processes is at the root of climate change. Myopic scientists now offer even more reckless interference as the solution, ignorant of the impacts of their technological fixes on the workings of the stratosphere and the ocean. For them further pollution is the solution.

There’s no shortage of such plans. A scientist in Germany has proposed the use of sub-micrometer-sized sulfate-based aerosols to reflect the sunlight in order to cool the atmosphere. In this plan high-altitude balloons and artillery cannon would blast sulfur dioxide into the atmosphere, where it would convert into sulfate particles. Blocking the sunlight would cost \$50 billion annually. Its ecological costs would be much higher. While any sane citizen would consider such reckless geo-engineering irresponsible, Ralph J. Cicerone, an atmospheric chemist and president of the US National Academy of Sciences, has supported such technology: “We should treat these ideas like any other research and get into the mind set of taking them seriously.”⁴⁷ Another idea Cicerone takes seriously is a proposal to put up trillions of lenses each about 2 feet wide to deflect sunlight. (With Nobel laureates promoting the idea of engineering the planet, the absurd becomes the acceptable.)

These proposals treat the sun, rather than industrial activity, as the problem. Blocking the sunlight will impact plant growth and photosynthesis. Particulate pollution from the sulfate will cause serious health hazards. Such reductionist

technological fixes will aggravate climate chaos, not create climate balance. The climate crisis we now face is the result of the unintended side effects of a fossil fuel-based economy. Adding more pollution to the atmosphere will only create more unintended side effects. It will not solve the problem; it will cause new ones. As the ETC Group, an organization focused on socially responsible development of technology, observes, “Geo-engineering the stratosphere makes it easier for industry to continue atmospheric pollution but compounds the potential problem by intentionally contributing massively to particulate pollution.”⁴⁸

Another set of proposals based on “pollution as solution” seeks to pollute the ocean with iron particles to sequester carbon. The theory is that the iron will induce the growth of phytoplankton, microorganisms on the ocean surface that capture carbon through photosynthesis. Phytoplankton account for half the carbon dioxide absorbed annually from the atmosphere. Proponents calculate that the spread of phytoplankton will sequester 2 to 3 billion extra tons of carbon dioxide every year—about 30 to 50 percent of the annual carbon emissions by industry and automobiles. ETCs reports that US-led geo-engineering experiments IRONEX I and IRONEX II, which carpeted 64 square kilometers of the ocean with iron particles near the Galapagos Islands did lead to a massive phytoplankton bloom. Another project, called SOIREE (Southern Ocean Iron Release Experiment Expedition), spread iron filings over 50 square kilometers, but the resultant phytoplankton bloom far exceeded the testing area and spread over 1,100 square kilometers of the ocean.⁴⁹

Millions of tons of iron may create phytoplankton, but it could kill ocean life. The SOIREE expedition had to conclude that “large-scale fertilization would be likely to cause substantial changes to the naturally occurring ecosystems of this pristine environment.”⁵⁰ John Martin of the Moss Landing Marine Laboratories in California, who first articulated the “iron hypothesis” in 1991, has stated that with half a tanker of iron he could create a new Ice Age. But is an engineered Ice Age a solution to global warming? While John Martin was proselytizing for iron filings, Dr. Kenneth Coale, the chief scientist of the experiments, announced that iron fertilization could theoretically sterilize portions of the Pacific.⁵¹ Irrespective of the risks, Green Sea Ventures plans to conduct a large-scale experiment of spreading iron over 5,000 square miles to sequester 100,000 to 200,000 tons of CO₂. Green Sea has even obtained patents on iron fertilization of the ocean to sequester carbon. Another California company, Climos, is selling carbon “offsets” through iron fertilization of the ocean.

The companies piloting this technology project annual returns of up to 75 billion euros. As with all reductionist technologies and market solutions, the calculus does not include the harm to Gaia's food webs and biogeochemical cycles. Geo-engineering the ocean or the stratosphere is another step in the privatization of the Earth's commons. The ETC group as part of its efforts to protect the earth from geo-engineering has called on the United Nations to reaffirm and expand the Environmental Modification Treaty adopted by the general assembly in 1978. At the 9th meeting of the UN Convention on Biodiversity in 2008, signatories unanimously agreed to a "de-facto moratorium" on ocean fertilization activities. Though 191 countries have agreed to the stoppage the US, who is not a signer to the convention, continues to support ocean fertilization research.⁵²

Playing with Gaia as if she were a LEGO set cannot be the appropriate response at a time when human intervention in the Earth's living systems is threatening our very survival. The mechanical mind cannot solve the problems of the mechanical age.

Globalization as Outsourcing of Pollution and the Politics of Climate Change

Trade liberalization and corporate globalization are causing climate change in two major ways. First, resource- and energy-intensive and polluting industries are moving to countries like India. In India, for example, we are seeing an explosion of steel, aluminum, and sponge iron production, automobile manufacture, and petrochemical industry, all of which lead to increased CO₂ emissions. Second, we are seeing the destruction of local economies and local production. As a result, to meet the same human needs, more carbon dioxide is being added to the atmosphere. The burden of global industrial production is now falling on countries like India, and, in a distorted paradigm, this pollution is presented as proof of India's development. This is part of a new global environmental apartheid in which environmentally destructive and polluting economic activities are relocated to the South. This was Lawrence Summers' plan when he was chief economist at the World Bank.

In 1991, Summers wrote in a memo to senior World Bank staff, "Just between you and me, shouldn't the World Bank be encouraging more migration of the dirty industries to the LDC [Less Developed Countries]?"⁵³ Summers had

justified the economic logic of increasing pollution in the third world on three grounds. First, since wages are low in the South, economic costs of pollution arising from increased illness and death are lowest in the poorest countries. According to Summers, the logic “of relocation of pollutants in the lowest wage country is impeccable and we should face up to that.” Second, since in large parts of the South, pollution is still low, it makes economic sense to Summers to introduce pollution: “I’ve always thought,” he writes, “that countries in Africa are vastly under polluted; their air quality is probably vastly inefficiently low compared to Los Angeles or Mexico City.” Finally, he argues, since the poor are poor, they cannot possibly worry about environmental problems: “The concern over an agent that causes a one in a million change in the odds of prostate cancer is obviously going to be much higher in a country where people survive to get prostate cancer than in a country where under five mortality is 200 per thousand.”

Summers has recommended the relocation of hazardous and polluting industry to the South because, in narrow economic terms, life is cheaper in the poorer countries. No matter how an economist might value life, all life is precious. It is equally precious to the rich and the poor, to the white and the black, to men and women.

The movement of dirty industry to India is increasing the pressure on land, water, and forests on which peasants and tribals depend. And as the rise of Indian industrialists Lakshmi Mitral, Mukesh and Anil Ambani, and K. P. Singh—now among the richest men in the world—shows, it is making the rich richer. These industrialists have accumulated their wealth through robbing land from the poor. While reducing wealth gaps between owners of corporations in the North and South, corporate globalization is increasing inequality across the world and within India.

Globalization, Equity, and Climate Change

Climate change today is global in cause and global in effect. Globalization of the economy has outsourced energy-intensive production to countries like China, which is flooding the shelves of supermarkets with cheap products. The corporations of the North and the consumers of the North thus bear responsibility for the increased emissions in the countries of the South.

In a globalized economy, addressing pollution by setting emissions levels for each country is inappropriate for two reasons. First, not all the citizens of a

country contribute to the pollution. As a result of becoming the world's factory, China has seen its CO₂ emissions rise from 13.8 percent of the world's total in 2005 to 22 percent in 2006, outstripping the US and putting it in first place worldwide. It produced 6.2 billion tons of CO₂ in 2006, compared with 5.75 billion tons for the US. However, per capita emissions of the US were 22 tons of CO₂, compared with 4.7 tons for China.⁵⁴

Further, much of China's 5.75 billion tons of CO₂ emissions could just as easily be considered US emissions because US companies have outsourced manufacturing to China of goods that are eventually consumed in the US. Wal-Mart's superstores procure most of what they sell from China. As Charles Fishman reports,

Wal-Mart, which in the early 1990s trumpeted its claim to "Buy American," doubled its imports from China between 1997 and 2002 to \$12 billion. In the next two years, Wal-Mart increased Chinese imports again by 50 percent, so that in 2004 the company and its suppliers landed Chinese made goods in the US with a wholesale value of \$18 billion.⁵⁵

Consumer spending accounts for two-thirds of the US economy, and Wal-Mart dominates retail. From 1997 to 2004, while jobs in US manufacturing fell by 20 percent, Wal-Mart imports from China increased by 200 percent.⁵⁶ Americans are shopping themselves out of jobs while Wal-Mart's everyday lowest price does not include the costs of climate change or the destruction of China's land, water, and air. As the Qin Gang, China's Foreign Ministry spokesman, said in June 2007, "China is now the factory of the world. Developed countries have transferred a lot of manufacturing to China. What many western consumers wear, live in, even eat, is made in China."⁵⁷ Outsourcing of manufacturing also outsources pollution, though it doesn't outsource profits. The carbon footprint should be assessed against the accounts of the company that is profiting, not the country that is bearing the burden of pollution.

As the Christian Aid report "Coming Clean: Revealing the UK's true carbon footprint" states, "While only 2.13 percent of the world's CO₂ emissions emanate from the UK's domestic economy, through the process of globalization CO₂ is emitted around the world on the UK's behalf in China, India, Africa, and elsewhere."⁵⁸ While the exact global footprint of UK companies is not known, estimates suggest that emissions associated with the worldwide consumption of the top 100 UK company products amounts to 12 to 15 percent of the global total."⁵⁹

In fact, the rural poor in China and India are losing their land and livelihood to make way for an energy-intensive industrialization. To count them as polluters would be doubly criminal. Corporations, not nations, are the appropriate basis for regulating atmospheric pollution in a globalized economy.

So far, emissions trading schemes have rewarded polluters by giving them quotas for pollution. And these quotas have allowed them to *increase* their emissions rather than decrease them.

What is needed is a carbon tax on corporations—both of their production systems, no matter where their facilities are located, and for transport. The global economy is destroying local production and promoting long-distance supply for even the smallest items. Long-distance transport is subsidized, while local, low-carbon-emissions production and distribution is penalized. And the high emissions of long-distance transport have been totally excluded in the Kyoto protocol. The WTO has spurred a huge increase in carbon dioxide emissions due to global transportation, an increase that is not even accounted for in emission reduction targets.

We need not concern ourselves with the inevitable corporate cry that regulations destroy markets. After all, even the “market instrument” of carbon trading requires government involvement. The Clean Development Mechanism requires the involvement of the United Nations. Instead of regulating through trade in pollution (which is like fiddling while Rome burns), governments and the United Nations could regulate through tax on the use of fossil fuel and incentives for the use of renewable energy. We can either destroy the conditions for human life on the planet while clinging to “free market” fundamentalism or we can secure our future and create climate justice by requiring that commerce work within the laws of ecological sustainability and within the laws of social justice.

Climate Change Begins on Land: the Global Ecological Footprint on India's Land

As the resource demands of globalization increase, land has emerged as a key site of conflict. In India, 65 percent of the people are dependent on land. At the same time a global economy, driven by speculative finance and limitless consumerism, needs the land for mining and industry, for towns, highways, and biofuel plantations. The speculative economy of global finance is hundreds of times larger than the value of real goods and services produced in the world.

Financial capital is hungry for investments and returns on investments. It must commodify everything on the planet—land and water, plants and genes, microbes and mammals. The commodification of land is fueling the corporate land grab in India, both through the creation of Special Economic Zones and through foreign direct investment in real estate.

Further, the global economy based on limitless consumerism has a rapacious appetite for natural resources. With trade liberalization, it has been increasingly turning to countries of the South and, within the South, to indigenous and peasant land. As globalization generates more luxury consumption by the rich the demand for cars grows. More cars means more automobile factories, more iron and bauxite mining, and more steel and aluminum manufacturing. The land grab for mining and industry has triggered major conflicts, including those in Orissa's bauxite and iron-ore mining and steel plants, and at Ford's car manufacturing plants in Tamil Nadu and Tata's in West Bengal.

In 2007, the Indian prime minister announced a ten-year automotive mission plan, with the goal of making India a global hub of vehicle manufacture, design, and component production with a turnover of \$145 billion by 2016. Special Automotive Zones, like Special Economic Zones, will be set up in Chennai, Mumbai, and Kolkata.

The plan calls for an increase in production from 10 million to 50 million vehicles per year (current auto production worldwide is 66 million).⁶⁰ In a time of climate crisis and peak oil, a time when we need to look at options beyond oil and motorization, globalization is driving India headlong into being a hub for the production of cars and automobiles by both making India's elite a market for cars and using India's land, minerals, manufacturing, and energy as an "environmental subsidy" for global production. If the mines and steel plants in Orissa and car factories in Kolkata and Chennai are already triggering major land conflicts, how many more conflicts will emerge when India increases its automobile production fivefold in the next decade?

Proponents of "industrialization" must recognize that if the entire world consumed resources and energy at US levels, we would need five planets. India's peasants cannot possibly carry the ecological footprint of a non-sustainable model of global manufacture and global consumption.

Threat to Food Security

Pro-industrialization forces invoke two false arguments to justify land grabs.

First, that agriculture has become unviable and farmers should abandon the land. For farmers, soil is not a prison from which they need to escape to an industrial job. It is their support, their means of livelihood, their security, their identity. The perilous state farmers find themselves in is a result of the corporatization of agriculture. The high costs of inputs like seeds and agrochemicals push up production costs. At the same time, trade liberalization is pushing down crop prices. The result is a negative economy, with costs higher than returns, leaving farmers facing debt and suicide. However, there are alternatives to this negative economy—lowering costs of production through seed sovereignty and ecological farming and increasing returns to the farmers through fair and just trade. We have achieved this in Navdanya, and there is no reason farmers should be denied prosperity and dignity through being *annadatas*, our food providers.

The second argument to justify land grab is that India must “industrialize” to have economic growth. Indian activist Jayaprakash Narayan wrote, “With the decline in share of GDP, greater industrialization and shift of occupations are both necessary and inevitable. India cannot continue to be a largely agrarian economy if we harbor ambitions of rapid growth and global power status.”⁶¹ The *Economist* concluded that Indian farmers could escape suicides if they “could escape the soil.”⁶² However, today’s economic growth is jobless growth, it is not labor-intensive and employment-generating. India will not be a “super power” with millions rendered destitute through land grabs. Furthermore, SEZs, a favored tool of industrialization, are for real estate companies, developers, and for the IT sector, which are already growing industries and do not provide basic needs. No society can become a post-food society. If India destroys her fertile farmlands for concrete jungles and uproots her small farmers for a speculative economy, there is not enough land in other countries to provide the food for India’s billion people. Land is inelastic. Fertile land is a very precious and very scarce resource. It needs to be protected and conserved as an asset of the farmers and as a national heritage to be passed on to future generations. It cannot be destroyed for the passing wave of short-term greed and speculation driving the corporate land grab. Climate change and peak oil should wake us up to the consequences of destroying our local food economies.

Though the relocation of dirty industries to the South is a Northern strategy, in countries like India, it is being presented as a “sovereign” and “autonomous” decision. Thus, Buddhadeb Bhattacharya, the chief minister of West Bengal, has stated, “That 63 percent of the population continues to depend on agriculture for its livelihood is a sign of backwardness. Should we be satisfied with 62 percent agricultural land and 24 percent urban and industrial land? Don’t we want new

cities and industries? From agriculture to industry, from villages to cities—this is what civilization is.”⁶³ There is in fact nothing inevitable or natural or evolutionary about people being moved from villages to cities. Brazil, for example, got rid of its small farmers while India instituted policies that maintained its own small farmers. That 60 percent of India’s population is still on the land is a consequence of conscious and enlightened public policy, not a “sign of backwardness.” In fact, in a period of climate change, peak oil, and the end of cheap oil, people in the industrialized world are talking of “re-ruralization” as a way to reduce emissions and dependence on oil by reducing food-miles. We cannot live without food. If we can no longer afford to transport food over thousands of miles, then localization of food production is the only alternative. Where local farmers have disappeared due to industrialization of agriculture, rejuvenating small local farmers is emerging as a core element of food security policy, environmental security policy, and energy security policy. India’s localized, decentralized, rural economies provide it with an advantage in making an energy transition beyond oil and an ecological transition to contain climate change.

India is today at an energy crossroads. Either this great civilization will build on its multiple renewable energies—its cultural energy, its people’s energy, its solar energy, its animal energy, its energy embodied in biodiversity and biomass, or India will be consumed by fossil fuel addiction, its elite joining hands with the global elite, the uprooting of rural communities, and the climate crisis this will aggravate.

Countries like India and China, by providing their people’s resources—land, water, biodiversity—and their people’s labor to produce for the global supermarket, are in fact subsidizing global production and consumption, ecologically and socially. This subsidy includes the land appropriated from peasants, the forests appropriated from indigenous people, the sovereignty and democracy robbed from the citizens. Yet because governments of the South—like those of the North—are highly influenced by corporate power and capital, on behalf of corporations they put forward the “right to pollute” as the “right to development.”

Ma Kai, head of China’s economic planning agency, has stated that mandatory caps on developing countries would hinder their development and hamper their industrialization. For China “the first and overriding concern was economic and social development and poverty eradication.”⁶⁴

In India too, “growth” is the first commitment. India is chasing 10 percent growth in GDP through economic globalization. Yet the climate change such

growth contributes too could wipe out 9 percent of India's GOP, erasing that economic growth and so much more.

India can lead the world to a post-oil future of low carbon dioxide emissions because it is still primarily an agrarian economy based on peasants and small farmers. India's energy is primarily renewable. However, with the push of globalization, India is rushing headlong into a fossil fuel economy by destroying the biodiversity economy of her rural areas and her peasantry. Instead of being a leader in the mitigation of climate change, India is hiding ostrich-like in the sand. As the *Indian Express* reports:

India has concentrated its energies in telling the world that its priorities are different. In December 2006 when Nick Stern presented a summary of his review showing the economic impact of climate change, India responded by presenting a paper by Indian economists to show that there is no need for the country to agree to any binding emissions and the focus must clearly be on achieving domestic goals like eradicating poverty.⁶⁵

The solution to climate change and the solution to poverty are the same—protecting, enhancing, and rewarding livelihoods, work, production, and consumption patterns centered on people, not on fossil fuels. Food, economic justice, and energy equity demand more small farms, not fewer. It demands more localization and less globalization of staple food production. And small ecological farms and local food markets are solutions for the North and South. In countries of the South such as India, the climate-change imperative is to protect small farms and small farmers. India must change the rules of trade that are pushing millions of farmers into debt and hundreds of thousands to suicide. It also requires an end to land grabs for Special Economic Zones.

In countries of the North the ecological imperative is to rejuvenate local farms and local markets.

We can reduce our ecological footprint, reduce economic inequality, and improve quality of life by protecting small farms and local food systems in the South and in the North. Food democracy paves the road to energy justice in a period of climate chaos.

How we grow our food and distribute it is part of the politics of climate change.

How land is used, and how its ownership is distributed, is part of the politics of climate change.

The politics of climate change is not just about what happens in the air. What happens in the air begins on the land. And those who are controlling and monopolizing the land and water of the poor are also privatizing the air, first through pollution, and then through “market” solutions to climate change.

Earth Democracy: Climate, Energy, and Resource Justice on a Small and Fragile Planet

To achieve genuine sustainability, energy systems need to be embedded in society and ecosystems. They can only be considered sustainable socially if they do not enclose and usurp the ecological space of the poor and lead to energy equity. They can only be considered sustainable ecologically if they facilitate the shift to decentralized, low-impact economies and do not introduce risks of atmospheric or nuclear pollution. Ecological democracy must be the context for sustainable energy choices.

Climate chaos is being recognized, even by the establishment, as a serious challenge for humanity. The IPCC has categorically established that the climate change we are living through is a result of human activity. The human activities responsible are those that require fossil fuels. We are releasing more CO₂ into the atmosphere than the planet can recycle and absorb.

There is a consensus that to avoid catastrophic climate change we need to contain the global temperature rise within 2 degrees Celcius. Yet most “solutions” for reducing emissions maintain and even expand energy-intensive production and consumption patterns. There is a deep disconnect between the aim to reduce carbon emissions by the necessary 80 to 90 percent and the steps prescribed to do so.

The problem of climate change is clearly related to the issue of energy. However, it is also an issue of the economy, how goods are produced and distributed, how our towns and homes are designed, how our food and clothing is produced. Yet globalization and industrialization are not even referred to in the climate change negotiations and discussion. The issue of overextraction and overconsumption of energy is being avoided, as is the issue of the injustice to the poor and other life-forms whose homes and habitats, lives and futures are being sacrificed to provide the energy and resources to run a globalized consumerist economy. The debate has been deflected from climate justice, energy justice, and ecological justice—diverse aspects of what I call Earth Democracy. It is being narrowed down to problems and solutions from the perspective of the rich, from the perspective of industrialized, urbanized societies.

From the perspective of the rich, the problem needing to be solved is how to control the world’s oil supply (through militarization) while simultaneously finding alternatives to oil before the oil runs out, and before climate change disrupts the current social order.

However, this is a reductionist analysis, leading to reductionist solutions. The first reduction is at the level of energy. Energy, a multidimensional category, is reduced to industrial consumptive energy alone—energy as electricity or energy as fuel. And society, in all its diversity, has been reduced to the society of the rich, especially in rich countries. What is left out is nonconsumptive energy—the creative energies of the universe. Gaia’s self-organizing energy, the creative human energy to work and to produce, to organize, and to transform. Also left out is the perspective of the poor—especially of the nonindustrial societies of the so-called third world—and the rights of other species.

The primary solutions this reductionism offers to the problems of peak oil and climate change are carbon trading and substituting other consumptive energies produced at high social and ecological costs for fossil fuels. The current favorites are nuclear and biofuels. That these solutions add new burdens on the poor by taking away their land and food, their water and air, is yet another externalized cost. The dominant analysis of the problem and the dominant solutions are centered on capital, and the laws made to enforce these solutions have been made by those who control capital. These laws are treated as immutable and unchanging. They define the boundary conditions for human actions and human choices. In effect, what is being said is that the system that has created the climate crisis must stay, it must be protected, even if the poor and the planet have to be sacrificed to maintain it a little longer, even if the laws of nature and of our common humanity have to be violated.

The climate crisis signifies a clash between the laws arising from the workings of the universe, the planet, and sustainable human communities and the laws of capital accumulation shaped by those who own and control capital. These are the so-called laws of the market, the laws imposed by the IMF and World Bank through structural adjustment, the laws embodied in the WTO and other “free trade” agreements.

The solution to climate chaos is not an energy shift—from fossil fuels to nuclear, biofuel, and big hydro. The solution is a paradigm shift:

- From a reductionist to a holistic worldview based on interconnections
- From a mechanistic, industrial paradigm to an ecological one
- From a consumerist definition of being human to one that recognizes us as conservers of the earth’s finite resources and cocreators of wealth with nature

A people- and planet-centered paradigm identifies laws to live by other than

the laws of the market. These are the laws of Gaia and the universe, the laws of physics, of energy and entropy, the laws of evolution and emergence, and the laws of justice, which offer equity on a finite and fragile planet. It is these ecological and natural laws that provide the real basis for human action and human transformation. And it is these laws that are being violated by the corporate-driven, globalized, high-energy economy that defines how we produce, what we produce, how we consume, what we consume, how we live, and if we live.

The climate crisis is at its roots a consequence of human beings having gone astray from the ecological path of living with justice and sustainability. It is a consequence of forgetting that we are earth citizens. It is acting like we are kids in a supermarket with limitless appetites for consumption and falsely imagining that the corporations that stock the supermarkets have unlimited energy warehouses. The real problem is the conflict between the economic laws that have reduced the planet and society to a supermarket where everything is for sale and the ecological laws that maintain the planet's ecological functions and social laws that distribute nature's goods and services equitably. The real problem is a global economy that has created a planetary ecological imbalance.

If these are the real problems, then the real solution cannot be replacing fossil fuels with other non-sustainable sources to power the same systems. The real solution must be to search for right living, for well-being, and for joy, while simultaneously reducing consumption. In Indian philosophy, right living is “dharma”—the bridge between resources, *arth*, and human needs, *kama*. Dharma is therefore based on the sustainable and just use of resources for fulfilling needs. Ecological balance and social justice are intrinsic to right livelihood, to dharma. “Dharanath dharma ucyat”—that which sustains all species of life and helps maintain harmonious relationship among them is “dharma.” That which disturbs the balance of the earth and her species is “adharma.”⁶⁶

Equity is about fair share. There are currently two paradigms of equity. One sets the overconsumption and waste of rich industrial societies as the model and measure of being human, being developed. Equity is presented as the entire world being pushed to that level of resource and energy consumption. However, this version of equity would need five planets. This non-sustainable paradigm inevitably produces inequity. Shall we choose a non-sustainable paradigm in which we affirm an equal right to pollute or a sustainable paradigm in which we affirm an equal responsibility to *not* pollute?

Earth Democracy and ecological equity recognize that because the planet's resources and capacity to renew resources are limited, a reduction in energy and

resource consumption of the rich is necessary for all to have access to land and water, food and fiber, air and energy. In an ecological paradigm, what works against Gaia works against the poor and works against future generations, and what works for Gaia works for the poor and for the future.

We need to define equity on the same ecological parameters locally and globally. If communities in India are resisting displacement and uprooting, if they define and experience their lives in the forest or small farms as the terms of their material and spiritual well-being, respecting their rights and freedoms is the first step toward equity. Equity needs to be grounded in the earth, in people's struggles and movements, not dropped in as an abstraction from remote conference halls. Those who would uproot farmers say the life of a peasant is "undignified." Those who would uproot indigenous people define life in a forest as "below the dignity line." Dignity is an experience and consequence of self-organization and sovereignty, of sufficiency and satisfaction. I have never found working the soil or lighting a wood fire lacking in dignity. It is disposability that robs people of their dignity and selfhood. That is why movements against displacement are so intense and widespread in contemporary India.

Real solutions will come from breaking free of the crippling world of mechanistic assumptions, industrial methods of producing goods with high energy and resource costs, and market mechanisms that make high-cost products appear cheap on supermarket shelves.

The eco-imperialist response to the climate crisis is to grab the remaining resources of the planet, close the remaining spaces of freedom, and use the worst form of militarized violence to exterminate people's rights and people themselves when they get in the way of an insatiable economy's resource appropriation, driven by the insatiable greed of corporations.

There is another response—that of Earth Democracy.

Earth Democracy recognizes that if the survival of our species is threatened, maintaining our ability to live on the planet is the only intelligent response. Chasing economic growth while ecosystems collapse is a sign of stupidity, not wisdom. Earth Democracy calls for a systemic and inclusive response to the climate crisis, not the fragmented and self-serving response that corporations and rich countries are making. Earth Democracy allows us to break free of the global supermarket of commodification and consumerism, which is destroying our food, our farms, our homes, our towns, and our planet. It allows us to re-imbed our eating and drinking, our moving and working, into our local ecosystems and local cultures, enriching our lives while lowering our consumption without impoverishing others. In Earth Democracy, everything is interconnected. To address the pollution of the atmosphere, we do not have to limit ourselves to

changes in the atmosphere. We can change agriculture; we can change the way we design buildings and towns; we can change the way we shop.

In Earth Democracy, solutions will not come from the corporations and governments that have raped the planet and destroyed peoples' lives. Solutions are coming from those who know how to live lightly, who have never had an oil addiction, who do not define the good life as "shop till you drop," but rather define it as looking after the living earth and their living community. Those who are being treated as disposable in the dominant system, which is pushing the planet's ecosystems to collapse and our species to extinction, carry the knowledge and values, the cultures and skills, that give humanity a chance for survival.

To mitigate and adapt to climate change we need to stop the assault on small farmers and indigenous communities, to defend their rights to their land and territory, to see them not as remnants of our past but as the path for our future. Earth Democracy begins and ends with Gaia's laws—the law of renewability, the law of conservation, the law of entropy, the law of diversity. In Earth Democracy, all beings and all peoples are equal, and all beings and all communities have rights to the resources of the earth for their sustenance.

In Earth Democracy, the solution to the climate crisis begins with the cultures and communities who have not contributed to it.

Earth Democracy is based on equal rights of all beings to ecological space, including atmospheric space. The atmosphere is an ecological commons. Climate justice demands that this commons not be enclosed by a handful of polluters. Climate justice also demands that people be compensated for the impact of climate chaos caused by the actions of others. But above all, climate justice demands that every person, every community, every society have the freedom to create and defend economies that cause no harm to the climate or to other people.

To prevent climate chaos and to avoid further increases in emissions we must stop the coercion of trade liberalization and rewrite the rules of trade to favor the local. WTO rules and World Bank structural adjustment programs are robbing sustainable local economies and sustainable communities of their freedom to be sustainable. Compelling them to import food from thousands of miles away and preventing them from having safeguards that protect the local are—in terms of climate change—ecological crimes.

Earth Democracy generates a radical shift in our paradigms and in our patterns of production. It offers real solutions to resource exhaustion, peak oil, climate change, disposability of people, and the erosion of democracy.

CHAPTER TWO

Sacred Cow or Sacred Car

India today is a zone of contest between a sustainable culture based on ensuring the living energies of biodiversity and the sacredness of species and a non-sustainable and inequitable culture based on fossil fuel. The sacred cow symbolizes a biodiverse culture and a living economy. The cow gives energy as draft power and fuel. She gives organic manure. She gives food in the form of milk, ghee, butter, buttermilk, and yogurt. And she provides pest control in the form of cow urine and *panchkavya*. Today cows are being banned from the roads to make way for the car.

The ficus tree is another sacred species in a biodiverse culture. Throughout India ficus trees have been planted along roads to provide shade. If a road had to be widened, it would go around the ficus trees. Today, millions of sacred ficus trees planted over hundreds of years are being cut down to make highways.

When something is sacred, it is inviolable. Cows and trees used to be inviolable in India. Humans had to adjust their activities to protect them. The livelihoods of the poor were inviolable under Article 21 of India's constitution, which guarantees the right to life. Today the car has become inviolable. Culture and constitution can be violated to protect the car. Humans and other species can be sacrificed to make way for the car. Globalization is literally being driven into India and corporate globalization's proclamation of the car as sacred silences other things held sacred. Cars need highways and overpasses, they need fossil fuels, and they need aluminum, steel, and petrochemicals. Cars redesign the countryside and the city.

Sacred species are being killed to expand the ecological space for the car. If "timeless India" rested on the gifts of the sacred cow—her living energy and her living manure, the foundations of a totally renewable economy—India's short-

term economic miracle rests on the sacred car. Nothing else is sacred anymore; nothing else is worthy of protection.

Cars Eat Men

During the enclosures of the commons in Britain, Sir Thomas More wrote, “sheep eat men.” He was referring to the diversion of land from providing for human needs and sustenance to providing wool as raw material for factories and profits for the landlords and factory owners. The peasants were uprooted; a new poverty was created. Land that fed people was now to feed the factories.

Today, cars eat men. Land is diverted for parking, roads, highways, overpasses, and car factories. The mining of the iron ore and bauxite that makes the steel and aluminum is destroying the land and ecosystem. The drilling for oil is eating up yet more land. Cars are eating up land and ecosystems. The atmosphere is being eaten up with fossil fuel emissions.

From 2006 to 2011 India is projected to be the fastest-growing auto manufacturer among the world’s top 20 car-making countries.¹ Most automobile companies are descending on India in their search for cheap land and cheap labor. General Motors, Honda, Volkswagen, and others will spend \$6.6 billion on new car factories. Fiat, Nissan, Renault, and others are linking with local manufacturers. Bloomberg, the economic news agency, describes this as “cashing in on the nation’s auto lust” but also recognizes that the “nation” is only 0.7 percent of the population. In India only 7 people in every 1,000 own a car, compared with 450 in the US and 500 in Europe.² The “market” is not 1.2 billion Indians but the 216 million members of the middle class. India will also be an export hub for automobiles.

And of course, India has to be made over for the car. The “\$14 billion highway program isn’t keeping up,” Bloomberg reports, concluding that in India car manufacturers will “find throngs of fervent car buyers, a surging auto-making and exporting market—and the burden of inadequate roads.”³

Nissan and Renault have announced they are considering producing a \$3,000 car to compete with Tata’s \$2,500 Nano, which is to be made in the controversial plant at Singur in West Bengal. Singur saw protests and bloodshed when the government of West Bengal forcefully acquired farmland for Tata’s Nano factory and the factory is still under siege. French carmaker Renault is joining up with Indian two-wheeler manufacturer Bajaj Auto whose products will include a \$3,000 car.⁴ Toyota, Honda, Fiat, and Volkswagen are also planning to build or

sell low-cost cars in India. Taxes on small cars have been reduced from 24 percent to 16 percent, further lowering prices. Tata Motors hopes to put 1 million small cars on the road each year.⁵ These “cheap” cars will add to India’s traffic congestion and increase her CO₂ emissions. In Dehli alone, 200,000 vehicles are added to the streets every year.⁶

The car has seriously divided India. People can no longer walk on the streets. Neighbors have turned into enemies over car parking. It has cut up rural India through land grabs for factories and highways. It has divided communities.

India’s automobile burden is not her own. It is imposed by the global automobile industry in order to increase their markets and their profits. It is also a burden imposed by servicing the outsourcing boom. BPOs (business process outsourcing centers) have exploded in India as airlines and banks, hospitals and law firms, every major business has shifted back-office work to India to lower costs. Now, hordes of call-center cabs shuttle BPO workers from residential areas to the exploding suburbs where global corporations have set up operations at all hours.

Population Explosion of Cars: an Ecological Catastrophe by Design

Since 1997, auto production has stagnated in Europe, North America, and Japan, but in Asia and India, production has doubled. China accounts for the biggest increase, with production going up threefold from 1.5 million vehicles in 1997 to 4.6 million in 2005.

In India, until 1984 cars were designated a luxury product and their production was regulated; there were quotas on imports. Six Indian manufacturers dominated the Indian automobile industry—Telco (now Tata Motors), Ashok Leyland, Mahindra and Mahindra, Hindustan Motors, Premiere Automobiles, and Bajaj Auto. In 1991, trade liberalization policies led to the delicensing of the passenger-car segment. In 2000 quotas were removed in the automobile sector, and 100 percent Foreign Direct Investment (FDI) was allowed in the automobile sector. India is increasingly producing for export, which over the last five years has increased at the rate of 39 percent annually.⁷

Along with exports the total number of vehicles in India has increased dramatically. In 1951 there were an estimated 300,000 cars in India. From there it grew to 5.4 million in 1981 and 37 million in 1997. In 2003 it reached 67

million and began increasing rapidly: 72 million in 2004, 85 million in 2005. The majority of this growth was for personal transport. With the efficiency, accessibility, and reliability of public transport systems in most urban areas lacking, people have turned to buying their own vehicles, and private vehicles are providing for a larger portion of travel. For this exploding car population, gasoline consumption is set to increase from 6 million tons in 1997 to 25 million tons by 2015. Diesel consumption will increase from 30 million tons in 1997 to 100 million tons in 2015. In a period of rising oil prices and peak oil, India can only meet this demand if land is diverted from food to biofuels.⁸

Historical data indicates that road transport grows at twice the growth rate of the overall economy. With an economic growth rate in India of more than 6 percent, road transport demand is expected to increase in the region of 12 percent every year. Environmental and safety aspects have become an area of increasing concern.

Any “Ayes” for Public Transport?

In 1950, buses and railways accounted for 70 percent of the total traffic volume in Western Europe; by 1997, their share had fallen to a meager 15 percent. In Australia, private vehicles accounted for 93 percent of total urban passenger transport in 2000. Despite extremely decrepit and ill-maintained vehicles in its fleets, public transport in India meets very high levels of travel demand. In Delhi, 64 percent of the demand is met by buses. In Mumbai, buses account for 69 percent of trips, using only 5 percent of the road capacity. However, more than 91 percent of on-road transport in Kanpur, 88 percent in Hyderabad, and 86 percent in Nagpur is composed of cars and two-wheelers, whereas buses constitute 0.5, 0.5, and 0.4 percent respectively.

Though an efficient public transportation system can provide mobility to all sections of society without the negative impacts of individual automobilization, transport planners have not given it due attention. Government policies are primarily responsible for public transport's losing the race: the majority of cities have, in effect, encouraged the use of the private car. Transportation expert Wendell Cox points to the essential ineffectuality of existing public transport options as another factor in their unpopularity: “People who are getting more affluent will not be content with a transport system that does not take them where they want and when they want in a competitive time.”⁹

Bicycles, bicycle rickshaws, and walking-modes of transportation that could work wonders to reduce congestion and pollution levels—are associated with the poor. And in some cities bicycles and bicycle rickshaws are being driven off the road as a matter of policy. Traffic departments in Lucknow and Guwahati have banned nonmotorized transport from most arterial routes for most of the day. Bicycle rickshaws have been banned in Chandni Chowk, one of Delhi's oldest and busiest markets. A disregard for walking is also quite evident in

transportation policy and planning. Planners focus on ensuring the fast movement of cars at the cost of pedestrians. Pavements and footpaths, where they exist, are either encroached upon or made narrower to increase road width for the benefit of motorized traffic.

Sans Car, Sans Crisis

The movement to make cities car-free is gaining momentum. The city of Freiburg in Germany has shown that it is indeed possible to virtually stop the growth of car use, even when car ownership is rising. Freiburg's car ownership rose from 113 cars per 1,000 people in 1960 to 422 per 1,000 in 1990. However, daily auto use remained constant between 1976 and 1991. Transit passengers increased by 53 percent and bicycle trips rose 96 percent over the same period. While total trips increased 30 percent, car trips increased just 1.3 percent over 15 years. In fact, the share of trips by cars fell 60 percent. Freiburg achieved this through mechanisms such as pedestrianizing the city center, traffic-calming programs, and a difficult, expensive parking system designed to discourage car use. The city also made improvements in its mass transit facilities, including extending and upgrading its light-rail system and using buses to feed the rail lines.

Examples of cities like Freiburg abound. Strasbourg (France) has provided its citizens with a 300-kilometer network of bicycle routes. Bogotá (Colombia), Singapore, and Hong Kong have made efforts to encourage walking and cycling as options. In Germany, public policies have been largely responsible for creating a suburban environment where bicycling remains a feasible travel option. Shanghai—possibly the world's greatest cycling city—has a population of 14 million; in 1995, of the 26.7 million person-trips of more than 0.4 kilometers undertaken each day, 42 percent were on bicycles. Almost all streets in Shanghai have spacious cycle lanes, many of them fully segregated from vehicle lanes.¹⁰

The automotive mission plan announced by the prime minister Manmohan Singh in 2007 projects tremendous growth over the next decade, but no lesson is learned about leaving space for cycles. The goal of the plan is for India “to emerge as the destination of choice in the world for design and manufacture of automobiles and auto components with output reaching a level of US \$145 billion accounting for more than 10 percent of the GOP and providing additional employment to 25 million people by 2016.”¹¹ These figures do not take into account the cost of uprooting peasants for automobile factories and highways, and destroying the non-automotive methods of transport that provide livelihoods to millions.

The plan is the ultimate expression of the petrol religion, in which the car is sacred and people, their livelihoods, and the land and the air can all be destroyed.

Nano-Mania

The launch of Tata's Nano, the Rs 1 lakh (\$2,500) car, in 2008 at the Delhi automobile fair unleashed Nano-mania. The Nano is 3.1 meters long, 1.5 meters wide, and 1.6 meters high. Tata's plant at Singur plans to produce 250,000 cars in 2008 and to increase to 1 million cars per year by 2011. The *Hindustan Times* announced "Small Wonder: It is green, it is global, it is Indian. It costs a lakh and it is called the Nano."¹² *India Today* ran a cover story on Ratan Tata, the owner of Tata, hailing "Ratan's Revolution." *Business World* talked of the "Nanolution." *Newsweek* ran a cover story on the Nano, "Small: It's the New Big," saying that India's new ultramini heralds the future of the car. The Nano became the answer to peak oil and climate change. "Poor countries are getting rich, gas costs are rising and our planet is heating up. The Nano is among the new breed of 21st century cars that are cooler, cheaper and more compact than ever."¹³

The Nano has been declared a "people's car." A full-page ad proclaimed: "It's here, the new Tata Nano to end all speculation, debate and talk and change the way India travels." However, the debates about the impact of millions of new cars are just beginning. Debates about cars eating into the ecological space of cities, causing traffic jams, and pollution. Debates about the need for improving public transport instead of increasing car ownership. Debates about the price paid by peasants and tribals as their land is grabbed for Tata's car factory in Singur and steel plants in Kalinganagar and Gopalpur. When these costs are taken into account, the Nano is not the world's cheapest car, but may be the world's most expensive car. And though *Newsweek* announced, "Poor countries are getting rich," the reality is that poor people in poor countries are losing what little they have, their land and livelihoods, while corporations get richer.

Is the Nano green? Journalist Adil Jal Darukhanawala said, "The 'Greens' should love it."¹⁴ Although the Nano gets good gas mileage, it is a fossil fuel-driven car. It is not an electric car; it is not a hybrid. It will, therefore, at 1 million new cars a year, contribute heavily to greenhouse gas emissions.

The Tata car is made cheap by land-grab subsidies, by mining subsidies, and through reduction in taxes. While the government is working on a National Urban Transport Policy on one side, it is encouraging projects like Tata's with tax benefits on the other. This will encourage private vehicles and conflicts with the sustainable transport system that government wants to promote. These small cars can do serious damage to the environment, especially in already highly

polluted cities.¹⁵

In a fast-developing economy like India, a \$2,500 car is affordable for many of the newly rich, and will result in millions of cars coming on the streets in a short span of time. There is a fear that these small cars will unleash chaos, leading to a mix of uncontrollable traffic and high pollution levels.

The High Cost of “Cheap” Cars

Cars have a limitless appetite for oil. They also have a limitless appetite for land—for car factories, for steel plants, for iron ore mining, for roads.

Kalinganagar in Orissa is the site of Tata’s steel plant. This land is home to the Ho, Munda, and Santhal tribal communities. The steel plant will displace the villages of Chandia, Gobaraghati, and Gadapur. Tata moved to Kalinganagar after its attempt to set up a steel plant in Gopalpur failed in the face of popular resistance there.

In Kalinganagar in 2004 tribals began to resist Tata’s land acquisition. In January 2006, the conflict escalated when Tata came with 400 police officers to start leveling the land. Four hundred tribals gathered to protest with their bows and arrows. The police fired stun shells, rubber bullets, and tear gas. By the end of the conflict, 13 tribals and one police officer had been killed and 37 tribals and four police officers had been injured. The villagers then sat on a dharna on the national highway. When the police returned the bodies of the dead, there was further outrage and anger when it was discovered that out of the eight bodies returned, five had their hands chopped off so that the bodies could not be identified by their fingerprints. In response the Visthapan Virodhi Jana Manch (WJM), the organization spearheading the anti-displacement movement, issued demands that included an end to displacement, compensation for those already displaced, prosecution of the government officials involved, the expulsion of multinational corporations from the region, and the granting of land and mineral rights to the tribal population.

Kalinganagar was just the beginning. Tata is now setting up a car factory on 998 acres of fertile land in Singur in West Bengal, where farmers grow three rotations of crops every year. The government started the physical takeover of the land in December 2006. About 5,000 police and Rapid Action Force (RAF) troops started erecting barbed wire on the land. The local villagers tried to resist the attempt of the government with mass mobilizations. In response the police force and RAF resorted to a widespread *lathi* charge (a lathi is a 6- to 8-foot-long

baton with a dull metal cap) and firing tear-gas shells and rubber bullets. Crops were destroyed, people were attacked, and women were molested and raped. Sixty people were arrested.

A few days later, the villagers spontaneously shut down shops and formed a procession of about 500 when the police forcibly took away six people from the Bajemelliya Santoshimatala hunger-strike camp. Hunger strikes followed all over Singur, with farmers and their families agitating against the state's acts. Farmers from 12 villages with 347 acres of land have pledged not to part with their land holdings. The numbers of hunger strikers increased day by day. Women and men of all ages—from 7 to 85 years old—participated in the fasts.

Such unruly transfer of agricultural land for industrial purposes is destroying the food-security paradigm, destroying the sociocultural-political-economic paradigm of the living community and the next generations to come. The disparity between agriculture and corporate industry is widening at a scale not seen before. It is happening through coercion and undemocratic and inhumane means. “Growth” and “development” must come through fair, just, and equal means, not under the shadow of the gun.

Highways to Dictatorship: Forgetting Gandhi, Following Hitler

Superhighways have become the perfect symbol of contemporary India's distorted identity. They are at the heart of the “India Shining” imagery that was launched as a way of marketing India internationally and used by the Bharatiya Janata Party (BJP) in their reelection campaign of 2004. Former prime minister Atal Bihari Vajpayee said: “Roads are like the lines on the palm. There is a line of destiny, which is going to join Srinagar to Kanyakumari. We want to see that day when we start from Kanyakumari and reach Srinagar with ease.”¹⁶ Prime minister Manmohan Singh, who heads the Congress-led government, has continued the love affair with cars and highways.

Highways are not lines on the palm, they are more like tattoos— marks imposed by external design on the landscape. The violence of this imposition is shaped by the World Bank and IMF. The redefining of India is a forgetting of Bharat, of India's history. By writing our fate in cement, we are erasing our fate from our soil, our land, and our ecology. In India, we have viewed our mountains and rivers as the “lines on the palm.” They are an intrinsic part of the ecology and geography of our motherland. They are our givens and our givers. We have

been intimately connected to our land, our rivers, and our mountains. The earth has shaped our destiny. And through this connection, we have been connected as a civilization from Srinagar to Kanyakumari.

The sources of the tributaries of the Ganges are the Char Dhams, and people across the country undertake the pilgrimage to the Himalayas and Yamunotri, Gangotri, Kedarnath, and Badrinath. It did not take a highway to bring the people of the South on this pilgrimage; it took the sacred bond with our mountains and rivers. And in fact, the pilgrimage had value because it was undertaken on foot. India has long celebrated the *padyatra* (the foot march). Gandhi's Dandi March was a *padyatra*. The Chipko movement to save trees spread in the High Himalayas through *padyatras*. Even today, thousands walk to take Gangajal home for Shivaratri. Most women in rural India walk to collect fodder, fuel, and water. The distances they walk will increase as more trees are cut and as the cement and coal tar of highways cover our soils, suffocating wells and streams.

Substituting sacred rivers with highways, substituting our connection to the sacred earth, her mountains and forests, with connection to automobiles, cement, and coal tar is rewriting India's ecology, culture, and history. It is adopting an obsolete, outmoded, unsustainable model of development imported from the West, one with high social and environmental costs.

Tagore reminded us that India is distinctive because it is an "Aranya Sanskriti." It derives its inspiration from the forest and the living world, unlike the West, which derives its cultural characteristics from dead brick and mortar. Yet, today across the country, ancient papal and sacred ficus trees, often hundreds of years old, are being chopped down to make way for the automobile.

As Gandhi stated:

Modern civilization seeks to increase bodily comforts, and it fails miserably even in doing so. ... This civilization is such that one has only to be patient and it will be self-destroyed. ... There is no end to the victims destroyed in the fire of [this] civilization. Its deadly effect is that people come under its scorching fumes believing it to be all good.

It is a charge against India that her people are so uncivilized, ignorant and stolid, that is a charge really against our strength. What we have tested and found true on the anvil of experience, we dare not change. Many thrust their advice upon India, but she remains steady. This is her beauty; it is the sheet anchor of our hope.¹⁷

India's tradition of leaving a small ecological footprint on the planet is being erased in a race by India's elite to imitate and outdo the industrialized West in consuming the earth's resources—in usurping the ecological space of other beings, indigenous and rural communities, and the urban poor.

The superhighway and the automobile are the ultimate cultural symbols of

non-sustainability and ecological exclusion. Our roads once had a place for the cow, the horse, the camel, the elephant, the pedestrian, the bicycle, and the car. We are now privileging the car owner. The sacred cow has disappeared from Delhi's roads.

Rickshaws and handcarts—the ultimate expression of fossil fuel-free, climate-friendly transportation—have been banned in order to “clean up” the streets for cars and automobiles. Kolkata and Delhi banned rickshaws in 2006 as a “cleanup” operation. In fact, getting rid of human-powered vehicles promotes highly polluting two-stroke motorized rickshaws. These mopeds are considered autos under Indian law and are thus allowed on the streets. Bicycle rickshaws generate no pollution and use renewable energy rather than fossil fuels. They account for 25 to 80 percent of vehicle trips in many cities and are responsible for more than 10 percent of the total employment in major cities. In Patna, 28,000 people are employed in public transport, most as bicycle rickshaw drivers. When indirect employment linkages are considered, it is 42,000 jobs supporting 150,000 livelihoods. A Rs 100,000 investment in a bus generates two jobs; investing that amount in auto rickshaws creates six jobs. Investing it in bicycle rickshaws creates 75 jobs.¹⁸ Ten to 20 percent of the urban traffic in India and Pakistan is made up of bicycle rickshaws. In Bangladesh, the bicycle rickshaw fleet has grown from two-thirds of a million in 1988 to over a million in 2000. Annually each bicycle rickshaw accounts for 30,000 passenger-miles and almost 100 ton-miles of goods. Bicycle rickshaws, ox carts, and country boats account for 75 percent of the economic value added to goods, 80 percent of the employment, and 40 percent of the vehicle assets of the transport sector.¹⁹

The highway and the automobile are symbols of totalitarian cultures: they deny people sustainable and equitable alternatives for mobility and transport. To go from Kanyakumari to Srinagar, India has a big network of railways. Yet, in the government's advertisements of the highway projects, it is suggested that there is no way for the people to get from one point to another without superhighways. Our leaders are blind to the experience of the West in giving up more sustainable and people-friendly methods of transport for road transport. In Germany, road transport accounts for 91 percent of all air pollution, 64 percent of all noise pollution, 56 percent of all construction and maintenance, and 98 percent of all accidents.²⁰

A study in Germany found that road transport causes eight times the pollution, destroys ten times the land, and results in 20 times the accidents that rail transport does.²¹ Road transport accounts for 17 percent of all CO₂ emissions. All these externalities are known. Yet, India's rulers are choosing an

obsolete, crude, and costly form of transport as a symbol of “India Shining.”

The *sadak* (road) was part of the BJP’s election slogan in the assembly elections. In the lead-up to the general elections in 2004, highway advertisements ensured that the highway and automobile became symbols of a brave new India. But we need to take lessons from history and other societies. We have a century of experience with the ecological and social violence of the automobile. We can save ourselves from becoming its slave. And we have lessons from Nazi Germany, where highways were designed as means of centralized control, fascism, and dictatorship, and not as routes to human freedom and democracy.²²

As Wolfgang Sachs writes in his classic, *For the Love of the Automobile*:

Dictatorships live not only by force but also by emotional appeal; the shining eyes of the man on the street are as much a part of the image of the time as the Gestapo at dawn. A history of this enthusiasm in the period of German fascism has yet to be written. Yet, whoever undertakes to eavesdrop at the corner bar and uncover that consent from below amid the oppression from above will have to make room for a chapter about the National Socialists’ motorization policy.²³

Former prime minister Vajpayee’s ground breaking for his pet highway projects recollects images of Hider’s ground breaking for the Frankfurt-Basel Hanseatic highway in 1933. Prime minister Singh continued the work Vajpayee started. The German Reich Automobile Law, which made the highways possible, took away road-building responsibility from the states and concentrated it at the national level. The 2003 Highways Act of India achieved similar centralization and concentration of power.

In both Germany and India “auto-only” roads have replaced the pluralism and democracy of transport. A German memorandum identified the countryside as the biggest impediment to the automobile:

It is expected to share the streets, with horse-drawn carriages, bicyclists and pedestrians.... The modern concept of traffic engineering is to introduce a network of special highways, to serve the needs of longdistance travellers and to be used by the fastest automobiles (for which it is meant).²⁴

The monoculture of the mind, which has destroyed biodiversity in farms and forests, which has fueled communal hatred, which fed Hider’s policies, is now being brought to India’s landscape and roads. The car owner and long-distance traveller is a privileged citizen. The ox cart, the bicycle, the pedestrian must be displaced for the automobile, which previously was just one among many modes of transport. The BJP’s highway project aims to rewrite India’s diverse and pluralistic fabric in a very basic way. Hitler too raised the flag of “national highways” as a way to create the Volksgemeinschaft, a national society

connected as “one people, one reich, one führer.” Nazi Germany used highways to wipe out diversity, autonomy, and decentralization and “to mould the German people into unity.”²⁵ India’s current rulers are also using the highway as a means of, and metaphor for, molding India into one monolith, serving the luxury demands of the rich, excluding the poor and the diversity that supports their lives.

One such highway ad from the 2004 election campaign proclaims the increase in highways built and people employed. Nazi advertisements used similar measures of achievement. The “India Shining on tarmac” advertisements find a parallel in the euphoria of the German Reich. The National Socialists presented road building as both a technical achievement and a cultural feat. As the inspector general of German roads, Fritz Todt stated after the first 1,000 miles were built: “It is once again a matter of pride to be a road builder. The German Reich is getting roads the levels of which, in their magnitude and beauty, have never been built in the history of human culture.”²⁶

India in the 21st century needs to be building on Gandhi’s legacy, not Hitler’s. It needs to avoid repeating the ecological and social mistakes of the West. India has offered alternatives based on sustainability and pluralism. Gandhi observed:

God forbid that India should ever take to industrialism after the manner of the West. The economic imperialism of a single tiny island kingdom [England] is today keeping the world in chains. If an entire nation of 300 million took to similar economic exploitation, it would strip the world bare like locusts.²⁷

We are today 1 billion. And we are being asked to adopt the lifestyles and economies of the 20 percent of humanity who have been using 80 percent of the world’s resources. If 100 million rich Indians want to live like their Western counterparts it would take more resources than the world has to offer and the attempt would force their brothers and sisters to give up their water, their land, their homes, and their livelihoods. The highway project is not uniting India; it is dividing India. It is creating an automobile apartheid in which the rich drive at high speeds on highways built by cutting through villages and forests, tearing down homes, farms, and trees. They drive through without even seeing the brothers and sisters whose livelihoods they are robbing. Superhighways are not our destiny or the lines on the nation’s palm. They are graveyards of cement and coal tar, which are burying our soils, our villages, and our freedoms.

Infrastructure for Which Structure?

An infrastructure holds together a structure. However, increasingly, the word *infrastructure* is being used without specification of the structure of society that it supports. In World Bank jargon, the assumption is that infrastructure should support a fossil fuel-based society for the elite, with the needs and rights of the poor not taken into account. Similarly, it's assumed that the infrastructure supporting local, decentralized economies based on renewable energies should be dismantled.

India already has one of the largest networks of roads in the world, totaling 3.32 million kilometers. The largest network connects villages, and the smallest are the expressways, which are part of the privatization project of the World Bank.

Although the national highways constitute only about 2 percent of the road network, they carry 48 percent of the total automobile traffic. Once you get off the national highways, types of transport expand to include the ox cart and the bicycle, the cycle rickshaw and the pedestrian, the *tonga* (horse cart) and the handcart. Yet the vehicles of millions find no place in the statistics that count only the automobile. Only when one ignores local economies and their diverse local transportation strategies can one say that 65 percent of freight and 80 percent of passenger traffic is carried by automobiles. When cycle rickshaws and ox carts are added, non-motorized vehicles provide 80 percent of local transport.²⁸

As the infrastructure of local economies is dismantled, and as the infrastructure for long-distance automobile commuting is increased, the number of automobiles increases. It has been growing by 10 percent a year from 1998 to 2003.

The infrastructure of decentralized, democratic, and renewable economies includes animal and plant biodiversity, which provide renewable resources and energy; it includes local markets for exchange; it includes the skills and knowledge that allow local production and local consumption to provide for people's needs and well-being. But have you ever heard of official statistics that recognize the tree or the cow as the infrastructure? Infrastructure in the fossil fuel paradigm is highways and overpasses, airports and harbors. And to build the structures for a non-sustainable fossil fuel-centered economy, other infrastructures must be rendered invisible and destroyed. The biodiversity infrastructure does not count; even the massive railway network is made to disappear. Mobility is reduced to cars and highways.

Highway Connectivity: Uprooted People

Land is the basis of agriculture. About 70 percent of the people in India depend on land and agriculture for their survival; of these 42.4 percent are landless and around 53.7 percent are partly landless. In addition to actual food, the timber, fuel, and fodder that access to land provides is of vital importance in creating individual, local, and a genuinely people-centered natural food security. The fertility of the land is of prime importance to meet individual, regional, and national food needs.

Privatization and globalization are undoing the land-reform laws that attempted to reverse the concentration of land ownership from the zamindari system in colonized India. Introduced by the British, the system wreaked havoc on the food security of the peasantry and resulted in the devastating famines of the 18th, 19th, and early 20th centuries.²⁹ Recognizing the importance of having access to land as the basis of survival for the majority of Indians and in response to the Tebhaga movement (1946–7) the Constituent Assembly unanimously agreed to abolish it. One of the major targets of the World Bank agricultural reforms is the abrogation of Land Ceiling Laws, a mechanism designed to prevent the continuation or emergence of monopolies in land ownership of the kind that the country faced throughout the 18th and 19th centuries under the old zamindari system. These changes were begun as part of a structural adjustment program.

Land is being taken over in the name of infrastructure for urban development and new roads and highways. The land being “acquired,” often through force and violence, is mostly agricultural. In addition to being supported through taxpayer money and international funding, the private sector is receiving numerous concessions from the government for the construction of the highways.

The participation of the affected people in planning and implementing resettlement programs should be ensured, and displaced persons should be assisted to improve and regain their former production levels, income-earning capacity, and living standards. No development project can be justified if it pauperizes a section of the society. The definition of “displaced persons” must go beyond landowners to include those whose sustenance depends on land owned by others or common property resources. Land should be compensated with land and resettlement for all the displaced persons should be mandatory.³⁰

For any rehabilitation policy to have teeth it must be able to be put into practice. The state is obligated to rehabilitate those affected. There should be no

displacement without rehabilitation. Those who customarily live on forest land and have customary rights to common property resources and have improperly been labeled “encroachers” in the property-oriented legal systems should be treated in the same way as those who have papers to their property. As of yet, neither state governments nor project authorities have shown either expertise in or commitments to the rehabilitation of displaced people. When assessing the economic value of assets lost for the purpose of compensation, the replacement cost of all quantifiable values should be used in place of the depreciated value of assets currently used. The implementation of the principle of land for land and—in many cases—even of rehabilitation itself has been left to the discretion of state governments. The responsibility for compensation and rehabilitation should be placed equally on the project authorities and the government. The displaced people have lost faith in the desire of the governments to honor their commitments.

The highways program is a privatization of space, of land, and of mobility. The land for six-lane highways comes at the cost of destroying settlements that have evolved along India’s ancient roads such as the Grand Trunk Road that connects Lahore to Kolkata. It comes from forcibly taking over the lands of the farmers.

Highway Robbery: Corporate Rule, Corruption, and Crime

Soil and oil come into deep conflict as India expands her highway and expressway network with funding and privatization conditionalities from the World Bank and the Asian Development Bank (ADB). In August 2000, the World Bank made a loan of \$516 million for the third national highway project and another loan of \$589 million in July 2001 for the Grand Trunk Road improvement project. The ADB made a \$245 million highway loan in 1995, and loans totalling \$740 million in 2002. Changes in the land acquisition act to facilitate appropriation of land from peasants for highway development and privatization of highways under pressure from the World Bank and the ADB have led to unprecedented land grabs and high-level corruption.

The Taj Expressway: This is the story of one of the biggest land swindles India has ever seen. The manner in which land worth millions of rupees was handed over by the Uttar Pradesh (UP) government headed by Chief Minister Mayawati to Jaiprakash Industries Limited (JIL), India’s biggest dam builder, led

to legal inquiries. The findings of Chief Justice Mishra are startling. He described the events as “fraud in its naked form.”³¹

It all began with the objective of constructing a six-lane, 160-kilometer expressway between Noida and Agra, called the Taj Expressway. Even though the Taj Mahal is well connected by train and road, and falls on the main railway line from Delhi to South India, the privatization of highways has created a frenzy for road building and has fueled new levels of corruption. The road was to be completed in seven years in a joint venture between JIL (75 percent share) and the state-owned Taj Expressway Authority (TEA). In return, the state government was to hand over 6,000 acres of land adjoining the expressway to Jaiprakash Industries, including 600 acres of prime land at Noida.³² Noida is part of Delhi, where land values are skyrocketing. According to the Mishra report, the purpose of the project, construction of the expressway, was totally sidelined. Instead efforts concentrated on ensuring that JIL won the bid and received the land. The highway project became a means of taking land from the poor.

The transaction was marked by many improprieties including bypassing the proper review process and last minute, one-sided changes drafted by JIL and agreed to by the government. Within 72 hours of signing the revised agreement, JIL asked the state government to release 600 acres in Noida (the market value of which would not be less than Rs 15 billion). The state government officials, who were hand-in-glove with the company, ordered the release of the land on the very same date.

The way these finances were arranged tells yet another story of how farmers' lands and public assets are being looted. The revised agreement, according to the chief justice's report, is so one-sided that it includes no safeguard to guarantee the successful completion of the project, nor was the transfer of land for development made subject to the successful completion of the project. There is no provision for return of land in the event that the project itself is found unviable or in default. The Taj Expressway remains unbuilt and the renamed Yamuna Expressway aims for completion by 2010. Even though the Taj Expressway project led to investigations and a change in government, the same chief minister is back in power and the same company, JIL, has won the bid for the Ganga Expressway.

Bangalore-Mysore Highway: The Bangalore-Mysore Industrial Corridor is India's largest private real estate/infrastructure project. It involves building an expressway between these two cities and creating five new cities along the corridor. The government allocated about 21,000 acres of land toward this project in 1997, of which one-third was government-owned land, such as forests,

village commons, and “wastelands,” the rest being small farms and common grazing pastures, which support small farmers and landless laborers. The developers admit that ousting people for a project is a thorny issue. The powers of over 170 village *panchayats* (village governments) to regulate development that fell within the acquisition area were made subservient to a state-led, high-level committee. The Rs 60 billion project to construct a toll expressway between Bangalore and Mysore, which entails the takeover of 8,000 hectares of land, is caught in allegations of a real estate swindle. The government paid farmers below-market rates (around Rs 200,000 to Rs 300,000 per acre when the market rate was Rs 5,000,000 to Rs 1,000,000 per acre) and gave the land to the developer, Nandi Infrastructure Corridor Enterprises Limited (NICEL), at an even lower rate (Rs 100,000 per acre).

Successive governments exempted Nandi Infrastructure Corridor Enterprises Limited (NICEL) from stamp duty and registration charges for the land transferred from the Karnataka Industrial Areas Development Board (KIADB) to NICEL, and from NICEL to its lenders, and exempted the company from industrial conversion charges, which, at the rate of Rs 160,000 an acre for over 20,000 acres of land, meant huge savings.

Upgrading the Bangalore-Mysore railway line would have been economically more cost-effective and environmentally less hazardous than the highway project. According to an independent study carried out by the Indian Institute of Science, Bangalore, doubling and electrifying the existing railway line would have taken only half the ten years it took to complete the expressway. The railway project would have cost a quarter of what the expressway cost, and the acquisition of the necessary land would have been negligible and would not have displaced anyone.³³

Honest engineers have even been killed over this highway project. Satyendra Dubey, a deputy general manager working on the World Bank-funded highway project in Gaya, was killed on December 27, 2003, allegedly by the construction mafia. In UP, construction contractors killed a deputy project manager working on National Highway 19.³⁴ Other engineers killed in UP include L.N. Singh, Indu Bhushan, S.C. Rai, Masood Alam Siddiqui, S.K. Bajpayee, and Anwar Mehndi Rizwi.³⁵

This is only part of the violence of creating automobile-highway culture in India.

Death Comes on Wheels: Trends of Road Accidents

According to the Institute of Road Traffic Education, Indian roads are witness to over 230 deaths and around 3,500 serious injuries every day. The death rate in road accidents is 2 per 100,000 vehicles in the US, and 32 in Pakistan; in India it is 140 per 100,000.³⁶ Most of those killed are in the age group of 15 to 44. A report by the Loss Prevention Association (LPA) states that nearly 130,000 people per year are crippled for life. A two-wheeler rider is five times more likely to have an accident than a person in a four-wheeler. There are nearly 47 million two-wheelers registered in India, which account for nearly 70 percent of the total motor vehicles in the country.³⁷

India has 1 percent of the world's motor vehicles but 6 percent of the vehicular accidents. An accident is reported every three minutes, and 100,000 people are killed every year in India on the roads, more than in any other country. Road accidents are highest on national and state highways, and heavy vehicles like buses and trucks are responsible for 43 percent of the accidents. It has been estimated that 6 million more people will die and 60 million will be injured over the next ten years in developing countries, with India experiencing 30 percent of those accidents, unless preventive action is taken.³⁸

More people are killed on the streets of Delhi than in the areas of the Kashmir insurgency. An article on the traffic in Delhi proclaimed, "There is a war in the capital's streets."³⁹

Building highways through India's ancient settlements leads to high rates of accidents. Originally a cowpath, the stretch of highway between Agra and Mathura has been identified as among the most dangerous for fatal accidents. A survey carried out by the National Highway Authority of India found that at least four commuters are killed or injured every day. The survey has marked 28 "black spots" on the 180-kilometer stretch of NH-2 where accidents take place frequently, claiming dozens of lives each month.⁴⁰ These "black spots" are where there are an "unnatural" number of pedestrians and small vehicles because there are "too many" villages along the roadway. People have been transformed into "black spots" as the landscape has been transformed from being centered on the sacred cow to being centered on the sacred car.

In India, pedestrians are a major proportion of road users killed in road accidents. In the year 2001, there were 924 pedestrians killed on Delhi roads—50 percent of the city's fatal road accidents. Elderly pedestrians, though less involved in accidents, are more likely to die after being in one. Who are the victims? Detailed breakdowns are available for only a few places like Delhi, but experts feel that they reflect the countrywide trend. Pedestrians are the majority of the victims, followed by people using motorized two-wheelers and then

cyclists.

Why do accidents occur in such frightening proportions? Fingers point to the inappropriate infrastructure design, which encourages mixing high-speed and low-speed traffic. And speed kills. Road design in India completely ignores the most-vulnerable road users—pedestrians and two-wheeler drivers. There is no space on the road for the slower traffic. Consequently, heavy vehicles like buses end up sharing the slow lane. The conflict between fast and slow vehicles leaves the slower ones vulnerable. The mix of vehicles is a social reality to which our planners are not sensitive. Even traffic-decongestion measures like overpasses and tunnels are conceived only for fast vehicles. As there are few pedestrian pathways and cycle tracks, the conflict is inevitable. The basic infrastructure for speed does not exist, while high-speed cars are the norm. High speed can be safe only with high cost, including the destruction of ecologically sustainable modes of mobility and transport.

Vehicular Pollutants Lead to Cancer

Climate change and auto accidents are not the only ways cars kill people; their pollution is also harmful on a more individual level. Acute respiratory infections and bronchitis are occurring more frequently in cities like Delhi because of the automobile explosion.

A study of the health hazards caused by Delhi's air pollution came up with alarming findings. The study found that 26 percent of Delhiites have undergone chromosomal changes that could be a precursor to cancer and that over 77 percent of Delhiites suffer from some negative reaction to vehicular pollution. The changes observed are more prevalent in people in the lower strata of the society, and women suffer more than men. High incidences are found among traffic police and street vendors, who work in traffic, where air pollution is highest. About 40 to 50 percent of the illnesses were attributable to high levels of respirable suspended particulate matter (RSPM) in the city.

Though carbon monoxide levels have gone down since the introduction of CNG in public transport, the RSPM level in Delhi is still the highest in the country. The RSPM level in Delhi has gone as high as 250 micrograms per cubic meter. The US Environmental Protection Agency says 65 micrograms per cubic meter is the peak safe level.⁴¹ The high levels are due to truck traffic, the soaring number of vehicles, and inversion condition in the winter.

Delhiites also face rising ozone levels that far exceed WHO standards.

Exposure to high levels of ozone for even a short duration can damage lungs and the rest of the respiratory system. Though ozone is not emitted directly by vehicles (or any other source), nitrogen oxide and other volatile organic compounds that result from vehicular traffic react with sunlight to form ozone.⁴²

Vehicular Pollution Triggers Asthma

Pollution and asthma are linked, as studies in India and around the world have shown. Delhi's polluted air is responsible for a 41 percent increase in asthma cases and a 39 percent rise in chronic bronchitis, according to a study by J. N. Pande, a former professor of medicine at All India Institute of Medical Sciences. The figures were established by correlating the amount of pollutants in the air with the number of emergency-room admissions on particular days. More people were admitted to emergency rooms on days when total suspended particulate matter and carbon monoxide levels were high.⁴³

Asthma places a financial burden on countries around the world. Directly and indirectly, the US spends more than \$6 billion on asthma. The average cost of treating asthmatic children in that country is \$3.2 billion. Britain spends about \$1.8 billion on health care for asthma and on days lost through illness. In Australia, annual direct and indirect medical costs associated with asthma reach almost \$460 million. The cost to India is not known, but it is high.⁴⁴

A shift from motorized transport to cycling and walking can reduce the costs associated with asthma. A 2001 report by the Economic Commission for Europe and the WHO's regional office for Europe points out: "The large health and environmental benefits that can be derived from encouraging cycling and walking in urban areas have not yet been given sufficient prominence in decisionmaking."⁴⁵ It goes on to say that each year air pollution causes 6 percent of all deaths in Austria, France, and Switzerland. About half of these are due to air pollution from motorized traffic, which is twice as many deaths as are caused by road traffic accidents in these countries. Traffic-related air pollution is also responsible for more than 25,000 new cases of chronic bronchitis in adults, more than 290,000 episodes of bronchitis among children, and more than 500,000 asthma attacks annually.⁴⁶ Asthma can't be cured, but the pollution that triggers it can be prevented.

Lead Poison Along National Highway

Not all is well with the health of National Highway 1 in Punjab. From Ambala to Amritsar (Wagah), the north-south highway is in the grip of toxic lead that has accumulated in roadside soil following years of exposure to motor vehicle exhaust. Despite a switch to tetraethyl lead-free fuel, the damage persists. Past studies show that, although lead is not readily translocated to edible portions of most plants, in areas of rapid urban expansion, where farms and homes are intermixed with heavily traveled highways, lead concentration in soils and plants can reach unusually high levels. Most of the lead from gasoline exhaust is deposited along the edges of highways as soluble chlorobromide.

A study on the problems associated with lead pollution along NH-1 passing through Punjab conducted by soil experts from Punjab Agricultural University (PAU), Ludhiana, made startling revelations. The researchers found that the soil samples taken from near the road have a much higher concentration of lead than samples taken from sites located just 50 meters from the highway.⁴⁷ Plant samples (leaf tissues of wheat crops and eucalyptus trees) collected from the edges of the national highway show that the concentration of lead is much higher than the permissible limit of 10 micrograms per kilogram. Concentrations in one sample were as high as 38.75 micrograms per kilogram. Such levels explain why people living close to roadsides suffer serious medical problems—sterility, miscarriages, chronic nephritis, discoloration of lips, and premature development of arteriosclerosis among them.⁴⁸

Lead pollution, due to its long persistence in soil and its association with disruptions in the cognitive development of children who live close to polluted areas, is a major concern. The use of tetraethyl lead as an anti-knock agent in vehicle fuel has been known to cause general increases in the content of lead in both atmosphere and soil. A study conducted in 2005 showed that even “unleaded” fuel contains traces of the heavy metal, which is then emitted in motor vehicle exhaust.

Animals: a Living Energy Alternative for Mobility

The automobile has emerged as a key symbol of the fossil fuel age. However, the fossil fuel paradigm of mobility ignores and eclipses the hundreds of sustainable options available in different cultures. Mobility has been provided by animals

and by rickshaws. Donkeys, mules, camels, elephants, yaks, llamas, buffaloes, oxen, dogs, and reindeer have helped humans move across diverse terrains in diverse ecosystems. Elephants have moved goods and people in India for millennia. I remember, as a child, going with my father, a forest conservator, on elephantback deep into forests where no roads could reach. And when the Forest Department would loan the elephant to our school for elephant rides during the annual fete, we would ride it to school.

When I was in the small mountain town of Nainital, we would walk the two miles to school. Later, in Dehradun, we would ride to school in a *tonga*—a horse-drawn carriage. And when we were older, the bicycle was our mode of transport.

During holidays, we would travel with our father—and in the mountains, travel means horseback. In the plains, we would travel in an ancient Austin, which would often have to be pulled out of ditches in *kutch*a (unpaved) roads by ox carts. I have never thought of the car as superior to other means of transport. The camel can provide mobility across the desert and the reindeer takes us across the snows.

Yet, the false economics of the mechanistic paradigm has treated biodiversity-based mobility as inferior and the automobile as superior. The mechanical age has privileged the car. All public policy instruments have been used to promote the car and exclude ecological transport. Speed is rewarded, even though speed is killing cultures, people, livelihoods, and the planet itself.

The camel has been called the “ship of the desert.” It can live where few other animals can live. The camel can tolerate an environment that ranges from 38 degrees Celsius to arctic cold. It can work six to eight hours a day, travel 25 miles, carry up to 1,000 pounds, and survive without water for several days. Imagine a car running without fuel for even one second!

The Bactrian camel originated in the ancient country of Bactria, which is now a part of Afghanistan. Today, it is in danger of extinction. Camels can survive in the desert environment because of unique characteristics such as the hump, which stores fat, allowing the camel to go without eating and drinking for five to seven days. The Bactrian camel has two humps; the Arabian camel has one hump. A large camel can drink 21 gallons of water in ten minutes and store it in its bloodstream. When it goes without food and water for many days, it can lose as much as a quarter of its body weight.

The total camel population in the world is about 19 million, out of which 14 million (74 percent) live in Africa, more than 4.5 million (25 percent) in Asia, and a quarter of a million elsewhere in the world. The so-called developing countries are home to almost 99 percent of the world’s camels. India stands third

in camel population, with 1.5 million.

In India, camels are used for providing energy for plowing, transport of people as well as agricultural produce, drawing water from deep wells, and extraction of oil from oilseeds. Camels also provide milk, wool, hide, dung, and bones.

The three major camel breeds of India are Bikaneri, Jaisalmeri, and Kutchi. The Bikaneri has excellent draft power and is the breed of choice in the Thar Desert. The Jaisalmeri was used in the military and continues to be used by the Border Security Force. The Kutchi breed has good milk potential. Other camel breeds are Mewari, Marwari, Sindhi, Shekhawati, Mwati, and Riverine.

Donkeys are an important source of energy in deserts and mountains. A donkey can pull 25 percent of its body weight for seven hours a day. An ox can only pull 18 percent of its body weight for four and a half hours per day. Donkeys work for 30 years, oxen for 15. And contrary to popular perception, donkeys are highly intelligent.

Former prime minister Chandra Shekhar used to have a statue of a donkey outside his home because he felt that if we can respect the donkey, we will respect humans who labor and work. It was his symbol of workers' rights. For the Greeks, the donkey was associated with the god of wine, Dionysus. The Greek word for ass or donkey, *hamor*, appears roughly 100 times in the Bible. In the Old Testament, God chose to speak through a donkey to show Balaam his folly, and in the New Testament Christ rode into Jerusalem on the back of a donkey.

Rajasthan has an annual donkey fair in Kanota, 25 kilometers from Jaipur. Donkeys are sold to washer men, potters, and construction workers. However, as the automobile has taken over, donkeys have been disappearing. Tractors and motor vehicles are replacing donkeys. *Asian News International* reports, "[The donkey fair] used to be for donkeys alone and around 4,000–5,000 donkeys used to be sold. But these days, only 800–900 donkeys are sold. The figures are falling day-by-day. It seems, the donkey fair will be finished and will become a horse fair."⁴⁹

Horses too have been a major source of energy. Indigenous breeds of horses include Marwari, Kathiawari, Manipuri, Spiti, Bhutia, and Zanskari. The Indian breeds Deccani, Chummarti, and Sikang are on the verge of extinction. The Marwari and Kathiawari are breeds selected for beauty and as racing horses. Bhutia, Spiti, and Zanskari ponies are slow-moving and used for transportation in the mountains. Manipuri horses were the original polo ponies because of their quickness and maneuverability.

To the extent that animals are part of the living energy system, animals will

be protected, as will their diversity. Draft animals provide a unique form of renewable energy—converting the sun’s energy through plant life into products and services. Draft animals’ power is self-repairing, self-propagating, and sustained by by-products of agriculture. In the third world, 2 billion people depend on animal energy, and 50 percent of the world’s food supply depends on animal energy. Animal energy provides 80 percent of the energy consumed in many countries and 95 percent in South Asia and the Far East. It is estimated that there are 300 million draft animals in the world. In India, there are 84 million draft animals, of which 72 percent are oxen. They plow two-thirds of the land and provide two-thirds of the rural transportation. There are more than 10 million animal-drawn vehicles in rural India.⁵⁰

When I was in Bangalore at the Indian Institute of Management in the early 1980s, our director, Dr. N. S. Ramaswamy, was lovingly called the “Bullock Cart Man” because of his life’s mission to make animals visible in India’s energy landscape. Even today, much of the data on animal energy comes through his publication, *Cartman*.

Animals plow 100 million hectares on India’s small farms and haul 25 billion kilometer-tons of freight in 15 million ox carts. It is estimated that 74 million oxen and 8 million buffaloes make available 40 million horsepower of energy worth Rs 100 billion per year. Animal energy saves 6 million tons of petroleum worth Rs 120 billion per year, mostly in foreign exchange. The asset value of draft animals is Rs 250 billion. Draft animals provide anywhere between 0.4 and 0.8 horsepower. Taking an average of 0.5 horsepower per animal, the power made available by 300 million work animals is 150 million horsepower. It would take 30 million tillers and tractors to replace their work. Thus the replacement of renewable animal energy by mechanized systems would require an investment of \$200 billion to \$300 billion.⁵¹

It has been calculated that 6.3 times more energy is needed to cultivate soybeans with a tractor than with draft animals.⁵²

Animals provide not just energy, but also food and fertilizer. The output of livestock in India—milk, work, dung, fiber, eggs, and meat—is Rs 1.2 trillion (\$24 billion). Livestock contribute 8 percent of the country’s GNP.

Cuba is an example of a country that renewed its renewable-energy base. From 1960 to 1990 the number of tractors increased tenfold in Cuba, from 7,000 to 70,000, while the number of oxen decreased. The sanctions against Cuba and the collapse of the Soviet Union in 1989 led to a serious shortage of fuel, spare parts, and fertilizer. Cuba rapidly made a transition from tractors to oxen through new agricultural policies and strategies. A new breeding program was started,

new animal-drawn implements were developed, and blacksmiths, artisans, and makers of yokes and harnesses were trained.

Today, Cuba has approximately 385,000 oxen, which have replaced 40,000 tractors.⁵³

After the oxen, comes the tractor,
After the tractor, comes the oxen,
After the car, comes the cow ...

CHAPTER THREE

Food for Cars or People

*Biofuels a False Solution to Climate Change
and a Threat to Food Security*

Ecological, Diverse, Decentralized Biofuels vs. Industrial Biofuels

Biofuels, fuels from biomass, continue to be the most important energy source for the poor in the world. The ecological, biodiverse farm is not just a source of food, it is also a source of energy. Energy for cooking the food comes from inedible biomass like the stalks of millets and pulses, farmed trees, and village woodlots. Managed sustainably, village commons have been a source of decentralized energy for centuries. The use of decentralized energy from biomass is a vital part of the transition from fossil fuels to renewable energies. Biomass can be used directly as a cooking and heating fuel. It can be turned into biogas, a decentralized-energy alternative promoted by Gandhi, among others. Biofuels can be used to generate electricity for decentralized use and can be part of a sustainable alternative to fossil fuels and nuclear power. If embedded in a democratic, decentralized framework of management and decision-making, biofuels can rejuvenate biodiversity, recycle carbon, enhance agricultural productivity, increase the resilience of agro-ecosystems in the face of climate change, and increase the food and energy security of the poor.

However, the current euphoria over industrial biofuels is promoting monocultures and destroying biodiversity; promoting continued luxury consumption by the rich at the expense of the basic food and domestic energy needs of the poor; and promoting centralized corporate ownership and control over land and biomass by grabbing them from the poor.

In 1995, there were 34 countries that relied on wood fuels for more than 70 percent of their energy needs; in 13 of these countries wood supplied more than 90 percent.¹ The diverse crop and tree species that have supplied rural energy in biodiverse agro ecosystems do not appear in the new lexicon of “biofuels.” Biofuels are no longer an agrarian product meeting the needs of the rural poor. In fact they are not even a complementary product to food. Instead industrial biofuels are in competition with food. They are not part of diversified, decentralized, sustainable, and equitable food and energy systems.

Industrial biofuels are not the fuels of the poor; they are the foods of the poor transformed into heat, electricity, and fuel for the rich. Liquid biofuels, in particular ethanol and biodiesel, are one of the fastest-growing sectors of production, driven by the search for alternatives to fossil fuels. Industrial biofuels have been promoted through legislation and policy. Laws are being enacted to promote and subsidize liquid fuels diverting land from food production. From the richest countries in the North to the poorest countries in the South, food security is being forgotten in order to keep the energy infrastructure of the fossil fuel age “well oiled.” The entire structure built on fossil fuels is seeking to be maintained and expanded on the basis of oil from plants.

President Bush is trying to pass legislation to require the use of 35 billion gallons of biofuels by 2017—a massive increase from the estimated 4 billion gallons used in 2005. Alexander Muller of the Sustainable Development Department of the United Nations Food and Agriculture Organization (FAO) has stated: “The gradual move away from oil has begun. Over the next 15 to 20 years we may see biofuels providing a full 25 percent of the world’s energy needs.”² Global production of biofuels has doubled in the past five years and is likely to double again in the next four years. Among the countries that have enacted pro-biofuel policies in recent years are Argentina, Australia, Canada, China, Colombia, Ecuador, India, Indonesia, Malawi, Malaysia, Mexico, Mozambique, the Philippines, Senegal, South Africa, Thailand, and Zambia. Former World Bank president Paul Wolfowitz once said:

Biofuels are an opportunity to add to the world supply of energy to meet the enormous growing demand and hopefully to mitigate some of the price effect. It’s an opportunity to do so in an environmentally friendly way and in a way that is carbon neutral. It is an opportunity to do so in a way that developing countries like Brazil can provide income and employment for their people.³

But are industrial biofuels carbon neutral? Are the poor gaining or losing with the explosive increase in the production of industrial biofuels? What are the soil and ecological implications of the new policy obsession with industrial biofuels? What are the implications for land sovereignty and food sovereignty of

the poor?

Industrial Biofuels: Green or GreenTM

Industrial biofuels are being promoted as a source of renewable energy and as a means to reduce greenhouse gas emissions. However, there are two ecological reasons why converting crops like soy, corn, and palm into liquid fuels can actually aggravate the CO₂ burden and worsen the climate crisis while also contributing to the erosion of biodiversity and the depletion of water resources.

First, deforestation caused by expanding soy and palm oil plantations is leading to increased CO₂ emissions. The FAO estimates that 1.6 billion tons, or 25 to 30 percent of the greenhouse gases released into the atmosphere each year, comes from deforestation.⁴

According to Wetlands International, destruction of Southeast Asian forests for palm oil plantations is contributing to 8 percent of global CO₂ emissions.⁵ By 2022, biofuel plantations could destroy 98 percent of Indonesia's rainforests.⁶ Every ton of palm oil used as biofuel releases 30 tons of CO₂ into the atmosphere, ten times as much as petroleum does.⁷ Ironically, this additional burden on the atmosphere is treated as beneficial and as a Clean Development Mechanism (CDM) by the Kyoto Protocol. Biofuels are exacerbating the same global warming that they are supposed to reduce.

Biofuels: A Greenhouse Threat

Two important studies published in February 2008 in the journal *Science* reveal that biofuels cause more greenhouse gas emissions than conventional fuels if the full emissions costs of producing these green fuels are taken into account. The studies follow a series of reports that have linked ethanol and biodiesel production to increased carbon dioxide emissions, destruction of biodiverse forests, and air and water pollution. The destruction of natural ecosystems, whether rainforests in the tropics or grasslands in South America, not only releases greenhouse gases into the atmosphere, but also deprives the planet of natural sponges to absorb carbon emissions. The new cropland, the study reports, also absorbs far less carbon than the rainforests or even scrubland it replaces.

Together the two studies offer sweeping conclusions: taken globally, the production of almost all biofuels resulted, directly or indirectly, intentionally or not, in new lands being cleared. Whether that land was rainforest or scrubland, the greenhouse gas contribution is significant.

Joseph Fargione, an author of one of the studies and a scientist at the Nature Conservancy, says, “The clearance of grassland released 93 times the amount of greenhouse gases that would be saved by the fuel made annually on that land.” In Indonesia and Malaysia, palm biodiesel, one of the most controversial biofuels currently in use, because of its connection to rainforest deforestation in these countries, has a carbon debt of 423 years. Soybean biodiesel in the Amazonian rainforest has a debt of 319 years. “Until the carbon debt is repaid, biofuels from converted land have greater GHG impacts than the fossil fuels they displace.”⁸ A “carbon debt” refers to the CO₂ released during the first fifty years from the land conversion. “People don’t realize there is three times as much carbon in plants and soil than there is in the air. While we cut down forests, burn them, churn the soil, we release all the carbon that was being stored,” says Dr. Fargione.⁹

According to Dr. Fargione, the dedication of so much cropland in the United States to growing corn for ethanol has caused indirect land-use changes far away. Previously, Midwestern farmers had alternated growing corn and soy in their fields, one year to the next. Now many grow only corn, meaning that soy has to be grown elsewhere. The studies show that the purchase of biofuels in Europe and the United States leads indirectly to the destruction of the natural habitats far afield. This has also been proven by the Navdanya study on food versus fuel, which found that the grasslands and common lands are being destroyed in Chhattisgarh and Rajasthan to grow jatropha for biofuel.

David Pimentel and Ted Patzek, professors at Cornell and Berkeley, respectively, have shown that all crops have a negative energy balance when converted to biofuels—it takes more fossil fuel energy input to produce biofuels than the resultant biofuels can generate.

It takes 1.5 gallons of gasoline to produce one gallon of ethanol.¹⁰ For each fossil fuel unit of energy spent producing corn ethanol, the return is 0.778 units of energy, 0.688 units for switchgrass ethanol, and 0.534 for soybean diesel.¹¹ Pimentel and Patzek were criticized by the US government for including the energy used for building new refineries. However, these are new energy investments that do generate emissions, and Pimentel and Patzek are right to include them when calculating the overall energy balance.

In 2006 the US used 20 percent of its corn crop to produce 5 billion gallons

of ethanol, which only substituted for 1 percent of its oil use. If 100 percent of the corn crop were used to make ethanol, it would be able to substitute for 7 percent of the total oil used.¹² Even if all US soy and corn were converted to fuel it would only substitute for 12 percent of the gasoline and six percent of the diesel. To satisfy the entire current oil demand of the US with biofuels would take 1.4 million square miles of corn for ethanol or 8.8 million square miles of soy for biodiesel, which is more than all the agricultural land in the US.¹³ All the solar energy collected by every green plant in the US in 2006—including agriculture, forests, and lawns—is only half as much as the fossil fuel energy consumed in that year.¹⁴ This is clearly not a solution to either peak oil or climate chaos.

In fact, ethanol is a source of other crises when you look at all the resources it demands. It takes 1,700 gallons of water to produce a gallon of ethanol. Corn uses more nitrogen fertilizer, more insecticides, and more herbicides than any other crop. Ethanol constitutes 99 percent of all biofuel production in the US. In 2004, there were 3.4 billion gallons of ethanol produced and blended into gasoline, amounting to about 2 percent of the nation's gas consumption.¹⁵

There has been a flood of subsidies in the West for production of biofuels. The cost of support of ethanol varies from \$0.29 to \$0.36 per liter in the US and \$1 per liter in the EU.¹⁶ Support for biodiesel varies between \$0.20 per liter in Canada and \$1 in Switzerland. In 2007 US taxpayers provided \$6 billion to ethanol producers through subsidies.¹⁷

In 2008, the government introduced a tax credit of \$0.51 per gallon on ethanol and mandated a doubling of the amount of ethanol to be used in gasoline by 2012, to 7.5 billion gallons. The total cost to the consumer of subsidizing corn ethanol is \$8.4 billion per year.¹⁸

Subsidization of biofuels is creating a deep impact on demand for foodstuffs from the United States. In 2007, for example, the increase in ethanol production will account for more than half of the global increase in demand for corn. Much the same is true in the US and EU for soybeans and rapeseed used in biodiesel. The rising price of food is good for producers. It is dreadful, however, for consumers, particularly for those in poor, food-importing countries. Increased production of biofuels also adds stress on existing land and water supplies.

These subsidies will distort agriculture policy and encourage farmers to divert their crops from food to fuel. They promote monocultures and industrial agriculture, which contribute to climate change. In effect, industrial biofuels will increase climate instability, rather than mitigating it.

According to Patzek, "the United States has already wasted a lot of time,

money, and natural resources pursuing a mirage of an energy scheme that cannot possibly replace fossil fuels. The only real solution is to limit the rate of use of these fossil fuels. Everything else will lead to an eventual national disaster.”¹⁹

For Italy to meet the EU requirement to have 5 percent of its gas and diesel be biofuel by 2010 will require 69 percent more land to be farmed than is available in the entire country and require 102 percent more water and 40 percent more chemicals.²⁰ The UK has set targets of 2.5 percent of fuel to be biofuel by 2008, rising to 5 percent by 2010. Compulsory biofuels are a recipe for disaster. It is a case of the cure being worse than the disease.

The planet and the poor are losing; the rainforests—the lungs, the heart, the liver of the planet—are being bulldozed to plant soy and palm. In Brazil, 22.2 million hectares have been converted to soy plantations, producing in 2004–2005 over 50 million tons.²¹ Brazil will clear an additional 60 million hectares of land due to the gold rush for soy. Since 1995 in Brazil, soy cultivation has been increasing at 3.2 percent (320,000 hectares) per year. Twenty-one percent of Brazil’s cultivated area is now soy, and 300,000 people have been displaced in Rio Grande do Sul.²² Since January 2003 nearly 70,000 kilometers of the Amazon rainforest have been cleared for biofuels production. Corporations like Cargill, ADM, and Bunge are at the heart of the destruction of the Amazon, according to Green peace.²³ Since 1990, Indonesia has destroyed 28 million hectares of rainforest for palm plantations.²⁴ The poor are losing because land and water that would have produced food for the hungry is being used to run cars.

Automobile companies and agribusiness are the ones who gain. Using liquid biofuels to run cars allows car manufacturers to keep selling cars despite peak oil and climate crisis. Biofuels gives them another way to avoid doing anything about fuel efficiency. As George Monbiot, the environmental columnist for the *Guardian*, reports:

The European Commission was faced with a straight choice between fuel efficiency and biofuels. It had intended to tell car companies that the average carbon emission from new cars in 2012 would be 120 grams per kilometer. After heavy lobbying by Angela Merkel on behalf of the car manufacturers, it caved in and raised the limit to 130 grams. It announced that it would make up the shortfall by increasing the contribution from biofuel.²⁵

Agribusiness is also benefiting from the expanded market for soy, corn, and palm. Monsanto can sell more herbicide-resistant seeds and collect more royalties from Argentina and Brazil. Cargill can make more profits selling fertilizers, agrochemicals, and agricultural commodities for biofuel, while also

increasing its profit margins on the food commodities it sells as prices rise.

Corporate Control of Biofuel Feedstock²⁶

	Top Corporations	Corporate Control
Corn merchants (US)	Cargill, ADM, Con- Agra	Top three control over 80% of US corn exports
Corn seeds (US)	Monsanto, DuPont, Syngenta	Monsanto controls 41% of global market
Sugar trade (Brazil)	Cargill, Louis Dreyfus, Cosan/Tereos/Sucden	Cargill is the largest shipper of raw sugar from Brazil.
Palm oil trade (Global)	Wilmar, IOI, Synergy Drive, Cargill	In Malaysia 60% is owned by corporations; only 9% by individuals.
Soy trade (Global)	Bunge, ADM, Cargill, Louis Dreyfus	three companies control 80% of European production.
Soy seeds (Global)	Monsanto, DuPont	Monsanto controls 25% of the global marker.

The expansion of industrial biofuels has also served as a convenient backdrop for the continued use and legitimization of genetically modified crops. Various companies have joined with Monsanto to modify crops to exhibit traits beneficial for increased production of biofuels and are particularly active in promoting second-generation biofuels derived from non-food biomass—husks, switchgrass, wood-chips. Work is being done to genetically engineer plants to have lower levels of lignin, a compound in plant cell walls that impedes the breaking down of cellulose.

The biotechnology industry is promoting second-generation biofuels as its response to those who say the use of food crops for fuel threatens food security. However, it is feared that the large-scale removal of organic residues from fields will undermine food security by depriving soils of the organic matter required for soil fertility and soil moisture conservation. Deprived of this organic matter future crops will require greater use of synthetic nitrogen fertilizers, increasing nitrous oxide emissions, which contribute to climate change.

Agribusiness and the auto and oil industries have set the ground for climate change. They will use the climate crisis they have created to increase their market opportunities, even if it comes at the expense of the starving poor and pushes the planet deeper into climate disaster.

Biofuels a Threat to Food Security

The biofuel sector has been growing rapidly. The United States and Brazil have established ethanol industries and the European Union is fast catching up. Governments all over the world are encouraging biofuel production by mandating that biofuel account for a percentage of their fuel supply. The United States, because of its voracious demand for energy, is promoting industrial biofuels in a big way. Fidel Castro has strongly criticized the use of biofuels by the US. Lashing out at the 2007 ethanol deal between Brazil and the US, he described it as “the internationalization of genocide.”²⁷

The deal, coming on the heels of President Bush’s widely protested tour of Latin America, aims to encourage the development of biofuels projects in poor countries, particularly in the Caribbean and Central America, and to promote a global biofuels market.

Brazil and the US will cooperate more closely on researching and developing biofuel technology.

Washington’s interest in ethanol accelerated after Bush admitted in January 2006 that the US was “addicted to oil,” which posed a “national security problem” because oil is “often imported from unstable parts of the world.” In 2006, the US produced 18 billion liters of ethanol from 53 million tons of corn. Increasing the use of corn for ethanol production has caused corn and other food-crop prices to rise. An internal World Bank report states that biofuels have forced global food prices up by 75 percent.²⁸

The increased use of corn and soybean oil for biofuel production has raised world food prices by about 10 percent, according to an IMF report.²⁹ Biofuel production has pushed up feedstock prices. The clearest example is corn, whose price rose by 23 percent in 2000 and by 50 percent in 2005 and 2006, largely because of the US ethanol program. Spurred by subsidies and the Renewable Fuel Standard issued in 2005, the United States has been diverting more corn to ethanol. Because it is the world’s largest corn exporter, biofuel expansion in the United States has contributed to a decline in international grain stocks and has put upward pressure on world cereal prices. Largely because of biodiesel production, similar price increases have occurred for vegetable oils (palm, soybean, and rapeseed). Cereal supply is likely to remain constrained in the near term and prices will be subject to upward pressure from further supply shocks.

Worldwide agricultural commodity price increases were significant between 2004 and 2006: corn prices rose by 54 percent, wheat by 34 percent, soybean oil by 71 percent, and sugar by 75 percent. This trend accelerated in 2007, due to

continued demand for biofuels and drought in major producing countries.³⁰ The World Bank reports that food prices increased by 83 percent from 2006 to 2008.³¹ Continued demand and floods in the US Midwest suggest that this trend will continue. Wheat prices have risen more than 35 percent since the 2006 harvest, while corn prices have increased nearly 28 percent. The price of soybean oil has been particularly volatile, due to growing demand in China, the US, and the EU.

The Hamburg-based oilseeds analysts of Oil World have predicted a substantial deficit of 17 to 18 million tons in the output of major oilseeds during 2007 and 2008 and a food crisis unless the use of agricultural products for biofuels is curbed or 2008 proves to be a bumper-crop year.³²

Inevitably, this massive increase in the demand for grains is going to price poor people out of the food market. The Brazilian Landless Workers Movement declared, “the expansion of the production of biofuels aggravates hunger in the world. We cannot maintain our tanks full while stomachs go empty.”³³

The diversion of food for fuel has already increased the price of corn and soy. There have been riots in Mexico because of the price rise of tortillas. More than 40 countries have had food riots, and this is just the beginning.

The agrarian crisis created by trade liberalization and globalization is being used in India to promote the conversion of food crops to ethanol. The argument is that farmers will get a better price for these crops. In India, the two target crops are sugarcane and sorghum. India grows as much sugarcane as Brazil, but because Brazil uses 55 percent of its sugarcane for ethanol, India is the largest producer of sugar. The sugar industry in India is now promoting the diversion of sugarcane from human nutrition to ethanol for cars. Sorghum, a cereal that grows in the semiarid tropics, is another food crop being promoted for ethanol. The International Crops Research Institute for the Semi-Arid Tropics (ICRISAT), one of the institutes run by the World Bank, is a major promoter of growing sorghum for ethanol production using public funds to promote the conversion of food into fuel. Corporations like Tata and Seagram are jumping onto the ethanol bandwagon, with Tata setting up a 60,000-liter-capacity plant in Nanded.

In 2002, the government of India introduced a policy of supplying gas with 5 percent ethanol. With the automobile explosion in India and high fuel prices, ethanol demand is growing. Projected demand is outpacing supply. Of the 132 million gallons of ethanol required, only 37 percent is in current production.

In Argentina, the pampas, the region’s biggest and most diverse ecosystem, is under threat from the expansion of herbicide-resistant soy cultivation. Soy has

become Argentina's biggest export. If deforestation continues at the current rate, the forest ranges of the Yungas will disappear by 2010. More than 2.3 million hectares of dry and humid vegetation have been cleared for soy since 1995.

Forests are disappearing and people are being displaced. Three out of five people in Chaco province have been driven out of rural areas to Argentina's slums.³⁴

Biofuels have in fact become a basis for legitimizing the spread of genetically modified (GM) soy. President Lula of Brazil has declared that GM soy is to be used for biofuels and "good" soy for human consumption. While the Amazon is being destroyed for soy—for cattle feed and biofuels—the rainforests of Borneo and Sumatra are being cut down for palm oil. Ninety percent of the world's palm oil is grown in Indonesia and Malaysia. Indonesia has 6 million hectares of land under palm cultivation and has cleared three times as much—18 million hectares of forests—to expand palm oil plantations. A further 20 million hectares have been allocated in Sumatra, Kalimantan, Sulawesi, and West Papua. Plans are also being made to establish a 1.8 million hectare palm oil plantation, the world's largest, in the heart of Borneo's rainforests. Biodiesel has become the most important reason for the expansion of palm cultivation, and palm oil has become the largest cause of tropical deforestation in Southeast Asia.³⁵ In Malaysia too, the increase in palm oil production is the reason for rainforest destruction and the disappearance of countless species of plants and animals. Fifty-four projects have been set up to create B100, a biodiesel made entirely from palm oil.

Biofuels Leading to Water Scarcity

An International Water Management Institute (IWMI) study warns that the ambitious plans in China and India to greatly increase domestic production of biofuels derived from crops will put greater stress on these countries' water supplies, seriously undermining their ability to meet future food and feed demands. In many areas where water is already scarce, biofuel production could threaten river and groundwater systems. China aims to increase its biofuel production by fourfold, from a 2002 level of 3.6 billion liters of ethanol to around 15 billion liters by 2020, or 9 percent of the country's projected gasoline demand. India is pursuing a similarly aggressive policy. To meet their targets, China would need to produce 26 percent more corn and India 16 percent more sugarcane. Doing so would require an extra 75 liters of irrigation water per

person per day in China and an additional 70 liters per person per day in India.³⁶ When the recommended WHO minimum supply of 20 liter per person per day often goes unmet, this increase is a high demand indeed.

Biofuel crops in India will compete with food crops for a limited supply of water and are likely to jeopardize sustainable water use. The IWMI landmark study, prepared in consultation with more than 700 water-management experts, concluded that more than a third of the world's population contends with water scarcity. The 2008 report predicts that, with incomes rising and diets diversifying, to feed another 2 to 3 billion people the demand for water for food will increase 20 to 55 percent. This increase does not take into account the demands made by biofuels, which will put greater stress on water systems and make it more difficult to achieve food security for all.

How the Poor in India Are Affected

The fears of a food crisis in India are coming true. According to the World Bank's *World Development Report 2008*, the country will have to increase its cereal production by 50 percent by 2030 to meet the escalating demand.³⁷ The increasing shift toward biofuels will further add to the crisis. India has once deferred wheat imports, with global prices hitting an all-time high. International wheat prices in early 2008 were 83 percent higher than they were a year before. The *WDR 2008*, focusing on agriculture and development, has warned that the long-term downward trend in food prices may have come to an end, and the world can continue to expect food prices to drift upward, adding to food insecurity. The government of India was forced to hike the gasoline and diesel prices by 10 percent in June 2008, which also impacts the price of food. Any increased dependency on imports and escalating food prices could be the source of economic and political trouble. The report also warns that the potential conflict between food and fuel is bound to increase. India's biofuel program is creating problems through its demands on the food supply, the land supply, and the water supply, yet it doesn't have the potential to solve either the fuel or the climate crisis.

In India, the availability of adequate food supplies could be threatened by bioenergy production if land and other productive resources are shifted away from food-crop cultivation. The degree of competition between energy crops and food and fodder production will depend on, among other things, progress in crop yields, livestock-feeding efficiency, and bioenergy conversion technologies.

Many of the crops currently used as feedstock in liquid biofuel production require high-quality agricultural land and significant quantities of fertilizer, pesticides, and water. India lacks in all of these. Furthermore, crop yields depend on the vagaries of the weather and rainfall. Increased production of biofuels will result in increased use of water, fertilizers, and pesticides, which will be beneficial only for the fertilizer companies. If biofuel production pushes up commodities prices in India, the households will not have access to sufficient food.

Price volatility can affect food security, particularly for the poor. Because of the importance of fertilizers and machinery as inputs, agricultural commodity prices have long been sensitive to changes in the cost of energy; increased biofuel production adds another element to this relationship and will increase the volatility of food prices.

Jatropha and Land Grab

One of the biofuel crops being promoted in India is jatropha. *Jatropha curcas* is a plant indigenous to Central America that was brought to Asia and Africa by Portuguese and Dutch mariners. It has been used as a hedge because animals do not eat it. The kernel of its plum-sized fruit is 60 percent oil. The cultivation of jatropha as a biofuel is being promoted on village common lands through high subsidies to farmers. Traditionally, the commons have protected food security by providing pastureland for livestock to graze on. Though jatropha is inedible and not a food crop, its use as a biofuel has serious consequences for food security, land security, and livelihood security of the poor.

The government wants to plant jatropha on tribal land in Chhattisgarh. The people there are being robbed of land on which they have been cultivating rice for the last 40 years. The villagers in Rajasthan are losing their grazing lands to the government in spite of opposition from the panchayat (village government). These pastures are the common resources in Rajasthan. Jatropha cultivation severely limits the ability of the commons to support rural livelihoods comprehensively. Livestock, a major source of livelihood for the poor, is heavily dependent on the common pastures for grazing. Planting jatropha decreases the available fodder.

The cultivation of biofuel crops in India has emerged as a threat to the social, economic, and political security of India. Unless the constitutional rights of the people to manage their natural resources and make decisions through

decentralized democratic processes are respected, there will be more conflicts in the name of land. The increased production of biofuels will take land and food away from the poor. In short, biofuels are a recipe for violence and civic breakdown, for hunger, for climate chaos and ecological catastrophe.

India's Biodiesel Program

The Indian government intends to replace 20 percent of India's diesel with biodiesel. Starting from 400,000 hectares of jatropha, yielding 1.5 million tons of oilseed a year by 2007 the plan calls for a continued increase in jatropha cultivation through 2012 and for the construction of more transesterification plants. To meet its biodiesel target, the planning commission called for the cultivation of jatropha on 11 million hectares of "unused" or "waste" land. The commission did not define what it actually means by "unused." In Rajasthan jatropha has become synonymous with land grabs.

"Biofuel" has become a buzzword, and innumerable companies have sprung up in a rush to cultivate jatropha and set up oil-processing factories. Many state governments, including those of Chhattisgarh, Rajasthan, and Maharashtra, have joined the bandwagon, instituting biofuel policies for their states and promoting jatropha cultivation in a big way. The governments want to mobilize a large number of stakeholders, including individuals, communities, entrepreneurs, oil companies, businesses, industries, and members of the financial sector to spur the growth of the nonedible oil industry.

Jatropha Biopiracy

D1 Oils, the world's largest commercial jatropha cultivator, has engaged in jatropha biopiracy. In 2005 D1 Oils hired Sunil Puri, head of the Department of Forestry at the Indira Gandhi Agricultural University (IGAU) in Raipur, Chhattisgarh. Puri then transferred varieties of jatropha considered to have high oil content and disease- and drought-resistance qualities from the university to D1's nearby farm. A high-quality Pendra variety was also stolen. D1 claims to have cultivated it in a rented farmhouse near Raipur. Puri had previously worked on two studies that found that Chhattisgarh's Pendra and Surguja regions had a high-yielding variety of the plant.

An inquiry committee set up by the university, accompanied by the police,

raided Puri's house and seized seeds of 43 accessions of jatropha germplasm. Another raid at the research farm of D1 in Panchdeori village yielded 1,540 plants of 28 accessions. All the plants seized had the accession and tag numbers of the university. A government inquiry concluded that both the scientist and D1, by accepting the plants without the necessary authority, had breached India's biodiversity laws designed to protect the country's bioresources from foreign exploitation. The government has blocked D1's application to do research on jatropha in the country.

Jatropha for Local Energy Needs: the False Promise

Proponents of jatropha production say it will provide villages with a local energy source. However, setting up large-scale, centralized, industrial plants for biodiesel from jatropha ensures that the oil will flow to cities, not to villages; it will make cars run while animals and people starve.

The National Development Council, in its approach draft to the 11th five-year plan (2007), has had a change of heart on jatropha cultivation, raising concerns about the socioeconomic and environmental impacts of jatropha cultivation.

The survival of pastoralism is crucial to sustainable land use. Besides conserving biodiversity, it is a means of producing food in dry lands without depleting groundwater resources. However, there are many constraints on expansion in this area. Grazing permits are denied in traditional grazing sites that have been converted to protected areas/wildlife sanctuaries/ national parks/ JFM [Joint Forest Management] programme areas. Original pasture lands or stipulated animal drinking water pots are encroached upon or used for other purposes. Biodiesel (Jatropha) planting is being promoted by state agencies without any thought to the consequences. Migration routes of animals are blocked or herd-passing pathways are encroached upon.³⁸

Based on our observations of the impacts of jatropha cultivation through the Navdanya studies in Chhattisgarh, Vidarbha, and Rajasthan, we are pleased that the council, which once strongly encouraged jatropha cultivation, has now expressed its apprehensions about the issue.

India is following a monocrop-based cultivation model for jatropha. Indiscriminate use of jatropha as a major biofuel crop in India not only harms ecology and agriculture, but also the interests of the farmers who grow it. There are a lot of risk factors for the farmers engaged in jatropha cultivation. Unlike other biofuel crops, the seeds of jatropha are inedible and can be used only to make biodiesel. If there is not enough demand, farmers will end up with a loss—they will not even be able to survive by eating their crop.

Though rural development is one of the objectives of switching to jatropha cultivation, there is little evidence that fuel made from jatropha is being used to meet rural energy demand. The main objective behind the greater use of biofuels is to decrease fuel-import bills.

On July 8, 2008, the government of India approved a national biofuel policy that prioritizes food crops over biofuel crops. It recognizes that giving biofuel crops precedence will destroy the country's food security. The national policy has said that the biofuel program will not compete with food security, and fertile farmlands will not be diverted for plantation of biofuel crops.³⁹ However, because jatropha is not an edible plant, and because it is supposedly cultivated on "wastelands," the promotion of jatropha will continue.

The conservation, propagation, and sustainable use of biodiversity for food security and energy security of local communities needs local decision-making. This is why the *gram sabha* must be strengthened, not undermined, by government policies or laws.

Toward Sustainable, Biodiverse, and Decentralized Bioenergy Alternatives for India

Diverting land and food to produce industrial biofuel undermines the land and food sovereignty of the poor, generating social conflicts and threatening the fragile fabric of democracy. Ethanol production has already contributed to an increase in food prices. With 1 billion people already going hungry, high food prices can only increase hunger around the world. Even when nonfood crops like jatropha are grown on nonagricultural lands, the poor lose. They are losing their commons, which supply them fodder and fuel. This in turn undermines their livelihoods, food security, and energy security.

The climate crisis and the end of cheap oil demand a shift to sustainable energy. However, energy supplied from plants and crops is not necessarily sustainable. Energy can only be considered sustainable if it does not compete with the food supply, does not divert organic matter from the maintenance of the essential ecosystem, is decentralized and based on decisions by local communities, and is based on biodiversity, not monocultures.

Local energy needs decentralized energy systems. Decentralization needs diversity, not monocultures. Biodiversity with multifunctional uses makes for the best local energy supply. It is complementary and not competitive with the local food supply. India has a rich diversity of oilseed trees and crops. Diverse oilseed-

bearing tree crops used locally can be an important source of village energy security. A Navdanya report on biodiversity-based organic farming shows that food production can increase with biodiversity intensification and ecological agriculture. And biodiverse organic farming also fixes more carbon in vegetation and the soil.

At the village level, multiple sources of biodiversity provide multiple sources of renewable energy—from animal energy to biogas to biomass for electricity. Biogas digesters were first developed in the 1950s in India, using cow dung to produce methane gas and a nitrogen-rich fertilizer. Asia has more than 15 million small digesters in rural areas to provide cooking fuel and fertilizer.

Alternatives to fossil fuel are limitless. Industrial biofuels are not an alternative because

- Their net energy efficiency is negative.
- They restrict themselves to liquid fuels, forgoing the many other forms of bioenergy needed at the village level.
- They promote non-sustainable industrial monocultures that serve to increase greenhouse gas emissions.
- They are becoming a major cause of hunger and landlessness.

Industrial biofuels threaten to impoverish the planet by reducing biodiversity and its benefits. This reductionism is leading to ecological and economic impoverishment. Biodiversity can lead to ecological and economic enrichment. A decentralized, biodiversity-based bioenergy policy can be a major component in rural development.

Democratic decision-making at the village level is the best process for determining the best mix of energy needed to meet local needs. Unfortunately, the current model of industrialized production of ethanol and biodiesel from plants based on monocultures fails the criteria of sustainability, justice, and democracy. It is centralized and driven by corporate greed, not community needs.

We need a new model, one that respects people's right to land and food, to their commons and biodiversity. A biodiversity-based, democratically evolved bioenergy program could enhance food security, energy security, and livelihood security of the poor.

CHAPTER FOUR

Soil Not Oil

Securing Our Food in Times of Climate Crisis

Industrialized agriculture and globalized food systems have been put forth as sources of cheap and abundant food. However, food is no longer cheap. The era of cheap food and cheap oil is over. The food crisis, mainly triggered by rising prices, that emerged in 2007 and 2008 has led to food riots in many countries. From 2007 to 2008 the price of wheat increased by 130 percent.¹ The price of rice doubled during the first three months of 2008.² Biofuels, speculation, destruction of local food economies, and climate change have all contributed to the rise in food prices. Climate change is aggravated by industrialized, globalized agriculture based on fossil fuels, and the resulting climate crisis in turn impacts food security in numerous ways, including intensified floods such as those Iowa experienced in 2008 and intensified and extended droughts like the one Australia witnessed in 2007. Globalization has also led to the destruction of local food economies and increased control by corporations like Monsanto and Cargill over our food systems. Global integration of agriculture in effect means global control over the world's food supply.

In India, the World Bank-imposed structural adjustment program of 1991 and the WTO rules that came into force in 1995 have jointly worked to dismantle the public framework for food sovereignty and food security and to force the integration of India's food and agriculture systems with those of rich countries. This has resulted in a deep agrarian crisis and an emerging food crisis, with farmers' incomes crashing as food prices go through the roof. The food and agriculture crises are a direct result of policies of corporate globalization. Yet globalization is what the government is offering as a cure for globalization's ills.

Food prices started to rise as a result of connecting India's domestic market

to global markets, especially the edible oil and wheat import markets. At first, in the early days of globalization, the agribusinesses that dominate trade lowered prices to grab markets. The dumping of soy in the 1990s is a prime example. Now that global corporations like Cargill have created import dependency, they are increasing prices. Additionally, speculation through futures trading is driving prices upwards. Climate change and the diversion of foods to biofuels are also adding an upward pressure on international prices. The increase in international prices highlights the need to focus on food sovereignty. It makes both political and economic sense to focus on self-reliance in food and agriculture.

While millions go hungry, corporate profits have increased. Cargill saw profits increase by 30 percent in 2007; Monsanto's profits increased by 44 percent.³ These profits will increase as corporate monopolies deepen. Monsanto increased the price of corn seed by \$100 per bag to \$300 per bag. For a 1,000-acre farm in the US, this means an increased cost of \$40,000.⁴

The solution to the food crisis is to reclaim food sovereignty and rebuild local food economies based on ecological farming. This path also frees agriculture from its dependence on fossil fuels while increasing mitigation and adaptation to climate change. A shift from oil to soil addresses the triple crisis of climate, energy, and food.

Eating Oil

Industrialized, globalized agriculture is a recipe for eating oil. Oil is used for the chemical fertilizers that go to pollute the soil and water. Oil is used to displace small farmers with giant tractors and combine harvesters. Oil is used to industrially process food. Oil is used for the plastic in packaging. And finally, more and more oil is used to transport food farther and farther away from where it is produced.

Fossil fuels are the heart of industrial agriculture. Fossil fuels are used to run the tractors and heavy machinery and to pump the irrigation water necessary for industrial farming. Industrial systems of food production use ten times more energy than ecological agriculture does, and ten times more energy than the energy in the food they produce.⁵

The *Stern Review on the Economics of Climate Change* has identified the following sources of greenhouse gas emissions responsible for climate change:

Greenhouse Gas Emissions, by Source ⁶

Power	24%
Industry	14%
Transport	14%
Buildings	8%
Land Use	18%
Agriculture	14%
Waste	3%
Other	5%

What the report does not mention is the particular kinds of agriculture, transport, and buildings that are responsible for the emissions. It fails to differentiate industrial, globalized agriculture, which is responsible for a large part of the 14 percent of emissions in agriculture, from non-industrial, biodiverse, ecological agriculture, which has much lower emissions and helps in carbon sequestration. It also does not break out the share of the 18 percent of emissions attributed to land use created when tropical forests are cut down to grow agricultural commodities, or the part of the 14 percent of transport emissions resulting from unnecessarily shipping and flying food around the world.

Localized, biodiverse ecological agriculture can reduce greenhouse gas emissions by a significant amount while improving our natural capital of biodiversity, soil, and water; strengthening nature's economy; improving the security of farmers' livelihoods; improving the quality and nutrition of our food; and deepening freedom and democracy. Instead of focusing on achievable solutions, the Stern report promotes the pseudo-solution of carbon trading, which translates into business as usual for the agrochemical and agribusiness corporations profiting from globalized, industrialized agriculture.

An analysis of energy in the US food chain found that on average, it takes 10 calories of energy to produce 1 calorie of food. This is a net negative energy production system.⁷ A shift to ecological, non-industrial agriculture from industrial agriculture leads to a two- to seven-fold energy savings and a 5 to 15 percent global fossil fuel emissions offset through the sequestration of carbon in organically managed soil.⁸ Up to four tons of CO₂ per hectare can be sequestered in organic soils each year.⁹

From field to table, the industrial, globalized food system is moving toward an increased dependence on fossil fuels. There have been dramatic changes in

how food is produced, processed, and distributed over the last 50 years. The most significant changes include the following:

- The mechanization of agriculture and increased reliance on external inputs such as synthetic fertilizers, pesticides, feed, plastics, and energy.
- A major shift to highly processed and packaged food.
- The globalization of the food industry, characterized by an increase in food imports and exports. Of particular note is the rise in imports of fresh fruits and vegetables, with more produce sourced from farther afield.
- Supermarkets emerging as sales leaders, accompanied by the loss of small shops, markets, and wholesalers. Parallel to this trend is the concentration of supply into the hands of fewer, larger suppliers, partly to meet supermarkets' preferences for bulk, year-round supplies of uniform produce.
- Major changes in delivery patterns, with most goods now routed through supermarkets' regional distribution centers, and a trend toward the use of large heavy goods vehicles (HGVs) and just-in-time delivery, sometimes referred to as "warehouse on wheels."
- A switch from frequent food shopping on foot at small local shops to shopping by car at large out-of-town supermarkets.¹⁰

David Pimentel and Mario Giampietro have focused on the relationship between endosomatic and exosomatic energy. "Endosomatic energy is generated through the metabolic transformation of food energy into muscle energy in the human body. Exosomatic energy is generated by transforming energy outside of the human body by mechanical means, such as by burning oil in a tractor."¹¹ Pimentel and Giampietro found that it takes 10 kilocalories of exosomatic energy to produce every 1 kilocalorie of food in the US. The remaining 9 kilocalories go to create waste and pollution, and increase entropy.¹² Part of this wasted energy is going into the atmosphere to contribute to climate change.

Industrial agriculture in the US uses 380 times more energy per hectare to produce rice than a traditional farm in the Philippines. And energy use per kilo of rice is 80 times more in the US than in the Philippines. Energy use for corn production in the US is 176 times more per hectare than on a traditional farm in Mexico and 33 times more per kilo.¹³ One cow maintained and marketed in the

industrial system requires six barrels of oil.¹⁴ A 450-gram box of breakfast cereal provides only 1,100 kilocalories of food energy but uses 7,000 kilocalories of energy for processing.¹⁵

Chemical industrial agriculture is based on the idea that soil fertility is manufactured in fertilizer factories. This was the idea that drove the Green Revolution, introduced in India in 1965 and 1966. In 1967, at a meeting in New Delhi, Norman Borlaug, the Nobel Prize-winning “father of the Green Revolution,” was emphatic about the role of fertilizers in the new revolution. “If I were a member of your parliament,” he told the politicians and diplomats in the audience, “I would leap from my seat every fifteen minutes and yell at the top of my voice, ‘Fertilizers! ... Give the farmers more fertilizers.’ There is no more vital message in India than this. Fertilizers will give India more food.”¹⁶ Today, the Green Revolution has faded in Punjab. Yields are declining. The soil is depleted of nutrients. The water is polluted with nitrates and pesticides.

The fertilizer industry has now found Africa.

The Rockefeller and Gates foundations have set up AGRA, the Alliance for a Green Revolution in Africa. However, AGRA will not be the site of a Taj Mahal for Africa’s agriculture. The new Green Revolution for Africa is in fact the old Green Revolution for Asia. And as the Punjab experience shows, the Green Revolution was neither green in terms of ecological sustainability and conservation of natural capital of soil-water-biodiversity nor revolutionary in terms of increasing equality and promoting justice for small and marginal peasants. This not-so-green revolution is now being proposed as a solution for hunger and poverty in Africa.

AGRA has a \$150 million Program for Africa’s Seeds Systems (PASS) that seeks to transform farming in Africa. The strategy is based on promoting private seed companies and commercializing the seed supply, which it assumes are necessary for improving Africa’s farm productivity. It is also based on increasing the sale of chemical fertilizer. Gary Toenniessen of the Rockefeller Foundation writes in *Securing the Harvest*, “No matter what efficiencies genetic enhancement is able to build into crop plants, they will always draw their nutrition from external sources,” and “No alternatives to the use of inorganic nitrogen currently exist for densely populated developing countries.”¹⁷ This ignores the successes in Asia, Africa, and Latin America of doubling and tripling farm productivity through biodiverse organic farming based on the farmers’ breeding, biodiversity conservation, and agro-ecology. Not only are chemical fertilizers not necessary for farming, synthetic fertilizers actually harm the living processes in the soil that are responsible for soil fertility, plant growth, and

production of healthy food.

Fertilizer advocates also ignore how the rising cost of oil affects fertilizer prices. Imported fertilizer costs from Rs 55,000 to Rs 60,000 per ton and is sold at Rs 9,350 per ton. Rs 45,000 per ton is paid through taxes collected to cover the subsidies. In India the shift to chemical agriculture has created the need for 4 to 4.8 million tons of synthetic Diammonium phosphate (DAP). As only around 2 million tons are produced in India, the rest must be imported.

Fertilizer protests are taking place in Karnataka, where a farmer was killed when police opened fire on hundreds of farmers waiting for fertilizers. This was an entirely unnecessary tragedy. Similar incidents have occurred in Amrati, Vidarbha, Latur, Marathwada, and Maharashtra. First the Green Revolution made Indian farmers addicted to chemical fertilizer. Now globalization is making them dependent on imports.

While the soil and farmers die, agribusiness corporations like Cargill are making a killing. Cargill's fertilizer profits doubled from 2006 to 2007, with India paying 130 percent more for fertilizers and China 227 percent more for fertilizers during that period.¹⁸

Baron Justin von Liebig, a German chemist, carried out research in the latter part of the 19th century on the elements and chemicals required by plants for growth. He determined that the principal ingredients for soil fertility were nitrogen (N), phosphorus (P), and potassium (K). This is how the NPK mentality was born.

In 1909, Fritz Haber invented ammonium sulfate, a nitrogen fertilizer made by using coal or natural gas to heat nitrogen and hydrogen. The manufacture of synthetic fertilizers is highly energy-intensive. One kilogram of nitrogen fertilizer requires the energy equivalent of 2 liters of diesel. One kilogram of phosphate fertilizer requires half a liter of diesel. Energy consumed during fertilizer manufacture was equivalent to 191 billion liters of diesel in 2000 and is projected to rise to 277 billion in 2030.¹⁹

Plants, however, need more than NPK. And when only NPK is applied as synthetic fertilizers, soils and plants, and consequently humans, develop deficiencies of trace elements and micronutrients. A pioneer of organic agriculture, Sir Albert Howard defined fertile soil as:

a soil teeming with healthy life in the shape of abundant microflora and microfauna, will bear healthy plants, and these, when consumed by animals and man, will confer health on animals and man. But an infertile soil, that is, one lacking sufficient microbial, fungous, and other life, will pass on some form of deficiency to the plant, and such plant, in turn, will pass on some form of deficiency to animals and man.²⁰

The millions of organisms found in soil are the source of its fertility. The greatest biomass in soil consists of microorganisms, fungi in particular. Soil microorganisms maintain soil structure, contribute to the biodegradation of dead plants and animals, and fix nitrogen. They are the key to soil fertility. Their destruction by chemicals threatens our survival and our food security. A Danish study analyzed a cubic meter of soil and found 50,000 small earthworms, 50,000 insects and mites, and 12 million round-worms. A gram of the soil contained 30,000 protozoa, 50,000 algae, 400,000 fungi, and billions of individual bacteria. It is this amazing biodiversity that maintains and rejuvenates soil fertility. To feed humanity we need to feed the soil and its millions of workers, including the earthworm.²¹

When I carried out research on the Green Revolution in Punjab, I found that after a few years of bumper harvests, crop failures at a large number of sites were reported despite liberal applications of NPK fertilizers. The failure came from micronutrient deficiencies caused by the rapid and continuous removal of micronutrients by “high-yielding varieties.” Plants quite evidently need more than NPK, and the voracious high-yielding varieties drew out micronutrients from soil at a very rapid rate, creating deficiencies of such micronutrients as zinc, iron, copper, manganese, magnesium, molybdenum, and boron. With organic manure these deficiencies do not occur, because organic matter contains these trace elements, whereas chemical NPK does not. Zinc deficiency is the most widespread of all micronutrient deficiencies in Punjab. Over half of the 8,706 soil samples from Punjab exhibited zinc deficiency, which has reduced yields of rice, wheat, and corn by up to 3.9 tons, 1.98 tons, and 3.4 tons per hectare, respectively. Consumption of zinc sulfate in Punjab rose from zero in 1969–70 to nearly 15,000 tons in 1984–85 to make up for the artificially created zinc deficiency. Manganese is another micronutrient that has become deficient in Punjab soils. Sulfur deficiency, which was earlier noticed only in oilseed and pulse crops, has now been noticed in cereals like wheat.

The Green Revolution has also resulted in soil toxicity by introducing excess quantities of trace elements into the ecosystem. Fluorine toxicity from irrigation has developed in various regions of India. Twenty-six million hectares of India’s lands are affected by aluminum toxicity. In the Hoshiarpur district of Punjab, boron, iron, molybdenum, and selenium toxicity has built up through Green Revolution practices and is posing a threat to crop production as well as animal health.

As a result of soil diseases and deficiencies, the increase in NPK application has not shown a corresponding increase in output in rice and wheat. Wheat and rice yields have been fluctuating and even declining in most districts in Punjab,

in spite of increasing levels of fertilizer use.

Experiments at the Punjab Agricultural University (PAU) are now beginning to show that chemical fertilizers cannot be substitutes for the organic fertility of the soil, and organic fertility can be maintained only by returning to the soil part of the organic matter that it produces. In the early 1950s, before the entry of the advisers of the Ford Foundation, when K. M. Munshi, India's Agriculture Minister at the time, referred to repairing the nutrient cycle, he was anticipating what agricultural scientists are today recommending for the diseased and dying fields of Punjab. And Howard's prediction, that "In the years to come, chemical manures will be considered as one of the greatest follies of the industrial epoch," is beginning to come true.²²

Fertilizers block the soil capillaries, which supply nutrients and water to plants. Infiltration of rain is stopped, runoff increases, and soil faces droughts, requiring ever more irrigation and ever more fossil fuels for pumping groundwater. Excess nitrogen in the root zone also denies nutrients to the plant. The negatively charged ions in the nitrates, the anions, take the cations, the positively charged ions of other elements, away from the root zone, thereby robbing the trees and plants of positive cations such as magnesium and calcium ions. Plants deficient in micronutrients create micronutrient deficiency in food and the human diet. And micronutrient deficiency leads to metabolic disorders.

Chemical fertilizers do not just destroy the soil and human health. They are also a major contributor to climate change because of pollution both from their production and from their use.

Long-distance globalized food systems, like the industrial food-production system they service, are contributing in a major way to greenhouse gas emissions. A study by the Danish Ministry of the Environment showed that 1 kilogram of food moving around the world generated 10 kilograms of CO₂. "Food-miles," which measure the distance food travels from where it is produced to where it is consumed, have increased dramatically as a result of globalization. As reported by environmental journalist Dale Allen Pfeiffer,

In 1981, food journeying across the US to the Chicago market traveled an average of 1,245 miles; by 1998, this had increased 22 percent, to 1,518 miles. In 1965, 787,000 combination trucks were registered in the United States, and these vehicles consumed 6,658 billion gallons of fuel. In 1997, there were 1,790,000 combination trucks that used 20.294 billion gallons of fuel. In 1979, David and Marcia Pimentel estimated that 60 percent of all food and related products in the US traveled by truck and the other 40 percent by rail. By 1996, almost 93 percent of fresh produce was moved by truck.²³

A study in Canada has calculated that in 2003 food in Toronto traveled an

average of 3,333 miles.²⁴ In the UK, the distance traveled by food increased 50 percent between 1978 and 1999.²⁵ A Swedish study found that the food-miles of a typical breakfast would cover the circumference of the Earth.²⁶

The increase in food-miles is related to fossil fuel and food subsidies, which allow food transported long distances to be cheaper than food produced locally. Thus, India imported 5.5 million tons of wheat in 2006, based on the argument that it was cheaper to import wheat from Australia and the US than to transport it from Punjab in the north to Kerala and Tamil Nadu in the south. We should be reducing food-miles by eating biodiverse, local, and fresh foods, rather than increasing carbon pollution through the spread of corporate industrial farming, nonlocal food supplies, and processed and packaged food. We need to reduce CO₂ emissions by moving toward economic localization and satisfying our needs with the lowest carbon footprint. Economic globalization, on the other hand, only serves to increase CO₂ emissions. This total disconnect between ecology and economics is threatening to bring down our oikos, our home on this planet.

Imports, which add unnecessary food-miles, are a direct result of free-trade agreements. Transport accounts for one-eighth of oil consumption, and a large part of it goes for food. Take, for example, the wheat imports that resulted from the US-India Knowledge Initiative on Agriculture. India is the second-largest producer of wheat in the world. Today, because of manipulation by the US and Indian governments, it has suddenly emerged as a big importer of wheat.

At the start of 2006 India's domestic production of wheat was projected to exceed domestic demand; it had been six years since India had needed to import wheat. However, by opening its domestic market to private corporations, foreign companies were able to buy so much wheat that the government found itself announcing that it would need to import wheat. It initially purchasing 0.8 million tons from the Australian Wheat Board, the only company able to meet India's import standards. The company had previously been implicated in the Volcker report for giving Saddam Hussein's regime a \$300 million kickback through Iraq's Oil-for-Food program.

As the year progressed and the Indian government continued to refuse a fair price for domestic wheat they found themselves, once again, forced to import wheat. This time they increased the price they were willing to pay and relaxed their import guidelines—allowing higher levels of toxins and pesticides. This meant that the big US agribusinesses, primarily Cargill and ADM, could sell their wheat to India. India imported another 2.2 million tons, corporate agriculture gained, and food security suffered.²⁷

From Food First to Export First

Until recently food has primarily been produced locally. Local food systems have evolved in accordance with local climates, and biodiversity, which in turn have shaped the rich cultural diversity of food. We need both the diversity and the decentralization of local food systems to mitigate as well as adapt to climate change. However, both the World Bank and the World Trade Organization are forcing countries to dismantle their local food economies, export what they produce and import what they need. The rise of “cash crop for export” policies are a result of World Bank structural adjustment policies. And the creation of import dependency is a result of World Bank conditionalities and WTO rules.

Sustainable agriculture is based on the sustainable use of natural resources—land, water, and agricultural biodiversity, including plants and animals. The sustainable use of these resources in turn requires that they are owned and controlled by decentralized agricultural communities, to generate their livelihoods and provide food. These three dimensions— ecological security, livelihood security, and food security—are essential elements of sustainable and equitable agriculture policy.

The current process of globalization of agriculture threatens to undermine all three of these dimensions. It is undermining ecological security by removing all limits on concentration of ownership of natural resources—land, water, and biodiversity—and encouraging non-sustainable resource exploitation for short-term profits. Trade liberalization of agriculture is not guided by the need to provide livelihood security for the two-thirds of India’s people who are farmers or to provide food security for the poorer half of Indians and for India as a whole. Just the opposite: it severely threatens food security at the household, regional and national levels.

The diversion of our natural resources from ecological maintenance, protection of livelihoods, and basic-needs satisfaction to luxury exports and corporate profits has been made possible because of the past three decades of agriculture policy. In that time agriculture has been made a state monopoly and run on massive debts and subsidies, while all ecological imperatives of sustainability have been ignored.

However, the new trade liberalization and globalization policies are not reducing the centralized control of agriculture; they are increasing it. Part of the reason people are not recognizing this new concentration and are misconceiving trade liberalization as a new freedom for farmers is because of the power shift from the nation-state to transnational corporations (TNCs). People have learned

to recognize the lack of freedom built into the rule of the nation-state. They have not yet learned to recognize the lack of freedom intrinsic to corporate rule. As the state withdraws from agriculture, it is not returning power to farming communities and autonomous producers. It is instead facilitating the transfer of control over natural resources, production systems, markets, and trade to global agribusiness, further disempowering and dispossessing small farmers and landless laborers.

The WTO and the World Bank are pushing countries like India to move from food-first to export-first policies. A nutritional apartheid is thus being created, with the scarce land and water of the South being used for growing fruits and vegetables for the rich North and the elites of the South, and leaving the people of the South dependent on imports of food staples such as wheat, rice, and corn. Both sides of the equation add food-miles to our daily bread. And while the destruction of local food systems and dependence on globalized food supply is made to look “natural,” it is a deliberate result of policy designed and driven by global agribusiness and supermarket chains. The step-by-step dismantling of India’s local food markets exemplifies just how artificial and violent the globalization of food systems really is.

Failure of “Export-First” Policies

It is a sad irony that the creation of Agricultural Export Zones (AEZs) intended to increase farm exports proved of no help to the vegetable grower. A bumper crop of potato did not bring farmers any profit; in fact, it ruined them, driving many to commit suicide. Despite the fact that the Indian government created three AEZs for potato cultivation, the potato could neither be exported nor utilized in the food-processing industry. Rather than increasing the exports of vegetables, the creation of AEZs has facilitated the import of vegetables.

- India is now the fifth-largest importer of raw vegetables, after the US, the EU, Japan, and Canada.
- The import bill for vegetables rose almost 20 percent in 2002; exports have been virtually static.
- India is spending three times more buying raw vegetables from the world markets than it is earning from exporting them. The bill came to a huge \$678 million in 2002. That was higher than the combined imports of

Russia, Hong Kong, and Brazil. In contrast, India sold only \$246 million worth of vegetables in 2002.²⁸

- Exports of processed vegetables, fruits, and nuts plummeted from \$70 million in 2001 to just \$58 million in 2002.²⁹

Experts have expressed fears that the large-scale diversion of land, capital, and other resources for crops like vegetables, flowers, and gherkins will severely affect food security. The very profitability of the cultivation of these crops needs to be properly assessed, taking fully into account the investment, the incentives given, and the value of the land and other forms of scarce natural resources diverted, or to be diverted. Since fruits and vegetables are perishable, they need to be transported in refrigerated trucks and by air. Trade in perishables is adding to the global carbon footprint.

The biodiverse, water-prudent, and drought-resilient agriculture of the South is being destroyed precisely when diverse and decentralized systems need to be conserved to reduce the impact of and increase resilience to climate change. On the one hand, drought is increasing as a result of climate change. On the other hand, it is increasing due to globalization of the food supply and diversion of the land and water to produce cheap food for the rich in the North. Peasants and pastoralists are pushed off the land and denied access to water as corporate farming for exports takes over.

A 50-gram bag of salad in the UK costs about 1 pound but wastes almost 50 liters of water. A mixed salad takes 300 liters.³⁰ As Bruce Lankford of the University of East Anglia has stated, “We are exporting drought.” Global retail chains like Tesco, Sainsbury’s, and Wal-Mart are increasingly sourcing fruits and vegetables from Africa and India. This is leading to the large-scale uprooting and impoverishment of farmers, and is contributing to drought and desertification, while increasing food-miles and undermining food security and food sovereignty. While India is being made to grow vegetables for Europe, we are also being forced to import pesticide-laden wheat in spite of sufficient domestic production, which is further threatening farmers’ livelihoods.

The poor are paying three times over—through increased vulnerability to climate change, through increased water scarcity as scarce water is used for export crops, and through the uprooting of communities from their land, villages, and homes to make way for wasteful globalized trade.

Globalized trade in food is hurting the poor and the planet. It is putting the future of our food at risk for short-term profits of global agribusinesses.

Soil Not Oil: Making a Transition to Biodiverse, Organic, Local Food Systems

The industrialized, globalized food system is based on oil. It is under threat because of the inevitability of “peak oil.” It is also under threat because it is more vulnerable than traditional agriculture to climate change, to which it has contributed. Industrial agriculture is based on monocultures. Monocultures are highly vulnerable to changes in climate, and to diseases and pests.

In 1970 and 1971, America’s vast corn belt was attacked by a mysterious disease, later identified as “race T” of the fungus *Helminthosporium maydis*, causing the southern corn leaf blight, as the epidemic was called. It left ravaged cornfields with withered plants, broken stalks, and malformed or completely rotten cobs. The strength and speed of the blight was a result of the uniformity of the hybrid corn, most of which had been derived from a single Texas male sterile line. The genetic makeup of the new hybrid corn, which was responsible for its rapid and largescale breeding by seed companies, was also responsible for its vulnerability to disease. At least 80 percent of the hybrid corn in America in 1970 contained the Texas male sterile cytoplasm. As a University of Iowa pathologist wrote, “Such an extensive, homogenous acreage is like a tinder-dry prairie waiting for a spark to ignite it.”³¹

Industrial agriculture is dependent on chemical fertilizers. Chemically fertilized soils are low in organic matter. Organic matter helps conserve the soil and soil moisture, providing insurance against drought. Soils lacking organic matter are more vulnerable to drought and to climate change. Industrial agriculture is also more dependent on intensive irrigation. Since climate change is leading to the melting of glaciers that feed rivers, and in many regions of the world to the decline in precipitation and increased intensity of drought, the vulnerability of industrial agriculture will only increase. Finally, since the globalized food system is based on long-distance supply chains, it is vulnerable to breakdown in the context of extreme events of flooding, cyclones, and hurricanes. While aggravating climate change, fossil fuel-dependent industrialized, globalized agriculture is least able to adapt to the change.

We need an alternative. Biodiverse, organic farms and localized food systems offer us security in times of climate insecurity, while producing more food, producing better food, and creating more livelihoods.

The industrialized, globalized food system is based on oil; biodiverse, organic, and local food systems are based on living soil. The industrialized

system is based on creating waste and pollution; a living agriculture is based on no waste. The industrialized system is based on monocultures; sustainable systems are based on diversity.

Living Soil

Every step in building a living agriculture sustained by a living soil is a step toward both mitigating and adapting to climate change. Over the past 20 years, I have built Navdanya, India's biodiversity and organic-farming movement. We are increasingly realizing there is a convergence between the objectives of conserving biodiversity, reducing climate-change impact, and alleviating poverty. Biodiverse, local, organic systems reduce water use and risks of crop failure due to climate change. Increasing the biodiversity of farming systems can reduce vulnerability to drought. Millet, which is far more nutritious than rice and wheat, uses only 200 to 300 millimeters of water, compared with the 2,500 millimeters needed for Green Revolution rice farming. India could grow four times the amount food it does now if it were to cultivate millet more widely. However, global trade is pushing agriculture toward GM monocultures of corn, soy, canola, and cotton, worsening the climate crisis.

Biodiversity offers resilience to recover from climate disasters. After the Orissa supercyclone of 1998, and the tsunami of 2004, Navdanya distributed seeds of saline-resistant rice varieties as "Seeds of Hope" to rejuvenate agriculture in lands that were salinated as a result of flooding from the sea. We are now creating seed banks of drought-resistant, flood-resistant, and saline-resistant seed varieties to respond to such extreme climate events. Climate chaos creates uncertainty. Diversity offers a cushion against both climate extremes and climate uncertainty. We need to move from the myopic obsession with monocultures and centralization to diversity and decentralization.

Diversity and decentralization are the dual principles needed to build economies beyond oil and to deal with the climate vulnerability that is the legacy of the age of oil. In addition to reducing vulnerability and increasing resilience, biodiverse organic farming also produces more food and higher incomes. As David Pimentel has pointed out: "Organic farming approaches for maize and beans in the US not only use an average of 30% less fossil energy but also conserve more water in the soil, induce less erosion, maintain soil quality, and conserve more biological resources than conventional farming does."³²

After Hurricane Mitch struck Central America in 1998, farmers who

practiced biodiverse organic farming found they had suffered less damage than those who practiced chemical agriculture. The ecologically farmed plots had on average more topsoil, greater soil moisture, and less erosion, and the farmers experienced less severe economic losses.³³

Fossil fuel-based industrial agriculture moves carbon from the soil to the atmosphere. Ecological agriculture takes carbon from the atmosphere and puts it back in the soil. If 10,000 medium-sized US farms converted to organic farming, the emissions reduction would be equivalent to removing over 1 million cars from the road. If all US croplands became organic it would increase soil-carbon storage by 367 million tons and would cut nitrogen oxide emissions dramatically.³⁴ Organic agriculture contributes directly and indirectly to reducing CO₂ emissions and mitigating the negative consequences of climate change.

Navdanya's work over the past 20 years has shown that we can grow more food and provide higher incomes to farmers without destroying the environment and killing peasants. We *can* lower the costs of production while increasing output. We have done this successfully on thousands of farms and have created a fair, just, and sustainable economy. The epidemic of farmer suicides in India is concentrated in regions where chemical intensification has increased costs of production. Farmers in these regions have become dependent on non-renewable seeds, and monoculture cash-crops are facing a decline in prices due to globalization. This is affecting farmers' incomes, leading to debt and suicides. High costs of production are the most significant reason for rural indebtedness.³⁵

Biodiverse organic farming creates a debt-free, suicide-free, productive alternative to industrialized corporate agriculture and brings about a number of benefits. It leads to increased farm productivity and farm incomes, while lowering costs of production. Pesticide-free and chemical-free production and processing bring safe and healthy food to consumers. We must protect the environment, farmers' livelihoods, public health, and people's right to food.

We do not need to go the Monsanto way. We can go the Navdanya way. We do not need to end up in food dictatorship and food slavery. We can create our food freedom. Biodiverse, organic, and local food systems help mitigate climate change by lowering greenhouse gas emissions and increasing absorption of CO₂ by plants and by the soil.

Organic farming is based on the recycling of organic matter; industrial agriculture is based on chemical fertilizers that emit nitrous oxides. Industrial agriculture dispossesses small farmers and converts small farms to large holdings that need mechanization, which further contributes to CO₂ emissions.

Small, biodiverse, organic farms, especially in third world countries, can be totally fossil fuel-free. The energy for farming operations comes from animals. Soil fertility is built by recycling organic matter to feed soil organisms. This reduces greenhouse gas emissions. Biodiverse systems are also more resilient to droughts and floods because they have a higher water-holding capacity, making them more adaptable to the effects of climate change. Navdanya's study on climate change and organic farming has indicated that organic farming increases carbon absorption by up to 55 percent and water-holding capacity by 10 percent.

Fertility Increase Due to Organic Farming ³⁶

percent increase over industrial farming							
Crops	Organic matter	Microbial activity	Microbial biomass	Water-holding capacity	N	P	K
Pearl millet	28-55	4-25	2-10	2-3	0-2	0-1	8-15
Clusterbean	32-44	22-54	12-25	4-9	12-34	2-4	25-47
Moth bean	31-47	11-23	8-15	4-7	7-21	1-2	4-9
Mung bean	27-41	28-59	11-33	4-8	11-27	2-6	5-11

The environmental advantages of small-scale, biodiverse organic farms do not come at the expense of food security. Biodiverse organic farms produce more food and higher incomes than industrial monocultures. Mitigating climate change, conserving biodiversity, and increasing food security go hand in hand.

The conventional measures of productivity focus on labor as the major input (and the direct labor on the farm at that) and externalize many energy and resource inputs. This biased productivity pushes farmers off the land and replaces them with chemicals and machines, which in turn contribute to greenhouse gases and climate change. Further, industrial agriculture focuses on producing a single crop that can be globally traded as a commodity. The focus on "yield" of individual commodities creates what I have called a "monoculture of the mind." The promotion of so-called high-yielding varieties leads to the displacement of biodiversity. It also destroys the ecological functions of biodiversity. The loss of diverse outputs is never taken into account by the one-dimensional calculus of productivity.

When the benefits of biodiversity are taken into account, biodiverse systems have higher output than monocultures. And organic farming is more beneficial for the farmers and the earth than chemical farming. When agroforestry is

included in farming systems, carbon absorption and carbon return increase dramatically. Date palm and neem increase the carbon density in the soil by 175 and 185 percent, respectively.

Studies carried out by the USDA's National Agroforestry Center suggest that soil carbon can be increased by 6.6 tons per hectare per year over a 15-year rotation and wood by 12.22 tons per hectare per year. Since both soil and biomass sequester carbon, this amounts to removing 18.87 tons of carbon per hectare per year from the atmosphere.³⁷

Soil and vegetation are our biggest carbon sinks. Industrial agriculture destroys both. By disrupting the cycle of returning organic matter to the soil, chemical agriculture depletes the soil carbon. Mechanization forces the cutting down of trees and hedgerows.

Organic manure is food for the community of living beings that depend on the soil. The alternatives to chemical fertilizers are many: green manures such as *sesbania aculeata* (dhencha), *gliricidia*, and sun hemp; legume crops such as pulses, which fix nitrogen through legume-rhizobium symbiosis; earthworms; cow dung; and composts. Farmyard manure encourages the buildup of earthworms by increasing their food supply. Soils treated with farmyard manure have from two to two and a half times as many earthworms as untreated soils. Earthworms contribute to soil fertility by maintaining soil structure, aeration, and drainage. They break down organic matter and incorporate it into the soil. The work of earthworms in soil formation was Darwin's major concern in his later years. Of worms he wrote, "It may be doubted whether there are many other animals which have played so important a part in the history of creatures."³⁸ The little earthworm working invisibly in the soil is the tractor, the fertilizer factory, and the dam combined. Worm-worked soils are more water-stable than unworked soils, and worm-inhabited soils have considerably more organic carbon and nitrogen than the original soil. Their continuous movement forms channels that help in soil aeration. It is estimated that they increase the air volume of soil by up to 30 percent. Soils with earthworms drain four to ten times faster than those without, and their water-holding capacity is higher by 20 percent. Earthworm castings, which can amount to 4 to 36 tons per acre per year, contain five times more nitrogen, seven times more phosphorus, three times more exchangeable magnesium, 11 times more potash, and one and a half times more calcium than soil.³⁹ Their work on the soil promotes the microbial activity essential to the fertility of most soils.⁴⁰

At the Navdanya farm in Doon Valley, we have been feeding the soil organisms. They in turn feed us. We have been building soil and rejuvenating its

life. The clay component on our farm is 41 percent higher than those of neighboring chemical farms, which indicates a higher water-holding capacity. There is 124 percent more organic-matter content in the soil on our farm than in soil samples from chemical farms. The nitrogen concentration is 85 percent higher, the phosphorus content 10 percent higher, and the available potassium 25 percent higher. Our farm is also much richer in soil organisms such as mycorrhiza, which are fungi that bring nutrients to plants. Mycorrhizal association makes food material from the soil available to the plant. Our crops have no diseases, our soils are resilient to drought, and our food is delicious, as any visitors to our farm can vouch. Our farm is fossil fuel-free. Oxen plow the land and fertilize it.

By banning fossil fuels on our farm we have gained real energy—the energy of the mycorrhiza and the earthworm, of the plants and animals, all nourished by the energy of the sun.

Biodiversity: Our Natural Capital, Our Ecological Insurance

Biodiversity is our real insurance in times of climate change. Traditionally, farmers have increased their resilience by growing more than one crop.

Sir Albert Howard saw in “mixtures,” or biodiversity, the secret of sustainability and stability of farming in India. As he wrote in the 1940 classic on organic farming:

Mixed crops are the rule. In this respect the cultivators of the Orient have followed Nature’s method as seen in the primeval forest. Mixed cropping is perhaps most universal when the cereal crop is the main constituent. Crops like millets, wheat, barley, and maize are mixed with an appropriate subsidiary pulse, sometimes a species that ripens much later than the cereal. The pigeon pea, perhaps the most important leguminous crop of the Gangetic alluvium, is grown either with millets or with maize. The mixing of cereals and pulses appears to help both crops. When the two grow together, the character of the growth improves. Do the roots of these crops excrete materials useful to each other? Is the mycorrhizal association found in the roots of these tropical legumes and cereals the agent involved in this excretion? Science at the moment is unable to answer these questions: she is only now beginning to investigate them. Here we have another instance where the peasants of the East have anticipated and acted upon the solution of one of the problems which Western science is only just beginning to recognize. Whatever may be the reason why crops thrive best when associated in suitable combinations, the fact remains that mixtures generally give better results than monoculture.⁴¹

At Navdanya we have built on this ancient, time-tested knowledge, farming

in nature's ways, based on biodiversity. Not only are we protecting biodiversity, we are increasing food production, farmers' incomes, and resilience in the face of climate change.

On our farm, we have fields of seven (*saptarshi*), nine crops (*navdanya*), and twelve crops (*baranaja*). *Navdanya* in fact means “nine seeds” or “nine crops.” Biodiverse fields always perform better than monocultures. They survive frost and drought, early rain and late rain, too much rain and too little rain.

Biodiverse vs. Monoculture Production⁴²

	Crop	Avg. Production (kg/ha)	Avg. Price (per kg)	Total Income (in Rs)
1	Baira	440	8	3,520
2	Maize	1,280	8	10,240
3	Safed Chemi	600	25	15,000
4	Ogal	360	20	7,200
5	Mandua	600	10	6,000
6	Jhangora	440	15	6,600
7	Urad	600	20	12,000
8	Navrangi	680	20	13,600
9	Koni N0.1	280	10	2,800
10	Lobia	600	20	12,000
11	Til	400	30	12,000
12	Koni N0.2	340	10	3,400
	Total	6,620		104,360
1	Maize	5,400	8	43,200
1	Til	400	30	12,000
2	Safed Chemi	720	25	18,000
3	Mandua	1,120	10	11,200
4	Dholiyia Dal	640	20	12,800
5	Safed Bhatt	760	15	11,400
6	Lobia	800	20	16,000
7	Jhangora	520	15	7,800
8	Maize	560	8	4,480
9	Wheat	480	25	12,000
	Total	6,000		105,680
1	Mandua	3,600	10	36,000
1	Urad	600	20	12,000
2	Moong	520	25	13,000
3	Mandua	560	10	5,600
4	Safed Bhatt	680	15	10,200
5	Dholiyia Dal	560	20	11,200
6	Maize	680	8	5,440
7	Lobia Dal	600	20	12,000
	Total	4,200		69,440
1	Urad	2,400	20	48,000

The baranaja (twelve crops) of bajra (pearl millet), maize (corn), safed chemi (beans), ogal (buckwheat), mandua (finger millet), jhangora (barnyard millet),

urad (black gram), navrangi (rice bean), two varieties of koni (horsetail millet), lobia (bean), and til (sesame) produced more food and earned more than twice that of the corn monoculture. The baranaja or navdanya system of farming is a guarantee against hunger and an insurance against crop failure due to climate variability. In diverse parts of the country, biodiverse agricultural systems outperform monocultures.

Symbiosis among plants contributes to an overall increase in productivity of the crops. In the Western Ghats, a small farm typically has 1.5 acres of paddy, 0.5 acres of areca nut, and a kitchen garden with vegetables that include eggplants, beans, cucumbers, chilies, and small gourds. Likewise, in the eastern Himalayas, especially in Sikkim, the dominant land use is the sustainable *Alnus*-cardamom agro-forestry system, in which cardamom plants and *Alnus* trees are intercropped to the benefit of cardamom production. In Rajasthan too, in the arid tract of Jodhpur and parts of western Rajasthan, neem-based agroforestry and khejri (*Prosopis cineraria*), wherein crops like bajra, sorghum mung, moth bean, and corn are grown together, have fulfilled the nutritional requirements of the communities.⁴³

A recent study conducted by Navdanya in four districts of West Bengal shows that multiple cropping (MC) is economically more efficient than modern intensive chemical farming systems that cultivate monocultures. The net value of the annual production of an average MC farm is uniformly more than that of an average monoculture farm. The MC farms of East Medinipur district are sown with a wide range of crops, both in a sequential rotation and intercropped. Some of these farms—mostly smaller than a hectare in size—grow over 50 types of crops, excluding rice. The rain-fed farms of Bankura district are comparatively less diverse, hardly exceeding 14 crops a year, including rice. The irrigated monoculture farms, by contrast, grow two rice varieties in Bankura district and three rice varieties in East Medinipur district (all high-yielding varieties, or HYV). The cost of all inputs (water for irrigation, seeds, agrochemicals, labor, and energy) was calculated to compare the relative gain in output value of the modern monoculture farms with that of the MC farms, and the remarkable finding was that the value of farm produce increases significantly with greater diversity of crops. Farmers explain this as “farm fatigue” from monoculture and intensive use of agrochemicals—an essential feature of modern agriculture.

These data contradict the prevailing mainstream agronomic view that intensive cultivation of a staple crop enhances productivity. A majority of farmers in Bankura and Medinipur have realized over years that the yield of monoculture farms is unsustainable. Many of these farmers have reverted back to traditional farming systems involving folk crop varieties. Some of them have

experimented with a hybrid system of rotational cropping of a large number of “secondary” crops and an HYV rice. However, most of these MC farmers reported that the cost of the inputs eats away at the extra production of HYV rice and that the best means to cut down on the extraneous inputs is to “give the land a recess” by growing vegetables and fruits for a few years before replanting it with rice.⁴⁴

Small biodiverse farms have higher productivity than monocultures, which are a necessary aspect of industrial agriculture based on external inputs. Higher biological productivity translates into higher incomes for small farmers. In Rajasthan, monocultures of pearl millet yielded Rs 2,480 of net profit per acre, whereas a biodiverse farm of pearl millet, moth bean, and sesame yielded a net profit of Rs 12,045 per acre, nearly five times the profit. In Uttarakhand, a monoculture of paddy yielded Rs 6,720 per acre, whereas a biodiverse farm yielded Rs 24,600 per acre, three and a half times the profit. In Sikkim, a monoculture farm of corn yielded Rs 4,950 per acre while a mixed farm of corn, radish, lahi saag, and peas yielded Rs 11,700 per acre. Navdanya’s rice and wheat farmers have doubled production by using indigenous seeds and organic methods. Jhumba rice in Uttarakhand produces 176 quintals per hectare compared with 96 quintals per hectare of Kasturi, a high-yielding rice variety. The paddy yields are 104 and 56 quintals per hectare, respectively. Farmers in West Uttar Pradesh have gotten yields of 62.5 quintals per hectare using a native wheat variety for organic production, compared with 50 quintals per hectare for chemically produced wheat.

Conservation of native seeds and biodiverse ecological farming have yielded incomes two to three times higher than monoculture farming, and eight to nine times higher than industrial systems using genetically engineered seeds.

Seeds of Freedom, Seeds of Life

Twenty-one years ago, in 1987, I started to save seeds to create a different future than the one envisioned by the biotechnology industry—in which all seeds are genetically engineered and patented. The vision for seed freedom evolved as Navdanya. *Navdanya* means “nine seeds,” and it also means “the new gift.” Through Navdanya, we have brought the new gift of ancient seeds to our farmers. Navdanya builds community seed banks based on rescuing, conserving, reproducing, multiplying, and distributing native varieties or farmers’ varieties—varieties evolved and bred over millennia. On the one hand, our seed saving

defends seeds as a commons—resisting through our daily actions the degraded, immoral, uncivilized idea that seeds are the “intellectual property” of corporations, and that saving them is a crime. On the other hand, Navdanya’s seed banks are the basis of another food economy, one based on biodiversity and cultural diversity, on sustainability, and on the future.

The dominant food economy is based on monopolies and monocultures, on industrialization of production and globalization of distribution of a handful of crops—corn, soy, rice, and wheat. This economy has pushed 1 billion people into hunger; another 2 billion into obesity. It is killing species and farmers. One hundred fifty thousand small farmers of India have committed suicide because they were forced to buy costly, unreliable seed every year from corporations like Monsanto, which collect exorbitant royalties.

Navdanya’s seed saving spreads seeds of life instead of seeds of death. We spread seeds of hope instead of seeds of hopelessness and despair. We spread seeds of freedom instead of seeds of slavery and seeds of suicide.

After the 2004 tsunami, our salt-resistant rice varieties rebuilt the devastated agriculture of Tamil Nadu. Our seeds of *Dehradun* basmati gave us the strength to fight RiceTec of Texas, which had patented basmati rice. Our seeds of native wheat varieties inspired us to fight Monsanto when it patented low-gluten wheat.

Our seeds teach us lessons in diversity and democracy. From our seeds we learn how to defend freedom of biodiversity and freedom of farmers in an age of corporate monopolies, terminator technologies, and globalized monocultures.

A false assumption is growing that we need genetic engineering to deal with climate change.

It is false for a number of reasons. First, nature and farmers have evolved, and continue to evolve, varieties of plants that are resilient to drought, floods, and salinization due to cyclones, three major impacts of climate change.

In Navdanya community seed banks, we have crops like millet that can withstand severe drought; we have rice that grows 18 feet tall and can survive the floods of the Ganges basin. We have rice that can tolerate salt, which we distributed after the Orissa cyclone and the tsunami. The salt-tolerant varieties we have saved, multiplied, and distributed include *Kalambank*, *Kartikpatini*, *Chakaakhi*, *Dhala patini*, *dudeshwar*, *lilabati*, and *luna* (which means “salt”). Flood-resistant rice varieties include *Jalaj*, *Abhiman*, *Bhutna*, *Sada dhepa*, *Sada pankul*, *Jal kalas* (which means “the water pot”), *Bagada*, *Betana*, *Bhundi*, *Champi*, *Fareka*, *Indrijiba*, *Madia*, and *Kala bagada*. In regions that face floods and the ingress of saltwater from the sea, these varieties offer security in the face of climate change.

But rice does not only grow in wet regions. We have also saved hundreds of

drought-tolerant rices, such as *Bhat kalon*, *Chaina*, *Gyarsu*, *Jhumka*, *Ramjawain ukhri*, *Asan leija*, *Bhut moni*, *Kaya*, *Loha*, *Cora*, *Nata*, and *Raja manik*. These are rain-fed rices that need no irrigation.⁴⁵ And there are many varieties of other crops that have the potential to evolve and help us face the growing water scarcity. The drought-resistant native wheats, and the millets like ragi, jhangora, koni, bajra, and jowar are “forgotten foods” that are the foods of the future.⁴⁶

Second, genetic engineering will only allow corporations to take these seeds, appropriate their traits, patent them, and prohibit their use by farmers who don’t make heavy royalty payments. Genetic engineering does not *create* the traits for drought, flood, and salt tolerance; it merely allows the *transfer* of traits across species.

In Navdanya we are creating community seed banks for climate emergencies so that the widest varieties of crops are available to communities to respond to climate-related disasters. And this diversity is available as a commons. Diversity and the commons are the two types of insurance we have in times of uncertainty and unpredictability. Diversity gives us the basis to evolve and adapt under changing conditions. Climate change is not a linear phenomenon that creates warming everywhere, or more rain or less rain. It is nonlinear, and it is better to talk of climate chaos than climate change or global warming. Our community seed banks of climate change-resilient varieties become even more important as the gene giants like Monsanto, DuPont, Syngenta, and Dow apply for patents on climate traits in crops such as drought tolerance and flood tolerance.⁴⁷

In the context of climate chaos, diversity is the basis of adaptation. Monocultures and uniformity are recipes for breakdown. While at the ecological level, we need diversity to respond to climate chaos, at the social and political levels, we need the commons. Monopolies and concentration of ownership of resources enhance vulnerability in periods of chaos.

The mechanistic paradigms on which genetic engineering, intellectual property rights and patents on seeds, and globalized corporate control over food systems are based have given us climate chaos. They cannot help us adapt and evolve. As Einstein said, you cannot solve a problem using the mind-set that created it. Mechanistic thought creates monocultures of the mind. We must move beyond monocultures to protect the earth’s rich diversity and use it to respond to climate chaos.

Humanity has eaten over 80,000 edible plants over the course of its evolution. More than 3,000 have been used consistently. However, we now rely on just eight crops to provide 75 percent of the world’s food. Monocultures are destroying biodiversity, our health, and the quality and diversity of food.

Monocultures have been promoted as an essential component of industrialization and the globalization of agriculture. They don't in fact produce more food. All they produce is more control and profits—for Monsanto, Cargill, and ADM. They create pseudo-surpluses and real scarcity by destroying biodiversity, local food systems, and food cultures.

In 1998, India's indigenous edible oils—made from mustard, coconut, sesame, linseed, and groundnut and processed in artisanal cold-press mills—were banned, with “food safety” used as an excuse. At the same time, restrictions on the import of soy oil were removed. The livelihoods of 10 million farmers were threatened. One million oil mills in villages were closed. More than 20 farmers were killed while protesting against the dumping of soy on the Indian market, which was leading to a fall in the price of domestic oilseed crops. Millions of tons of artificially cheap GM soybean oil continue to be dumped on India.

Women from the slums of Delhi formed a movement to dump soy and bring back mustard oil. “Sarson bachao, soya-bean bhagao” (save the mustard, drive away the soybean) was the women's call from the streets. We did succeed in bringing back mustard through our *satyagraha* (non-cooperation with the ban).

The same companies that dumped soy on India—Cargill and ADM—are destroying the Amazon to grow soy. Millions of acres of rain forest—the lungs, the liver, the heart of the global climate system—are being burnt to grow soy for exports. Armed gangs take over the forest and use slaves to cultivate soy. When people like Sister Dorothy Stang oppose the destruction of the forests and the violence against people, they are assassinated.⁴⁸

While people in Brazil and India are being threatened directly by these agribusiness monocultures, people in the US and Europe are also at risk. Eighty percent of soy production is being used as cattle feed to provide cheap meat. Cheap meat that is, in effect, destroying both the Amazon rainforest and people's health in rich countries. One billion people are without food because industrial monocultures robbed them of their livelihoods in agriculture and their food entitlements.⁴⁹ Another 1.7 billion are suffering from obesity and food-related diseases. Monocultures lead to malnutrition—for those who are underfed as well as those who are overfed.

Corporations are forcing us to eat untested GMO food. Soy is in 60 percent of all processed food. It has high levels of isoflavones and phytoestrogens, which produce hormone imbalances in humans. Traditional fermentation, as in the food cultures of China and Japan, reduces the levels of isoflavones. The promotion of soy in food is a huge experiment promoted with \$13 billion in subsidies from the

US government between 1998 and 2004, and \$80 million a year from the American soy industry.⁵⁰ Nature, culture, and people's health are all being destroyed. Local food cultures have rich and diverse alternatives to soy. For protein we have thousands of varieties of beans and grain legumes—the pigeon pea, chickpea, mung bean, urd bean, rice bean, adzuki bean, moth bean, cowpea, lentil, horse gram, and fava bean. For edible oils we have sesame, mustard, linseed, saffola, sunflower, groundnut.

With the spread of monocultures and the destruction of local farms, the food system has become dependent on fossil fuels—for synthetic fertilizers, for running giant machinery, for long-distance transport. We are increasingly eating oil, not food, threatening the planet and our health.

Moving beyond monocultures of the mind has become an imperative for repairing the food system. Biodiverse small farms have higher productivity and they generate higher incomes for farmers. And biodiverse diets provide more nutrition and better taste.

Bringing back biodiversity goes hand in hand with bringing back small farms. Corporate control thrives on monocultures. Citizens' food freedom depends on biodiversity. Human freedom and the freedom of other species are mutually reinforcing, not mutually exclusive.

Rebuilding Local Food Communities

The globalized food system is causing destruction at every level. Biodiversity is being destroyed in favor of monocultures of corn, soy, and canola. Food has been reduced to a commodity. And the commodity can run a car, feed animals in factory farms, or feed people. Uniqueness, distinctiveness, quality, nutrition, and taste are no longer in the equation.

Farmers are being destroyed because prices of farm products are driven down through a combination of monopolistic buying by global corporations and dumping of subsidized products. In the meantime food prices keep rising for the poor, and hunger grows. The long-distance transport of food pollutes the atmosphere with carbon dioxide emissions from fossil fuels. No one is gaining from globalized trade in food except the corporations. Localization of food systems to reduce food-miles is a climate-change imperative. It is also a food-sovereignty and human-rights imperative. Small farmers will only survive in the context of vibrant and robust local food economies.

Localization is also a food-security imperative. Short supply chains ensure

better democracy in distribution, better-quality food, fresher food, and more cultural diversity. In India, the movement for retail democracy is a vital part of keeping local markets alive. Across the world, farmers markets are reappearing. The search for local foods to reduce food-miles and create more intimate food systems has created a new dichotomy between “organic” and “local.” In my view this is a false dichotomy. To be organic means to be whole and wholesome—for the earth, for our bodies. Food that could have been grown next door but has been imported from thousands of miles away is not organic by any ecological standards. If we care about getting rid of toxins in our food, we should also care about the atmospheric pollution that is causing climate change. They are two facets of ecological destruction. A nonviolent, wholesome food system should have place for neither. Organic that leaves out food-miles is not fully organic. Organic that leaves us feeling strangers on the land is not truly organic.

As Michael Pollan observes in his book, *The Omnivore’s Dilemma*,

One of the key innovations of organic food was to allow some more information to pass along the food chain between the producer and the consumer—an implicit snatch of narrative along with the number. A certified organic label tells a little story about how a particular food was produced, giving the consumer a way to send a message back to the farmers that she values tomatoes produced without harmful pesticides or prefers to feed her children milk from cows that haven’t been injected with growth hormones. The word organic has proved to be one of the most powerful words in the supermarket: Without any help from government, farmers and consumers working together in this way have built an \$11 billion industry that is now the fastest growing sector of the food economy.

Yet the organic label itself—like every other such label in the supermarket—is really just an imperfect substitute for direct observation of how a food is produced, a concession to the reality that most people in an industrial society haven’t the time or the inclination to follow their food back to the farm, a farm which today is apt to be, on average, fifteen hundred miles away. So to bridge that space we rely on certifiers and label writers and, to a considerable extent, our imagination of what the farms that are producing our food really look like. The organic label may conjure an image of a simpler agriculture, but its very existence is an industrial artifact. The question is, what about the farms themselves? How well do they match the stories told about them?⁵¹

Organic farming is based on ecological processes and principles of agroecology. It is also based on human communities working in cooperation and with dignity and freedom.

There was an old conflict between chemical-industrial agriculture and organic farming. There is a new conflict emerging between *authentic organic*, based on small, biodiverse farms, and *pseudo-organic*, based on large-scale, monoculture corporate farms that grow for export. Authentic organic farming is based on biodiversity, small family farms, local markets, and fair trade. Organic farming emerged as a systemic alternative to industrial agriculture, which destroyed biodiversity, polluted ecosystems and food with agrochemicals,

uprooted and displaced small farmers, and undermined local markets through subsidized long-distance transport.

Pseudo-organic farming destroys small farms and uproots small farming communities to create large export-oriented industrial farms in which farmers are viewed as laborers and serfs, instead of sovereign producers. Pseudo-organic farming is based on destruction of biodiversity and creation of monocultures. It does not abide by the essential ecological processes of renewal of soil fertility, rejuvenation of water, and biodiversity. It merely substitutes chemical inputs with “organic” inputs. This is input substitution, not agroecology.

Agroecology is the scientific basis of authentic organic farming. Authentic organic practices are based on principles of self-organization—from the level of the organism to the farm and agroecosystem to the community. Ecologically, self-organization refers to the capacity of living organisms and agro-ecosystems to renew fertility by rejuvenating soil microorganisms and recycling organic matter; to manage pests through building resilience and maintaining a pest-predator balance; to conserve water; and to conserve and renew biodiversity. Seed giving rise to seed and earthworms rejuvenating soil fertility are examples of the self-organizing capacity of nature and living systems, which are the basis of a sustainable agriculture.

Socially, self-organization is encapsulated in Gandhi’s *swaraj* (self-rule, self-governance, self-organization). It is the basis of food sovereignty—the right to produce in freedom. Social and ecological self-organization reinforce each other. Only small farmers working in cooperation with the soil and plants can provide the care and attention required to facilitate nature’s self-organization. Food sovereignty, therefore, rests on agroecology. Both are built on the principle of self-organization. Self-organized production rests on the principles of agroecology, and self-organized distribution rests on the principles of localization—local consumption through local markets. Such economic self-organization ensures that local food needs are met and local food security and livelihoods strengthened, preventing malnutrition, hunger, poverty, and unemployment. It also provides the ground for cultural diversity in food systems, supported by biodiversity in agricultural systems.

Pseudo-organic agriculture is built on the destruction of the self-organizing capacity of human communities and agroecosystems. It mimics industrial agriculture, focusing on largescale production for export, uprooting small farmers, and undermining people’s food security and sovereignty. Large-scale, industrial-style, export-oriented pseudo-organic farms are run by giant corporations for profits at the expense of the health of the earth, diverse species, and local communities. The entry of multinational corporations in organic

agriculture is based on land reforms for the rich, which usurp the lands of poor and marginal farmers. This is what is happening in Punjab, where the government is taking over land by force from small farmers and handing it over to corporations planning to export “organic” vegetables and fruits. Just as chemical farming and GM seeds are driving farmers into debt and suicide, pseudo-organic farming, which is corporate and export-driven, is also killing farmers by taking away their land, their livelihoods.

An agriculture that destroys biodiversity, uproots local farmers, and leaves local communities without food is not worthy of the label “organic.” To be organic is to be just and fair. An agriculture that turns rural areas into graveyards for farmers cannot be called organic. *Organic* means life-giving. Authentic organic farming gives life. Pseudo-organic farming ends life. To remain authentic, organic farming must be biodiverse, it must stay in the hands of small farmers, and it must deepen food sovereignty.

In Navdanya, we work on the following principles of organic and local:

- Food for the soil and her millions of microorganisms

Organic can be organic only if the food rights of millions of soil organisms are protected. This involves the law of return, of growing food for the soil, not just growing commodities for the market. In fact all “developments” in industrial agriculture are methods of increasing commodity production at the expense of the soil. The Green Revolution, with its chemical-intensive dwarf varieties, killed the soil organisms and used techniques that did not return organic matter to the soil. Genetically engineered herbicide-resistant crops, like Roundup Ready soy and corn, deliberately kill vegetation that would have gone back to feed the soil. Feeding markets while starving the soil is a recipe for hunger and desertification. If we feed the soil, we will also feed people, and even have quality production for the market.

- Food and nutrition for the farming family

The tragedy of industrialized, globalized agriculture is that while commodity markets grow, people starve. More than 1 billion people are now permanently hungry. Most of them are from rural areas. Many of them are food producers. They are denied food either because their soils have been desertified or because chemical agriculture and costly seeds have got them into debt or because they are growing cash crops like cotton and coffee, which bring insufficient returns because globalized trade has pushed down farm prices, or because they have been pushed off the land. It is criminal that our *annadatas*, our food providers, should themselves be hungry. That is why we ensure that every producer family that is a member of Navdanya first grows healthy and nutritious foods for the household and only trades any surplus.

- Food for local communities

Everyone must eat. If food is not grown locally, local communities will have to import their food from somewhere far away. That food will be more contaminated and adulterated and less safe. If local communities do not eat local produce, biodiversity will disappear from our farms and cultural

diversity will disappear from our diets, making both the land and its people poorer.

- Unique products for long-distance trade and exports

Every part of the earth is productive. Every culture on the earth has evolved its diet according to the particular ecosystem it inhabits. As much as possible, food staples must be grown locally, both to produce what the ecosystem is best suited for, and to produce what local cultures have adapted themselves to.

Trade in food must be restricted to what cannot be grown locally; it must be restricted to foods with both a high value and a small ecological footprint in terms of land and water use.

Different vegetables and fruits grow in different climates. It is wrong to grow temperate-zone vegetables in the tropics and fly them back to rich consumers. This uproots local peasants, creates hunger and poverty, and destroys local agrobiodiversity. It also blocks the potential for localization in importing countries. Since vegetables and fruits are perishable, transporting them long distances is highly energy-intensive, contributing to climate change.

In India, the home of the mango, the Alphonso is only traded and eaten in Maharashtra and Goa, where it grows, and the Dasherri is largely eaten in the northern regions where it grows.

Global trade in perishables destroys the biodiversity of fruits and vegetables. One kind of Chiquita banana, one kind of Washington apple ends up on every table. Local production for local consumption is the best way to conserve biodiversity, taste, and quality.

Spices are a perfect candidate for long-distance trade. Tiny quantities are needed to add flavor to food. Spices grow in very specific ecosystems. They cannot be grown everywhere. They give high value with low volumes. This benefits the producer, who can also grow food. In Karnataka, spice growers use 10 percent of their land for spice gardens of pepper, cardamom, and areca nut and 10 percent for paddy for local consumption. These gardens have existed for centuries and are a model for farming that supports trade but is not destroyed by trade.

“Spice of life trade” is justified when it enriches the giver and the receiver.

Re-localization of our food systems has become an ecological and social imperative. Richard Heinberg, one of peak oils’ preeminent theorists, has pointed out that this will require the deindustrialization of agriculture.

The general outline of what I mean by de-industrialization is simple enough: this would imply a radical reduction of fossil fuel inputs to agriculture, accompanied by an increase in labour inputs, and a reduction in transport, with production being devoted primarily for local consumption. Fossil fuel depletion almost ensures that this *will* happen. But at the same time, it is fairly obvious that if

we don't *plan* for de-industrialization, the result would be catastrophe.⁵²

Rob Hopkins, the inspiration behind the new transition-culture movement, elaborates on how energy-descent plans, or “powering down” of fossil fuel use, can be a “powering up” of quality of life.

The essence of an energy descent plan is that it creates a vision of an abundant low energy future. While the transition away from fossil fuels will be a task of unprecedented proportions, at the same time it offers the potential for a society which is better in many ways, more connected to nature, healthier with more meaningful work, access to nutritious food, enhanced social capital, and more cooperation.⁵³

Climate Change and the Two Carbon Economies: Biodiversity vs. Fossil Fuels

Reductionism seems to have become the habit of the contemporary human mind. We are increasingly talking of climate change in the context of “the carbon economy.” We refer to “zero carbon” and “no carbon” as if carbon exists only in fossilized form under the ground. We forget that the cellulose of plants is primarily carbon. Humus in the soil is mostly carbon. Vegetation in the forests is mostly carbon. It is living carbon. It is part of the cycle of life.

The problem is not carbon *per se*, but our increasing use of fossil carbon that was formed over millions of years. Today the world burns 400 years' worth of this accumulated biological matter every year, three to four times more than in 1956. While plants are a renewable resource, fossil carbon for our purposes is not. It will take millions of years to renew the earth's supply of coal and oil.

Before the industrial revolution, there were 580 billion tons of carbon in the atmosphere. Today there are 750 billion tons. That accumulation, the result of burning fossil fuels, is causing the climate-change crisis. Humanity needs to solve this problem if we are to survive. It is the other carbon economy, the renewable carbon embodied in biodiversity, that offers the solution.

Our dependence on fossil fuels has broken us out of nature's renewable carbon cycle. Our dependence on fossil fuels has fossilized our thinking.

Biodiversity is the alternative to fossil carbon. Everything that we derive from the petrochemical industry has an alternative in the realm of biodiversity. The synthetic fertilizers and pesticides, the chemical dyes, the sources of mobility and energy, all of these have sustainable alternatives in the plant and animal world. In place of nitrogen fertilizers, we have nitrogen-fixing

leguminous crops and biomass recycled by earthworms (vermi-compost) or microorganisms (compost). In place of synthetic dyes, we have vegetable dyes. In place of the automobile, we have the camel, the horse, the bullock, the donkey, the elephant, and the bicycle.

Climate change is a consequence of the transition from biodiversity based on renewable carbon economies to a fossil fuel-based non-renewable carbon economy. This was the transition called the industrial revolution.

While climate change, combined with peak oil and the end of cheap oil, is creating an ecological imperative for a post-oil, postfossil fuel, postindustrial economy, the industrial paradigm is still the guiding force for the search for a transition pathway beyond oil.

That's because industrialization has also become a cultural paradigm for measuring human progress. We want a post-oil world but do not have the courage to envisage a postindustrial world. As a result, we cling to the infrastructure of the energy-intensive fossil fuel economy and try and run it on substitutes such as nuclear power and biofuels. Dirty nuclear power is being redefined as "clean energy." Non-sustainable production of biodiesel and biofuel is being welcomed as a "green" option.

Humanity is playing these tricks with itself and the planet because we are locked into the industrial paradigm. Our ideas of the good life are based on production and consumption patterns that the use of fossil fuels gave rise to. We cling to these patterns without reflecting on the fact that they have become a human addiction only over the past 50 years and that maintaining this short-term, non-sustainable pattern of living for another 50 years comes at the risk of wiping out millions of species and destroying the very conditions for human survival on the planet. We think of well-being only in terms of human beings, and more accurately, only in terms of human beings over the next 50 years. We are sacrificing the rights of other species and the welfare of future generations.

To move beyond oil, we must move beyond our addiction to a certain model of human progress and human well-being. To move beyond oil, we must reestablish partnerships with other species. To move beyond oil, we must reestablish the other carbon economy, a renewable economy based on biodiversity.

Renewable carbon and biodiversity redefine progress. They redefine development. They redefine "developed," "developing," and "underdeveloped." In the fossil fuel paradigm, to be developed is to be industrialized—to have industrialized food and clothing, shelter and mobility, ignoring the social costs of displacing people from work and the ecological costs of polluting the atmosphere and destabilizing the climate. In the fossil fuel paradigm, to be

underdeveloped is to have nonindustrial, fossil-free systems of producing our food and clothing, of providing our shelter and mobility.

In the biodiversity paradigm, to be developed is to be able to leave ecological space for other species, for all people and future generations of humans. To be underdeveloped is to usurp the ecological space of other species and communities, to pollute the atmosphere, and to threaten the planet.

We need to change our minds before we can change our world. This cultural transition is at the heart of making an energy transition to an age beyond oil. What blocks the transition is a cultural paradigm that perceives industrialization as progress combined with false ideas of productivity and efficiency. We have been made to believe that industrialization of agriculture is necessary to produce more food. This is not at all true. Biodiverse ecological farming produces more and better food than the most energy- and chemical-intensive agriculture. We have been made to falsely believe that cities designed for automobiles provide more effective mobility to meet our daily needs than cities designed for pedestrians and cyclists.

Vested interests who gain from the sale of fertilizers and diesel, cars and trucks, have brainwashed us to believe that chemical fertilizers and cars mean progress. We have been reduced to buyers of their non-sustainable products rather than creators of sustainable, cooperative partnerships—both within human society and with other species and the earth as a whole.

The biodiversity economy is the sustainable alternative to the fossil fuel economy. The shift from fossil fuel-driven to biodiversity-supported systems reduces greenhouse gas emissions by emitting less and absorbing more CO₂. Above all, because the impacts of atmospheric pollution will continue even if we do reduce emissions, we need to create biodiverse ecosystems and economies because only they offer the potential to adapt to an unpredictable climate. And only biodiverse systems provide alternatives that everyone can afford. We need to return to the renewable carbon cycle of biodiversity. We need to create a carbon democracy so that all beings have their just share of useful carbon, and no one is burdened with carrying an unjust share of climate impacts due to carbon pollution.

CONCLUSION

Unleashing Shakti: Our Power to Transform

The triple crisis of climate change, peak oil, and the food and agrarian crisis are creating an imperative for change, to make a transition to an age beyond oil.

However, if that transition is driven by the same paradigm and powers that created our current climate crisis, it will only perpetuate the problems of ecological non-sustainability and social and economic injustice. Our food is being converted to fuel to run an industrial infrastructure. The lands of the poor are becoming the next oil fields. But there is not enough land to fuel the ever-increasing population of cars and ever-increasing demand for energy. And when the rights of the poor are taken into account, there is only one way forward—reducing the energy demands of the rich and the non-sustainable patterns of production and consumption that are the legacy of industrialization and globalization.

The powerful corporations, governments, and elites will, of course, try to avoid any reduction of their profits and power, preferring instead to make the poor pay. A top-down model for sustainability results in pseudo-sustainability and eco-imperialism. A bottom-up search for sustainability creates an Earth Democracy based on living economies. Unleashing our living but latent energies can create new economic and political possibilities. But recognizing the emergent possibilities requires a paradigm shift from a mechanistic worldview and its limited and limiting categories of mechanical energy. The transition to a post-fossil fuel age needs to focus on living energies—our energy to creating living democracies and living economies.

A universe run on the paradigms and fossil fuel energy of the mechanical, industrial age dispossesses and displaces billions. In the final analysis it

displaces humanity from being a part of the future evolution of the planet.

The climate change discussion has focused exclusively on two constructs, “growth” and “industrial development.” Both are artificially restrictive. The mechanistic paradigm holds that there must be growth to eliminate poverty, that there must be “industrial development” for the poor of the South to achieve the living standards of the rich of the North. However, as the examples in this book show, both people and nature are being impoverished to make the economy grow. India’s 9 percent growth is based on pushing small farmers to suicide; uprooting tribals for mines, factories, and Special Economic Zones; and damming every inch of every river. If measured in terms of nature’s economy and people’s economy, this growth would register as destruction. We need to move beyond the mechanistic paradigms of industrialization and neoclassical economics if the rights of the poor are to be defended and if future generations are to have a chance to live on this planet.

For too long the very instruments that threaten the poor have been proposed as solutions to poverty. Industrial production systems, which displace people and lay waste to the resources of the earth, are currently reaching the last village, the last mountain in India. And the globalized economy, based on the assumption that perpetual growth is possible, is outsourcing its pollution and resource burden on the poor. To address the issues of energy change and climate change, we need to look at what is happening to people and the soil.

Living Systems, Living Energies

In the dominant paradigm, “energy” refers to oil and coal, which are mined from the earth, shipped thousands of miles, and transformed into electricity to light up neon signs or into fuel to run SUVs. It is fruitful to remember that energy has other meanings and other forms. From Shakti, the generative force of the universe, to the sun that powers our lives, to the water that comes to us as bountiful rain or a flood or a tsunami, to the air and the wind that move the clouds and create the climate. Energy is not just oil and gas. Energy is an all-pervasive element of life.

The broader our paradigm of energy, the wider our choices as human beings. Fossil fuels have fossilized our imagination, our potential, our creativity. We need to break free of this fossilization to choose life-enhancing pathways for our selves, our species, and the planet.

The mechanistic paradigm has robbed us our freedom and creativities. It has

replaced living energy with fossil fuels; the wealth created by nature and people with capital; the freedom of citizens and communities with the coercive power of the corporate state, which imposes the rule of capital in every dimension of our life—the thoughts we think, the food we eat, the settlements we shape.

In this life-threatening period of globalization and climate crisis, we need to unleash our hidden energies to make a transition to a post-fossil fuel economy. To do so, we need to reinvent democracy. A renewable-energy economy will only be built through the renewable energy of free and self-organized citizens and communities. The transition beyond oil is not merely a technological transition—it is above all a political transition in which we stop being passive and become active agents of transformation by recognizing that we have the capacity, the energy, and the creativity to make the change.

Life is based on the self-organizing energies of the universe, from cells to Gaia, from communities to countries. We as living systems are networks of chemical and energetic flow and transformation. Thus life is energy—not fossil fuel energy, but *living* energy.

Rather than bounty, mechanistic science, industrial technology, and the market economy have created scarcity—scarcity of food and water, scarcity of work and energy, scarcity of happiness. They have worked in concert to increase human dependence on energy from fossil fuels and, in the process, have destroyed life on earth and meaningful, creative work for all. A false paradigm of growth that allows the rich to get richer has sucked out the resources from the planet and the joy of living from humanity.

Reductionist, mechanistic science creates scarcity by blinding itself to the connections that support and maintain the cycle of life, the energy flows that are based on self-organization. Industrial technology creates scarcity by using ever more resources and energy while externalizing the social and ecological costs of its appetite. By substituting fossil fuels for people (without providing access to sustenance), industrial technology also creates a scarcity of work and well-being.

The market economy creates scarcity by promoting a consumerist culture based on overextraction and overconsumption of energy, robbing the poor, other life-forms, and future generations of their share of resources and energy. The vision of those who promote this paradigm is based on the illusion of endless growth. They don't acknowledge that endless growth requires endless extraction of resources and endless generation of waste.

Shakti: The Creative Energy of a Self-Organizing

Universe

Energy is Shakti—the primordial power of creation, the self-organizing, self-generative, self-renewing creative force of the universe in feminine form. Shakti comes from the root “sak,” meaning “the capacity to do,” “to have power.” Shakti is power. Shakti is force. Shakti is the personification of primordial energy and the source of all divine and cosmic evolution and also the source and controller of all forces and potentialities of nature. The universe is an expression of Shakti and an infinite reservoir of power.

The ancient Indian text, an ode to the creative energy of the universe, *Devi Bhagwat Puran* has referred to Shakti as the power inherent in all beings, the force under whose control the world is working and is compelled to work according to her wishes. “There is nothing that is not joined to me. I am the power behind everything.”¹

Sister Hildegard of Bingen, a 12th century abbess, writer, and musician, in a very different context, wrote:

I am the supreme and fiery force, who kindled every living spark, and I breathed forth no deadly thing—yet I permit them to be. ... And I am the fiery life of the essence of God. I flame above the beauty of the fields; I shine in the waters, I burn in the sun, the moon, and the stars. And with the airy wind, I quicken all things vitally by an unseen, all-sustaining life. For the air is alive in the verdure and the flowers; the waters flow as if they lived; the sun too lives in its light; and when the moon wanes it is rekindled by the light of the sun, as if it lived anew.²

When we think of energy as Shakti, we embed ourselves in Gaia’s laws. We unleash our own inner and outer energies for the radical transition needed to avert climate disaster.

In Indian philosophy, we define energy as *satvik* (or *sattva*), *rajsik* (or *rajas*), and *taamsik* (or *tamas*). These energies define the quality of foods we eat and the personalities we become. The three energies reflect three qualities. Vinod Verma writes,

Sattva is the quality of truth, virtue and equilibrium; rajas denotes the notion of force and impetus; tamas restrains, obstructs and resists motion.... In relation to the body tamas, rajas and sattva represent the physical, the subtle, and the spiritual aspects of human existence. In the cosmic view, rajas denotes the creative principle of the universe, tamas is the devouring principle, and sattva is the principle of energy and life.³

A former Jain monk, the editor of *Resurgence* magazine and founder of Schumacher College in the UK, Satish Kumar is the author of *Spiritual Compass: The Three Qualities of Life*, which frames satvik, rajsik, and taamsik

as guides to our journey to sustainability and living lightly on the planet. He translates satvik as elegant, simple, natural, and real; that which rings true at the deepest level. Rajsik is translated as extravagant and excessive, and taamsik as dark and depressing.⁴

Humanity has allowed the devouring taamsik energy to overtake our minds and the way we live. Our consumption and production systems are devouring the planet. This entropic degradation is putting our very existence at risk. The laws of thermodynamics state that energy will naturally flow from a state of higher organization to a state of lower organization. To move in the opposite direction requires special energy; Satvik energy can move us from lower to higher levels of self-organization, from ecological degradation to ecological renewal.

Rejuvenating Work for Living Economies

Work is energy. Two crises of our times are intimately connected—the climate crisis and the unemployment crisis. As long as we address these crises separately, we will not solve either. The climate crisis is normally addressed scientifically in diagnosis and economically in terms of solutions. However, economic solutions are offered only within the global market and commodification paradigm. The most recent example of this is the Stern report. In it, people do not figure anywhere. However, substituting the work of people with work done by machines running on fossilized carbon is a major contributor to the pollution of the atmosphere and the climate crisis. Bringing work to people and people to work can be a significant solution of the crisis of human disposability and the crisis of climate disruption.

As the fossil fuel economy has grown, it has substituted energy for humans. On the one hand, this has rendered humans redundant to the economic enterprise of production. It has created the crisis of poverty and unemployment, of dispensability and disposability. On the other hand, it has led to the problem of carbon pollution. Whereas humans are sustained by renewable carbon embodied in plants and biomass, industrial energy consumes fossil fuel, adding more CO₂ to the atmosphere than the planet can recycle.

Industrialization replaces renewable energy of humans and other animals with non-renewable energy. Even though industrial production requires more resources than feeding humans, replacing people with fossil fuel-driven machines is defined as a gain in productivity.

Scientific and technological progress is guided by this narrow and distorted

concept of productivity, which externalizes the social costs of livelihood destruction and the ecological costs of ecosystem destruction.

Economic policy seeks to subsidize the spread of industrialization and fossil fuel addiction. Thus fertilizers are subsidized in industrial agriculture and exports are subsidized in global food systems. Petrochemicals are subsidized; coal and oil are subsidized. Besides the direct financial subsidies are indirect subsidies such as funding the construction of the massive infrastructure for the fossil fuel economy, such as highways. The ultimate subsidy to the fossil fuel economy is militarized support for the extraction of fossil fuels, with the oil wars being an example. Socially, ecologically, and economically high-cost systems of food production, energy production, and transportation are made artificially cheap. Subsidies for globalized, industrialized food make local, organic food appear more expensive; subsidies for private cars and air travel make train travel appear more expensive; subsidies to supermarkets make local shops appear more expensive.

These economic distortions lead to social and cultural distortions. Creative and productive human beings are laid to waste, and creative work is perceived as second-class work, as obsolete. Craftspeople are expected to disappear for industrialized production of petro-products; small farmers are expected to disappear to make way for industrial agriculture. And with the disappearance of creative work comes the disappearance of the very knowledge and skills we need to shape economics beyond oil.

We need to shift the way we define and perceive physical work. Replacing people has been defined as liberating people from work. Physical work has been defined as drudgery and as degrading. When the communist government of West Bengal passed a law to ban hand rickshaws, it justified it on the grounds that it was putting an end to “degrading” work. The rickshaw pullers organized protests and said there was nothing degrading in their work. In fact, it is being without work that is degrading. When Monsanto introduced Roundup, a broad-spectrum herbicide that “kills everything green that comes in contact with it,” it proclaimed it was liberation from weeding. Destroyers of work and employment always present destruction as liberation. Yet there is no liberation in being robbed of one’s productive capacity. There is no dignity in being treated as disposable.

To make the energy transition beyond oil, we need to bring people back into the economy, bring human energy back *into* production, respect physical work, and *give it* dignity. Climate change is giving us adequate warning that substituting people with fossil fuel is ecologically non-sustainable. Social and cultural breakdown should likewise be recognized as a warning about the social

non-sustainability of displacing people from work.

There are multiple levels at which the human capacity to do work needs to be put back into the energy equation and into human destiny. First is the internal energy of human beings in all its dimensions—spiritual, cultural, emotional, intellectual, physical. If it is tapped, this is the largest energy source we have—inexhaustible, replenishable, and ever enlarging. Human energy, combined with the energy of the sun, the plants and animals, the wind, the air, the sea and water, the soil and the earth, is both the basis of work and livelihoods and the source of sustainable and renewable energy.

The Law of Entropy and the Principle of Emergence

In the larger ecological context, mechanistic growth based on mechanical engineering is leading to impoverishment, both material and spiritual. It is entropic growth—growth that increases waste and pollution. Climate change is, in fact, a result of increasing pollution and increasing entropy.

In *The Entropy Law and the Economic Process*, Nicholas Georgescu-Roegen, a former professor at Vanderbilt University and the inspiration for all ecological economists, wrote,

Neoclassical economics is mechanistic, like classical mechanics is. It can neither account for the existence of enduring changes in nature nor accept this existence as an independent. Nature on which the economic enterprise and industrial enterprise rests is totally discounted. Neoclassical economists left nature out of their calculus because the “no deposit, no return” analogue befits the businessmen’s view of economic life. For if one looks only at money, all he can see is that money just passes from one hand to another: except by a regrettable accident it never gets out of the economic process. Perhaps the absence of any difficulty in securing raw materials by those countries where modern economies grow and flourished was yet another reason for economists to remain blind to this crucial economic factor. Not even the wars the same nations fought for the control of the world’s natural resources awoke economists from their slumber.⁵

The mechanistic worldview and the industrialization process have promoted the overextraction and overconsumption of resources, and yet it externalizes the energy and resources it uses, as well as the waste and pollution it generates, from the calculus of productivity and profit. This has led to three interrelated effects: the exhaustion of natural resources; the build-up of pollution; and the destruction of people, communities, and cultures.

While the dominant paradigm can ignore the creation of waste, the human activities that it is promoting are subject to the law of entropy. The term *entropy* was first introduced by German physicist Rudolf Clausius more than 100 years

ago. Essentially it describes the movement of any system toward disorder.

Georgescu-Roegen illustrates entropy with a simple example of coal being used to run a railway engine. The burning coal creates steam, which runs the engine, and makes the train move. This conversion of coal, into motion on the one hand, and ashes on the other, is subject to the first law of thermodynamics—the conservation of matter and energy. But there is another change taking place, which is governed by the law of entropy, or the second law of thermodynamics. Georgescu-Roegen writes,

At the beginning of the process before burning, the chemical energy of the coal is free, in the sense that it is available to us for producing some mechanical work. In the process, however, the free energy loses this quality, bit by bit. Ultimately, it always dissipates completely into the whole system where it becomes *bound* energy, that is energy which we can no longer use for the same purpose.⁶

Entropy is a measure of the bound energy of a system. In a system with a high level of entropy, most or all of its energy is bound; in one with a low level of entropy, the opposite is true. The degradation of energy from free to bound has also been referred to as a turning of order into disorder. In an ordered structure, energy is free, while a structure with bound energy is chaotic, disordered.

Life has the capacity to oppose the qualitative degradation to which inert matter is subject. In fact, life may be characterized as the capacity to create order, to create free energy, to create diversity, to evade the law of entropy.

Georgescu-Roegen has called the possibility of life to move from high to low entropy “entropic indeterminateness.”

Without the entropic indeterminateness it would not be possible for a living creature to maintain its entropy constant. Nor would it be possible to reverse entropy from high to low. Above all, it would be impossible for the living forms to offer environmental low entropy and use it in manners as strikingly diverse as that of a bacterium, a lobster, a butterfly, tumbleweed, a *Homo sapiens*, and so on down the potentially limitless list.⁷

Georgescu-Roegen has called the principle that allows life to new patterns “the principle of emergence of novelty by combination.” Living energies allow us to break out of the predictable destructiveness and dissipation inherent to mechanical energy. They create new material possibilities for well-being without ecological destruction. We become cocreators and coproducers with nature’s creative renewing processes.

Reductionism, which focuses only on parts, cannot see or understand the unexpected creativity of natural processes. The unexpected, orderly patterns created from disorderly elements are known as emergent properties. As eminent

development biologist Brian Goodwin points out:

It is now recognized that emergent properties are very widespread in nature, particularly in living systems. Many of the most intriguing characteristics of life, such as the way a complex organism emerges from the interaction of many cells during embryonic development, or the pattern of species extinctions during evolution, are unexpected results of particular patterns of interaction between components in complex systems.⁸

Such emergence is characteristic of all living systems. When society is organized on the basis of the living energy of its citizens, living democracy is an emergent phenomena, making the impossible possible, creating hope out of hopelessness, unleashing our creative energies in the midst of ecological and social ruin.

Human beings, as living beings, have a choice between two alternatives—the entropic option or an emergent option. The former locks us in a mechanical worldview, based on mechanistic science, mechanical production, and a mechanistic economics whose myth of perpetual growth leads us to death, decay, and disintegration. The latter relies on ecological science, ecological production, and ecological, living economies that provide new levels of human enrichment through biological and cultural diversity. The choice we make will decide whether or not we survive as a species. And the possibilities are not restricted by the present range of choices; in society, too, there are emergent properties, unpredictable forms that emerge through new associations, new networks, new solidarities.

The high-entropy journey humanity has undertaken under the illusion of growth and progress does not have a future. We will have to change the road we are on, and we will have to change our goals. The goals cannot be set by reductionist science, industrial technologies, and neoliberal economies. The goals cannot be narrowly defined as economic growth or consumerism. The goals have to be the preservation of the earth, her diverse species, and future generations.

As Edgar Cahn, creator of time dollars and time banking writes,

Human beings inhabit an ecological niche on this planet. None of us “pay” full fare. We all draw upon a common stock: the cumulative knowledge and discoveries of our ancestors, the genetic pool that has demonstrated superior survival capacity, and the ecosystems that provide us with a vast array of essential goods and services. Our ability to survive this next millennium will depend upon whether we function to exploit and deplete or, as trustees, to conserve, develop, and enhance that legacy. ...

The challenge our species now faces is to create a healthy ecological niche: communities that nurture, space that is non-toxic, exchanges that do not deplete, relationships based on love and caring, transactions powered by the renewable energy of compassion and empathy and reciprocity.⁹

The multiple crises of climate insecurity, energy insecurity, and food insecurity create an imperative and an opportunity to transcend the limits of the mechanistic-industrial-capitalist paradigm that has been systematically shrinking our potential even as it peddles progress.

The paths out from this crisis are not being blazed in the boardrooms of the global corporations who dominate our world today and are largely responsible for crimes against nature and humanity. Industrialization of food and agriculture has put the human species on a slippery slope of self-destruction and self-annihilation. The movement for biodiverse, ecological, and local food systems simultaneously addresses the crises of climate, energy, and food. Above all, it brings people back into agriculture and reclaims food as nourishment and the most basic source of energy. New ways of thinking and acting, of being and doing, are evolving from the creative alternatives being employed in small communities, on farms, and in cities.

It is this renewable energy of ecology and sharing, of solidarity and compassion, that we need to generate and multiply to counter the destructive energy of greed that is creating scarcity at every level—scarcity of work, scarcity of happiness, scarcity of security, scarcity of freedom, and even scarcity of the future.

Climate chaos, brutal economic inequality, and social disintegration are jointly pushing human communities to the brink. We can either let the processes of destruction, disintegration, and extermination continue unchallenged or we can unleash our creative energies to make systemic change and reclaim our future as a species, as part of the earth family. We can either keep sleepwalking to extinction or wake up to the potential of the planet and ourselves.

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